



**KTH Industrial Engineering
and Management**

Title: Energy Modelling in Residential Houses: A Case Study of Single Family
Houses in Bahir Dar City, Ethiopia

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Summary/Abstract

Several studies have been conducted and revealed that household is the major energy consumer sector in developing countries like Ethiopia. This study focuses on evaluation of the existing residential energy consumption and projection of the energy demand.

The energy consumption has been studied by conducting survey on 350 households using stratified random sampling technique. Then the analyzed data have been used to model the energy demand and to project the future energy consumption till 2030 using LEAP (Long Range Energy Alternative Planning) simulation software. The model is done based upon baseline scenario and energy efficiency improvement scenario (mitigation scenario).

The total energy consumption in Bahir dar in 2014 is nearly 330 Giga watt hour per year, and of this value about 83.8% is used for cooking and TV, lighting, refrigerator, and water heater consume 7%, 4.5%, 3.5%, 1% of the total energy and remaining 0.2% is consumed for other auxiliary appliances. The projection of the energy consumption in 2030 will be more 450 Giga-watt-hour with business as usual scenario compared to just less than 350 Giga-watt-hour with mitigation scenario.

As the result of the poor consumption efficiency, households that use traditional biomass tend to have more primary energy intensity than household that use electricity. The consumption of electricity is projecting rapidly while charcoal and firewood will still be the significant energy sources. The potential for energy saving is from improving the efficiency of stoves. Comparing with developed countries, for example Sweden, where the energy in dwellings is mostly used for space and water heating and the energy saving mostly target on improving wall insulations, the energy saving on Bahir dar is based mostly on cooking.

The findings obtained in this shows options to improve household energy efficiency intervention planning and to enhance the effectiveness of policy interventions. Further studies could be done on modeling of other sectors.

Key words: energy modeling, efficiency, LEAP, energy intensity, scenario, residential buildings

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List of Nomenclatures, Abbreviations and Terminologies

Nomenclature

E_{f,hs_i}	primary annual energy consumption of household i (kWh/year)
$E_{f,per\ capita,useful}$	the end-use per capita energy consumption from a fuel f
$E_{f,total,useful}$	end-use total energy consumption of a fuel f
$E_{f,total,}$	total energy consumption of the city from a fuel f (kWh/year)
$E_{f,hs,average,enduse}$	useful household energy consumption of a fuel (kWh/year)
$E_{f,hs,average}$	the average household energy consumption (kWh/year)
$E_{f,hs}$	household's annual energy consumption from a fuel f (kWh/year)
$E_{f,per\ capita}$	average per capita energy consumption of a fuel f
f	stands for fuel type like charcoal, firewood, etc.
LHV_f	Lower heating value (LHV) of the fuel (MJ/kg)
m_f	amount of the annual fuel consumption by mass (kg/year)
N_t	total number of the households in the city
η_{st}	device or stove efficiency (%)
n_{fam}	average number of family per household
N_s	the total number of sample households
n	number of sample households
Sh_f	the share of fuel type f (%)
X_i	the number of households using particular fuel type f

Abbreviation

BAU	Business as Usual
CO	Carbon Monoxide
CRGE	Climate Resilience and Green Economy
GIZ	Deutsche Gesellschaft für Internationale Gesellschaft
GWh	Giga-Watt-Hours
kWh	Kilo-Watt-hours
LEAP	Long-range Energy Alternatives Planning system
LHV	Lower Heating Value of the Fuel
LPG	Liquidified Petroleum Gas
MJ	Mega Joule
MWh	Mega-Watt-hours
PM	Particle matter
SIE	Stockholm Environment Institute

Terminologies

End Use Energy

Are the final energy consumption sectors in the household such as cooking, lighting space heating, air conditioning, water heating, refrigeration, and so on

Energy Efficiency

Energy efficiency can be defined as the ratio of the amount of energy actually used to operate the device to the amount supplied as input multiplied by 100.

Energy Intensity

The amount of energy used to perform a certain activity on a device. In this study the energy intensity is expressed

Injera

It is popular bread in Ethiopia. It used as part of the meal on which other appetizer foods are eaten. It is thin and circular made in injera baking stoves.

Primary Energy

Primary energy consumption is the amount of energy from conventional or renewable fuels directly consumed by either one of the major end-use sectors.

Chapter 1. Introduction

1.1 Background

Energy consumption trend is an important statistics to plan the future energy generation and distribution of a nation. The energy consumption in households in the same way is important for planning and policymaking. Most developed countries have a national statistics that describe the trend in energy consumption. In addition, several studies have been conducted on the reasons that cause the change in residential energy consumption such as, prosperity, consumption behaviors, settlement types, climatic conditions, etc.

In developing countries like Ethiopia, residential sector takes the dominant share of the total national energy consumption. The majority of the residential energy is used for cooking or preparation of food, and the energy carrier is mostly traditional biomass; however, the usage of modern fuels like electricity is increasing (Adria & Bethge, 2013). In developed countries, in contrast, the residential energy consumption is about a third of the total energy consumption. The residential energy is mostly used for heating (space heating and water heating), space cooling, ventilation, lighting, cooking and for appliances; less than 10% of the energy is used for cooking (Santín, 2010).

Our world has shown tremendous modernization and technologies which have been transferring to developing countries at alarming rate (Pandey, 2002). This changing is leading to transition of life-styles including transition from traditional fuels to modern fuels. The economic growth in Ethiopia will lead to the rate of change of energy consumption to be dynamic and faster as it is clear that it has great impact on the energy consumption pattern (Boris, 2010). The increase in income will help the residents to own extra appliances with larger size, which has more power wattage. Having various mix of people with ranging of income level and ways of living, it is highly important to perform studies on the current energy consumption and evaluate future energy consumption figures based up on various scenarios. This result helps to provide an energy policy that meets the demand and to investigate the energy saving options in the residential sector.

Internationally there have been various studies conducted about modeling of energy consumption particularly in residential sector. Taking Tehran, Iran as a case study, Abbaspour et al. (2013) studied about previous trend and future consumption projection. They identified the possible projection of consumption of various fuels and also analyzed the possible amount of energy that could be saved if policy implication is conducted. Ren Z. et al. (2013) tried to develop a model for predicting the total residential energy consumption and associated GHG emissions in Australia. They tried to include space heating, space cooling, water heating, lighting, and household appliances in their model. Those parameters in conjunction with the house type and occupancy behavior had enabled them to predict the energy consumption of houses. Oberascher et al. (2011) studied about the possible reduction of the energy required for preparation of food based upon the change in behavioral aspects of the consumers coupled with using alternative cooking appliances. They considered the activities like boiling water, brewing coffee, cooking potatoes and boiling eggs as relevant cooking processes. They investigated the specific energy consumption for boiling water using various types of appliances like: ideal pot with and without lid, microwave and electric kettle and they compared the corresponding energy savings. Manipulating data from nine OECD countries, Urban and Scasny (2012) studied about effect of dwellers' environmental concern on residential energy-saving reductions and efficiency measures. They found out that inhabitants with greater environmental concern have more chance to achieve energy-saving targets.

Several studies have been done to study household energy consumption in Ethiopia (Alem et al, 2013; Taddese, 2013; Abebe & Steven F, 2011). It could be examined that most studies focus primarily on household energy use or just on the determining factors for household fuel choices. In addition the studies usually focused only on the two major end use energy sub sectors, cooking and baking, and few studies have dealt on energy consumption for non-cooking sectors like appliances, lighting and water heating. Moreover, modeling the long term residential energy consumption based upon various scenarios hasn't been studied in depth.

This research deals about modeling energy consumption in single-family urban households of Bahir City, Ethiopia. It mainly focuses on common energy consumption areas within households of Bahir Dar city (i.e., for lighting, cooking, baking, hot water heating,

appliances, etc.). To undertake the energy modeling, there are various approaches to be used. The primary step would be to understand the level of the energy consumption in single family houses. This would be performed by primary data collection in sample households. Obviously, people might deliver information about their fuel consumption with unit of kg for wood or charcoal or litters for kerosene, etc. Changing these units to usable and more meaningful value like kWh as annual household energy consumption is also performed in this thesis. This helps to quantify and compare the energy consumption of various fuels with consistent unit.

1.2 Problem Statement

Household energy consumption trend in developed countries is well documented and Energy modeling in residential sector is also common in the developed world. However, in developing countries like Ethiopia, although varied studies in general exist, there are hardly any recent information exists on the nature of energy use, and any attempt for long-term residential energy modeling. As the result of their economic tie and different reasons energy, consumption in other sectors like commercial or industry has been studied being given more emphasis. However, residential sector's energy consumption can be affected by various reasons like the behavior of occupants, availability and cost of the fuel, settlement type, etc. In urban households of Ethiopia, as in other developing countries, there has been a dramatic rise in the number of consumer electronics and electronic appliances, most studies however have given less attention for the rising in energy consumption due to appliances. The main issue that is addressed in this study is the household energy consumption in single family residential houses for cooking, water heating, Lighting, Television, refrigerators including other appliances like washing machines, dishwashers, juicer, etc. by using Bahir Dar, Ethiopia as a case study city.

1.3 Objective

1.3.1 General objective

To evaluate and model the residential energy consumption in single family houses in Bahir Dar City, Ethiopia for cooking, water heating, Lighting, Television, and other appliances.

1.3.2 Specific objective

- To gather recent data and information on energy consumption patterns of single residential households in Bahir Dar, Ethiopia.
- To analyze the collected data and evaluate the existing energy consumption by its energy intensity and energy share.
- To model long-term energy consumption of residential sector based upon various scenarios.

1.4 Scope of the Research

In order to make the research controllable for the given time frame, energy consumption in single family residential houses will be studied and modeled. The scenarios on reducing energy consumption in residential buildings will also be considered to some extent. The study mostly considered dominant areas of household energy consumption particularly on cooking, lighting, on appliances, heating etc.

1.5 Research Questions

The research questions of the study are:

- What types of fuels are being used in single family houses?
- What type of stoves and other appliances are being used in single family houses?
- How much energy is being used in the houses and what is the share of fuels?
- How do the type of buildings and occupancy affect the fuel type and the amount of energy consumption?
- How will the future energy consumption pattern be in business as usual scenario and mitigation scenarios?
- What measures can be taken to reduce the future energy consumption in urban households?

1.6 Organization of the Thesis

Chapter 2 will provide a literature review which will illustrate the background of the energy usage in the case study area, reveals about previous works and energy modeling tools and also determine the parameters used in the selected long term energy modeling tool. Chapter 3 will present about the methodology that means about the data collection and post processing of the data. Chapter 4 will present the results of the statistical analyses and the energy modeling tool, primarily presenting the types of fuels and devices like stoves being used in the selected households and differences in energy use for different types of houses, the fuel prices, etc.; and secondly, presenting the results of the forecasting analysis for various types of scenarios. Chapter 5 will deliver the conclusions of the study and provide recommendations for further research.

Chapter 2. Literature Review

2.1 Residential Energy Consumption

Residential sector is one of the major consumers of energy in the world, statistics show that it accounts for about 30% of the total energy consumption. Figure 2-1 shows the comparison of household energy in various countries (Swan & Ugursal, 2009). Households in USA are the largest consumer of energy in developing country like Ethiopia the major portion of the total energy is consumed by residential sector. Various studies show that the energy consumed by household in the world basically depend on traditional inefficient biomass, and more importantly the larger part the in habitants relying upon the forest land and crop residue to their energy demand (NKWATOH et al, 2009). In Africa, nearly 660 million people (80% of the population) depend on the traditional biomass for cooking in households (IEA, 2010).

Several literature shows that in developing countries like Ethiopia the largest energy consumption is observed to be characterized by the predominance of the household sector, for example, Mekonnen et al, 2011 stated that it accounted for 88% of total final energy consumption (73% by rural and 15% by urban households) in Ethiopia. The main reason for this is cooking, which is mainly performed by using traditional fuels like fire wood, animal manure or agricultural residue. The usage of energy for hot water, for tea coffee, and rarely for shower and other similar activities also consumes fuel mainly firewood or charcoal.

Table 2-1 shows that globally more than 2.5 billion people still use traditional biomass, like wood, waste or dung for cooking and. More than 83 % of the rural and 23 % of the urban population still use biomass. The figure is more in sub-Saharan Africa (which includes Ethiopia) with more than 76 % of the total population rely on biomass more than 93% of the rural population and 58% of the urban population is dependent on biomass for cooking. Worldwide if no any other policies designed to addressing this problem, due to the rapid population growth, the number of people who use biomass will increase significantly to more than 2.7 billion people by 2030 (Adria & Bethge, 2013).

For comparison, households of developed countries rely on gas and electricity as their primarily fuel for cooking. For example in UK of the amounting to a total energy consumption of an estimated 8 TWh/year in 2009 for cooking about 55 percent of the households use gas as their cooking fuel while the remaining 45 per cent rely on electricity (Defra, 2009). In EU-27, the use of electric stoves and ovens is about 60TWh/year, which is 7.5 percent of the residential total electricity consumption (Adria & Bethge, 2013).

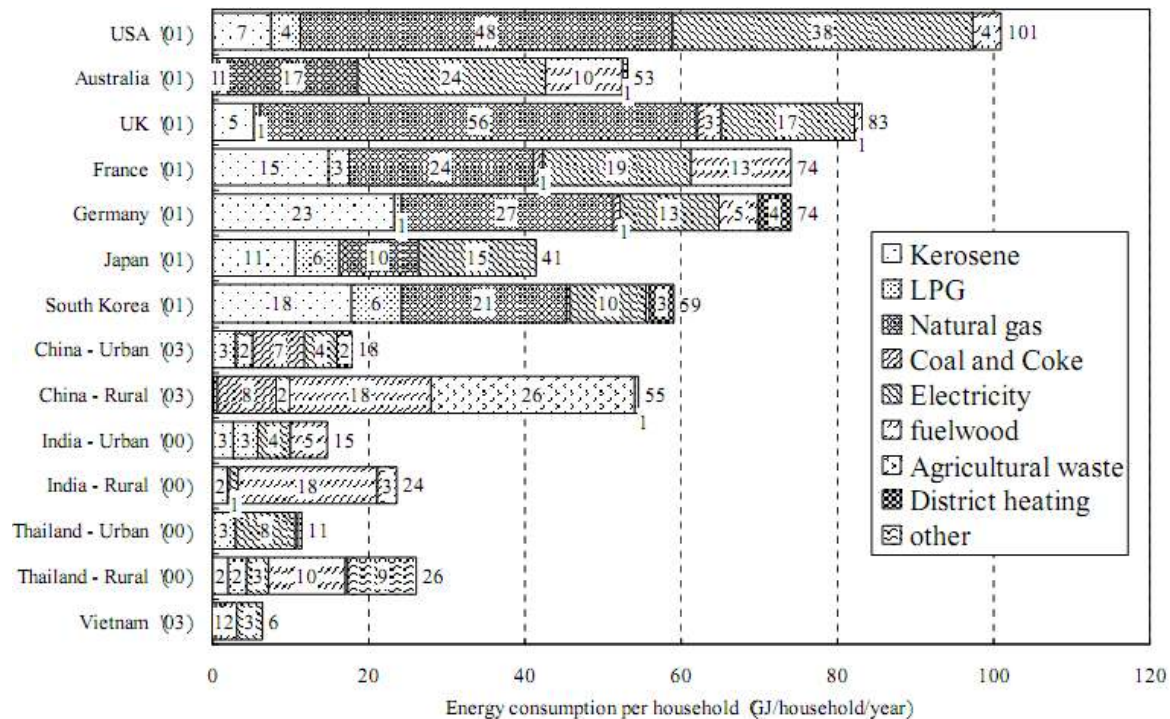


Figure 2-1 Comparison of household energy consumption by fuel type in various developed and developing countries around the world, Source: Hidetoshi et al. (2008)

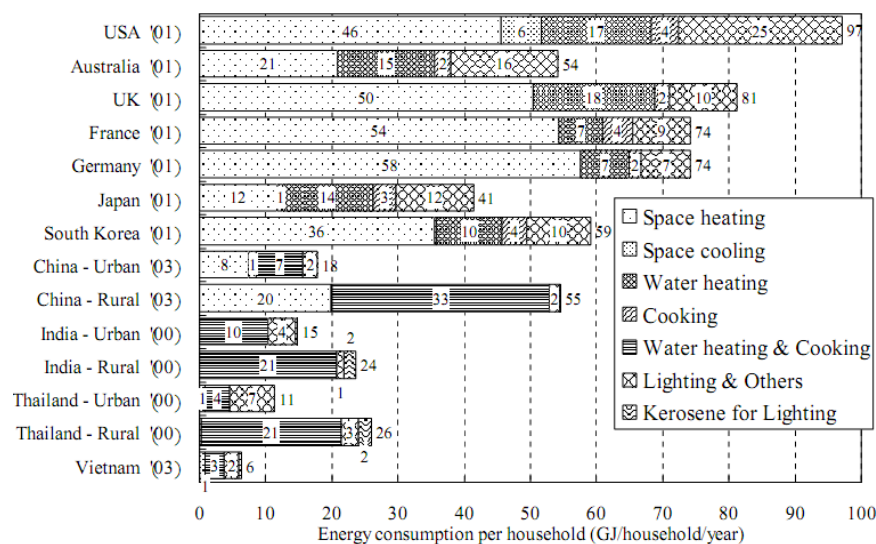


Figure 2-2: Comparison of Energy Consumption per Household by Final End Use Sectors in various developed and developing countries around the world. Source: Hidetoshi et al. (2008)

Table 2-1 People in Developing Countries Relying on Biomass Resources as their Primary Fuel for Cooking, 2004 (source: OCED/IEA 2006)

	Total population		Rural		Urban	
	%	million people	%	million people	%	million people
Sub-Saharan Africa	76	575	93	413	58	162
North Africa	3	4	6	4	0.2	0.2
India	69	740	87	663	25	77
China	37	480	55	428	10	52
Indonesia	72	156	95	110	45	46
Rest of Asia	65	489	93	455	35	92
Latin America	23	83	62	75	9	33
Total	52	2,528	83	2,147	23	461

Although there are various studies that have been done to understand the energy consumption trend in Ethiopia, most of these are performed for rural areas or for the

whole Ethiopia. Ejigie (2007) studied the factors that affect household use of fuel wood in the Jimma town of Ethiopia. The results provided evidence of the existence of significant variation in fuel wood consumption across the sample households and it appears that households with larger families use less per-capita firewood and charcoal than households with smaller family members. Households with more educated heads tend to be less dependent on firewood. The result indicates per-capita fuel wood is expected to rise with wealth and income (Ejigie, 2007).

2.2 Review of Previous Studies about Modeling of Household Energy Consumption

Internationally there have been various studies conducted about modeling of energy consumption particularly in household sector. Abbaspour et al. (2013) studied about previous trend and future consumption projection as a case study in Teheran. They used LEAP as modeling software by considering Iran's long term development policy and included various mitigation scenarios. They identified the possible projection of consumption of various fuels and also analyzed the possible amount of energy that could be saved if policy implication is conducted. Lutzenhire, et al. (2010) conducted literature review and examined the suggestion about what they called common "sticky points" in modeling residential energy consumption. Ren Z. et al. (2013) tried to develop a model for predicting the total residential energy consumption and associated GHG emissions in Australia. They tried to include space heating, space cooling, water heating, lighting, and household appliances in their model. Those parameters in conjunction with the house type and occupancy behavior had enabled them to predict the energy consumption of houses. Pukšec et al. (2013) conducted long term energy demand model of Croatian households sector considering the domestic sectors like heating, cooling, electrical appliances, cooking and hot water. They tried to analyses various mechanisms that have potential of affecting future energy demand scenarios of Croatian households sector. Through their studies, they illustrated out significant possibilities for energy efficiency enhancements and reducing energy demand for households in the future, if careful and rational energy planning is implemented. Taking 2011 to 2050 as their study period, Jia-Jun Ma et al. (2014) have calculated energy demand for China's urban sector (embodied and operating energy). They

designed six scenarios to quantify the influences of the factors like new building floor areas, energy efficiency, etc. and showed the possibility of building energy consumption effectively.

There are limited number studies conducted about the household energy consumption in of Ethiopia. It could be examined that most studies focus primarily on household energy use or just on the determining factors for household fuel choices. In addition, the studies usually focus only on the two major end use energy sub sector cooking and baking, few studies have dealt with energy consumption in appliances as well. Moreover, modelling the long term residential energy consumption based upon various scenarios hasn't been studied in depth.

Taddese (2013) studied about energy consumption in urban and rural households in Mekelle city, Ethiopia and the nearby rural areas. The result showed that most of the households still use biomass as their primary fuel for cooking.

Most of the studies that have been conducted on examining the factors that determine the fuel choices showing the variation in household energy use within the communities depending on income, education, way of life, type of building, living standards, etc. Other factors are country or region specific, such as climate or cultural practices (Taddese, 2013). Access to modern fuels like electricity is also another factor that determines the choice of fuels and appliance usage. Studies show that households with higher income or higher level of education have more interest towards shifting to electricity and other clean fuels (Taddese, 2013).

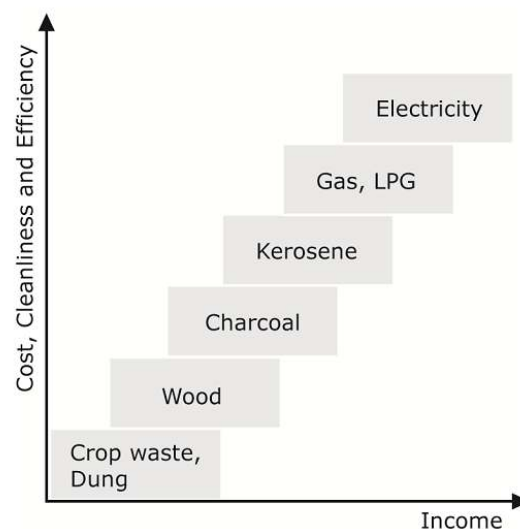
Gupta and Kohlin (2006) argue that fuel choice is mostly dependent on availability, affordability and ease of use. It is also dependent on region of residence and tradition. In similar way, with their regression results from a random effects multinomial logit model, Alem et al (2013) argue that households' economic status, price of alternative energy sources, and education are important factors of fuel choice in urban Ethiopia. Narasimha and Reddy (2007) examine the residential fuel choice decisions for rural and urban households of India and find that the factors that affect fuel choice are entirely different in each case.

Taking India and China, the two most populous and rapidly developing countries in the world, as a case study, Hubacek et al (2007) studied about the scenario analysis on Changing lifestyles and consumption patterns in developing countries. They found out that population, wealth and technology to be the main driving forces for the environmental pollution of the developing countries (Hubacek, Guan, & Barua, 2007).

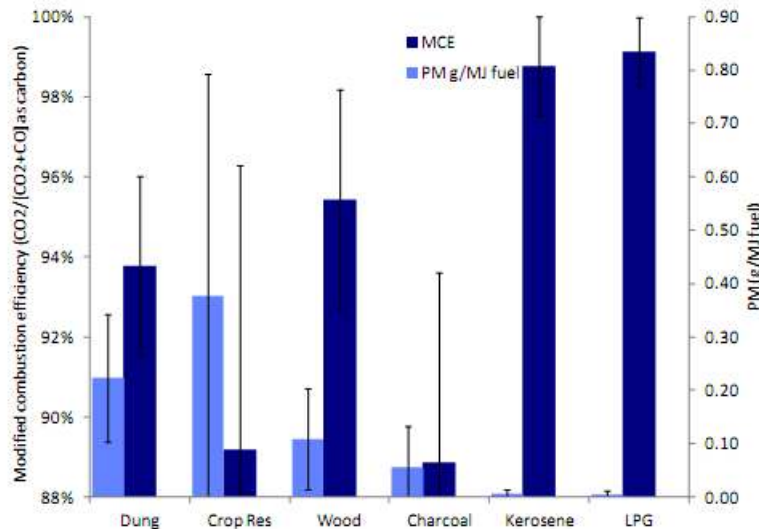
2.3 Determinants of Household Energy Consumption and Fuel Shift

Several scholars have outlined the determinants of household energy consumption. Recent results from Carlos et al. (2014) listed factors like demographic, socioeconomic, and housing characteristics results in variation of household energy consumption. Kaza (2010) discovered that housing type has a more nuanced impact on residential energy consumption compared to house size and housing density. He used data from Residential Energy Consumption Survey from the Energy Information Administration.

There is a strong positive relationship between growth in per capita income and growth in household demand for fuels. This concept is referred as “Energy ladder” which indicates the fuel type used in households change as their income changes (Link, Axinn, & Ghimire, 2012). Figure 2-3 illustrates the process of energy ladder which simply shows that when households with less income cease to use traditional biomass fuels and when their income improves they adopt modern alternatives (Leach, 1992).



(a)



(b)

Figure 2-3: Energy ladder: (a) Schematic representation of the energy ladder (Leach, 1992)., (b) Combustion efficiency and PM emission factors across the energy ladder (source: Berkeley Air Monitoring Group, 2012)

The energy ladder model thus suggests income as determining factor for fuel choice, and thus the rationale for transitioning up the energy ladder. The lowest income group being user of traditional biomass such as wood, cow dung and agricultural residue etc., and the middle class is believed to be user of fuels such as kerosene and charcoal. When the income level increases people will join the third and final phase which is categorized by using modern cooking fuels such as LPG, natural gas, or electricity or other cleaner ' sources of energy (Boris, 2010).

Various studies have shown that age of householder is strongly related to the type and quantity of household energy consumption. Baker et al. (1989) showed that household energy expenditures varied among the ages based on the kind of energy consumed. This may result that households in the neighborhood of younger generation will consume differently than those in the neighborhood of senior generation (Carlos, et al., 2014).

Several studies suggest that household size, or the number of occupants in a household, has a positive relation to energy consumption (Gatersleben et al., 2002). It is clear that the

larger the household size the larger would be the quantity of energy; however, there is also variation in the type of energy consumption.

Energy efficiency relies on type of fuel used and the characteristics of particular appliances or devices on which the fuel is being used. In many developing countries, where traditional fuels, such as firewood, charcoal and agricultural waste, are the predominant of the total household energy consumption, the low energy efficiency of the devices is the main factor that causes the end use efficiency to drop. The thing that worries energy consumers is not the overall amount of energy used, but the final energy used. Traditional fuels like firewood or charcoal are mostly used in ways that yield very low efficiencies that causes more consumption of the primary fuel (Taddese, 2013).

Therefore, households that use the modern fuels, which are commonly used in modern efficient devices, consume less amount of fuel perhaps expends less money on fuels than the households that use traditional fuel.

2.4 Energy Modeling Tools

Energy analysis involves the designing of energy systems and their assessment through the application of energy- economic models, usually referred to as energy models, which usually involves the interactions of the multiple components of the energy system based on mathematical formulations (Nakata et al, 2011). Energy modeling tools have been used for energy planning and have been mostly implemented in by governmental bodies, academic organizations, and researchers (Nakata et. al, 2011).

Nakata et al., 2011, defined Energy system analysis as a comprehensive approach that brings together several aspects and stakeholders to solve energy planning issues that could be complex to solve otherwise. Energy system analysis has been applied to address energy issues characterizes the flow of energy supply and demand in a society based on the systems approach.

2.4.1 Classification of energy modeling tools for energy system

The energy system in the real world is complicated so, in order to represent this limitation models should be prepared as a tool to explain, predict or control the behavior of the actual

energy system in simplified way (Nakata et. al., 2011). An energy model can be taken as approximate representation of the real energy situation.

Several scholars have categorized energy model in to two broad approaches, top-down approach and bottom up approach (Swan & Ugursal, 2009); .Top-down approach is applied to analyze and compare the macro-economic interactions between the energy sector and the rest of the economy (Nakata et al, 2011). The other one, the bottom-up approach, deals only on the energy sectors, it analyzes the choice of technologies with the best performance, and it is usually referred to as the engineering approach (Nakata et. al., 2011).

According to literatures forecasters normally use several methods to forecast the future energy consumption, to list: trend projections, econometric analysis, scenario analysis, end-use analysis and systems dynamics. Trend projections and economic analysis are categorized as top down methods they consider the past and the existing trend to predict the future. They are straightforward extrapolations of time arrangement information, relating energy utilization to a picked variable, for example, GDP, utilizing standard slightest squares investigation. They can be direct, semi-log or log-log. Their drawback is use to the fact that they require reasonable and consistent previous information.

Scenario analysis and End use analysis forecast the future by considering what is likely to happen over time. In those methods new technologies can be considered although they are not used previously or currently. Therefore in scenario analysis is suitable and flexible tool deal with uncertainties in demand and is particularly perfect for modeling energy consumption in developing countries.

Energy models can be classified into three types considering their general purposes (Van Beeck, 1999); those are forecasting models, back casting models and scenario analysis models. Forecasting models are mostly used to predict or forecast the future by taking some input variables from previous or current data.

The second type is Scenario analysis model, which is mostly used to discover exploring the future situation (Nakata, Silva, & Rodionov, 2011). "Scenarios are self-consistent story-

lines that show how a future energy system might evolve over time. It considers particular socio-economic setting and particular set of policy conditions” (Amare, 2007; SEI, 2011). Scenarios can be used to consider several "what if" type questions, some example of those questions could be what if the system undergoes change in terms of using more efficient appliances, what if the current electricity is changed towards more renewable resources, etc. (SEI, 2011). When dealing with scenario analysis models, it requires considering a number of intervention scenarios which affects the future. Reference scenario also called “Business as Usual” scenario or BAU Scenario is the one that considers the future conditions if no intervention like policy change happens (SEI, 2011) . Mitigation scenarios must be made considering parameters like policy changes, economic growth, infrastructure, population growth, etc. The other one is called Back casting models; its purpose is to construct visualizations of a desired future target, by considering the view of professionals to look at what measures might be changed to be able to achieve such a future (Nakata, Silva, & Rodionov, 2011).

Swan & Ugursal (2009) classified the purpose of energy modeling tools in to macro-scale and micro-scale. The former one being the evaluation of particular regional or national energy supply requirements by considering various scenarios and the later one being evaluation of energy consumption of particular building for example to assess its ability of meeting the national code. Macro-scale energy modeling is used for national level energy modeling to be used for policy making.

There are several studies conducted on modelling the energy demand of various sectors or specific building. Using the Household Energy Demand (HED) bottom-up model, Pukšec et al. (2013) estimated the Croatian Residential Sector energy demand from 2007 to 2050. They applied efficiency improvement methods to the end-uses like space heating and cooling, water heating, preparing food (cooking), and for appliances and lighting that use electricity. In their model, the energy demand of end-uses were analyzed thoroughly, e.g., the spacing heating demand was described using ISO 13790 standard. In addition the effect of floor area expansion of the buildings was integrated into the model.

Farahbakhs et al. (1998) studied about the characteristics of the residential energy consumption in Canada using end use model called CREEM. Using the model they studied

the impact of two upgrading options on the Canadian residential energy consumption. They showed that due to the upgrading significant amount of energy could be saved in the residential sector. In their study they considered only single-detached and single-attached dwellings as the result of the data limitation.

EnergyPlus, tool developed by United States Department of Energy has been used for building simulation based up on the demand of end use sectors like space and water heating, cooling, lighting, ventilating, appliances, etc. in buildings. EnergyPlus does not have user friendly' graphical interface that it reads input files and writes output as text file too.

LEAP (Long-range Energy Alternatives Planning), for example, is a tool that is used to predict the long-term energy consumption of a nation or part of it. Its design is based for energy policy making or energy management tool. LEAP is developed at the Stockholm Environment Institute (SEI) targeting of energy policy analysis and climate change mitigation assessment (SEI, 2011). It is particularly suitable for long-range analyses of energy systems. A key benefit of LEAP is its availability free for low-income country users and technically, it requires low initial data requirements. The data input in leap is

Several scholars have used LEAP to model the energy demand of a region, country or city. Park et al. (2013) performed an assessment of long-term scenarios for the transition to renewable energy in the Korean electricity sector. Their scenarios are baseline and governmental policy scenarios that focus on the electricity supply by nuclear expansion. Through their model, they performed comparison of electricity generation outlook by the scenarios between 1990 and 2050.

Using LEAP model application, Huang et al. (2011), performed the long-term forecasting of energy supply and demand of Taiwan. They compared future energy demand and supply patterns of Taiwan including the Greenhouse Gas Emission for various scenarios such as Business as Usual Scenario, Efficiency improvement Scenario, retirement of nuclear power plant, etc.

Using leap model, Shin et al. (2005) studied the environmental and economic impact of landfill gas (LFG) electricity generation in Korea. In their analysis various scenarios were

considered including business as usual scenario with existing electricity services, technological enhancement of gas engine and landfill gas maximum operation potential. They compared the electricity output and the corresponding level of GHG emission for those scenarios.

Similarly LEAP has been used in various studies to forecast the long-term energy demand: Taiwan's energy supply and demand (Yophya, Jeffrey, & Chieh-Yuc, 2011), Panama's electricity sector (McPherson & Karney, 2014). In this paper LEAP (Long-range Energy Alternatives Planning) is used to model the energy demand of Bahir Dar area focusing on analysis of household energy consumption.

2.4.2 Modelling of cook stoves

Bryden et al. (2005) described design principles for wood burning cook stoves, proposing an insulated combustion chamber with an optimum short chimney between the fire source (fuel) and the underneath of pot surface. They said that this is important in sustaining a strong draft to ensure high temperature flame and complete combustion reducing emissions. Their analysis is based on simple steady state fluid flow and heat transfer analysis in the form of zone modelling over different parts of the stove. Similar study based on zonal modelling is performed by MacCarty (2013) who described a zonal model for a cookstove. For zonal modeling purposes, he classified the cookstove system into three zones: the solid phase packed bed zone, the gas phase combustion or flame zone, and the heat transfer zone. He used correlations for various losses including fluid friction, and presented convective heat transfer relations for various parts of the geometry.

Kshirsagar et al. (2015) developed spreadsheet of a heat-transfer model for natural draft stove with unshielded pot targeting prediction tool to the stove makers and artisans. Their analysis is based up on simplified heat transfer fluid flow and combustion equations by applying in to various stove zones. Simplified model for understanding natural convection driven biomass-cooking stoves has been done by Agenbroad (2011a) and Agenbroad (2011b).

Schumack (2015) used standard explicit finite difference method to solve the fluid flow and heat transfer equation in the J tube and adiabatic chimney of rocket mass heater. However, the heat input is model as a single point high temperature resulted from calorific value and mass flow rate

of fuel. Although their model could be taken as an improvement of simple model, the model did not analyze the detail combustion issues and the turbulence flow as CFD modelling has not performed.

2.5 Background of Household Energy Usage

As previously described, like many other areas in Ethiopia, household is the dominant sector in energy consumption in Bahir Dar city. Most energy is consumed for cooking, which could be classified in to baking traditional bread called *injera*¹ and preparing a traditional appetizer stew called *wot*². Preparation of coffee and tea is also categorized as cooking. The majority of Ethiopian households including Bahir Dar use Firewood used for baking injera which is mostly done on the traditional inefficient stove called three stone stove open fire stove. Cow dung is also used in few households as fuel for baking injera. The end use residential energy consumption subsectors are: baking injera, cooking wot, Water Heating, lighting, TV and its accessories, refrigerator, and other appliances. Cooking wot and baking injera are the two most energy consuming activities in the households that consumes significant portion of the household energy. In urban households, charcoal is one of the main fuels used for cooking wot while firewood is used for baking.

In urban households of Ethiopia, including Bahir Dar, the majority of the households are connected to the grid either with own electric meter by borrowing electricity from neighboring households mostly just for lighting. Therefore the energy used for lighting, water heating (mainly for shower) and to operate appliances comes from electricity.

2.5.1 Cooking

As stated above cooking is the major end use sector for energy consumption in Ethiopia including Bahir Dar City. There are various devices used to prepare food. The efficiency of those devices is highly important parameter that affects the fuel consumption.

As discussed previously, in many developing countries like Ethiopia, more than 90% of the household energy consumption is spent for cooking. The large amount of the energy

¹*Injera*(also spelled as '*injera*') is a traditional bread baked over local plate called mitad

²*Wot* is a traditional liquid food cooked on pots, it is used with injera and acts as appetizer. There are several type of wots

consumption is as the result of the very low levels of energy efficiency of traditional biomass stoves. Biomass cooking stoves are devices, in which biomass (wood, agricultural residuals, etc.) is used to prepare food or to heat water (Adria & Bethge, 2013). In Ethiopia and most development countries, the most basic type of cooking stove used for biomass fuels is the so called “*traditional three-stone fire stove*” or “*traditional open fire stove*”. It is made by arranging three stones in such way that it is possible to place a pot for cooking above it and to pass the firewood or any biomass underneath the pot. Although the low efficiency and other risks associated to this type of biomass cooking stoves is known for long time, it is still being used in various households in Ethiopia and other developing countries because of its low initial capital. Beside the traditional three-stone fire, there are various traditional cooking devices in Ethiopia or most developing countries. In Ethiopia, most biomass cooking stoves are not manufactured through factory production processes but rather in homemade or with simple local workshop with limited technical knowledge about the efficiency. The following sections describe about the stove types that are used for cooking in Ethiopia.

Traditional three stones (open fire) stove

Traditional three stone fire stove (Figure 2-2) is one of the traditional stoves being used in Ethiopia. This stove has been used for several function either baking injera or normal bread or preparing wot. Several report shows that the thermal efficiency of this stove being in the rage of 5-10%. That means about 90% of the thermal energy is lost to the environment.



(a)



(b)

Figure 2-4 Traditional three stone fire stove being used for cooking (b) An improved three-stone fire.

The main drawback of traditional three stone cooking stove is its large heat loss to the environment by radiation and convection through the large opening. This in addition the heat transfer from the flame or smoke or the char to the baking circular plate so called '*mitad*' is very inefficient that reduces the overall performance. Therefore with its existence in numerous households and wide application, it causes wastage of firewood and in turn to excessive deforestation.

Traditional open fire three stone stove is used in significant portion of the households for various noon baking activities like preparation of food items such as: wot, coffee, tea, porridge etc. as shown in Figure 2-4 (b). The stove is simple to use just three stones over which the pot rests and firewood is instated three the spacing between the stones. The stove apparently has high thermal losses due to radiation and convection. According to the test report made by GIZ it has thermal efficiency between 9% and 17% depending on the usage.

Improved Stoves

As the result, many actions have been made in order to improve the energy efficiency of traditional cooking stoves and reduce the health effects associated to human beings due to smoke exposure. These efforts have led to the invention of several stoves also called "improved cooking stoves" which again vary each other in terms of their construction, performance and costs from countries to countries. The performance of biomass cooking stove is mostly rated by their capability of reducing of harmful emissions like carbon

monoxide (CO) including particulate matters (PM), and increasing their thermal efficiency (their ability to convert the chemical energy of the fuel into useful heat energy).

According to Adria & Bethge (2013), the first attempt to modify the traditional biomass stove was made in India in the 1950s. It was done by adding chimney as an additional component to remove smoke from the cooking room. Rise of the cost of biomass fuels, excessive deforestation, and health problem associated with cooking has forced several stakeholders to consider the issue of improving cooking stoves seriously. More promising technical analysis like thermodynamic and heat transfer analysis of biomass cooking stove combustion have also followed targeting to raise the efficiency (Adria & Bethge, 2013). The improvement of the cooking stoves comprises the following parameters (Adria & Bethge, 2013):

- Integrating chimney to extract smoke and associated heat from the kitchen
- Enclosing the combustion chamber to retain the heat and improve combustion
- Careful design of pot holder to maximize the heat transfer from fire to pot and adding skirt to reduce heat loss from the side of the pot
- Creating turbulence by using baffles and hence improve heat transfer
- Controlling the fresh air flow dampers to control and optimize the air flow
- Inserting insulating materials like ceramic or creating air pocket on the wall of the combustion chamber to minimize the rate of heat loss to the surrounding.
- Adding fuel supporting structures like grate to allow ash to be removed and air to be entered underneath the fuel.
- Accurately calculating just the required amount of fresh air and sizing the air inlet and smoke outlet or chimneys.
- Simultaneously using several pots serially multi pot systems to maximize extraction of heat from the flame and smoke.

Several attempts have been done to develop improved firewood and coal stoves. The Kenya Ceramic Jiko (KCJ) is one of the improved stove projects conducted in the Eastern region of

Africa. It is reported to have significant enhancement in the efficiency compared to the traditional three stone stoves (Samuel, 2009). Lorena stove was developed by Approvecho team. It is a rocket stove having an integrated smoke exhaust chimney and it has multiple pot seats to cooking several foods at a time this is intended to maximize the extraction of heat from the flame or hot gas. The exhaust chimney was reported to have double advantages: firstly it acts as vent for the pollutant created during combustion and release it outside. Secondly, it enhances the draft of fresh air towards the combustion chamber. The stove construction was from materials with wood ash, bricks and earth to improve the insulation.

Most stoves are designed not only to alleviate their problems in terms of efficiency, but also the behavior of the people who are going to use the stove is highly considered as understanding of the user's needs and requirements is important for the success of the improvement (Adria & Bethge, 2013). For example although its concept is the same worldwide, the rocket stove being produced in Ethiopia as improved stove has been made according to Ethiopian culture. Possibility to be maintainable by local people is considered. In addition, the type and size of the biomass fuels determine the design of the combustion chamber of the cooking stove and air supply required which in turn determine the stove size and construction methods. In summary, the behaviors of people and kind of locally available cooking fuel are critical parameter when it comes to designing and selecting appropriate biomass stoves for a specific region (Adria & Bethge, 2013).

It is known that open fire also have large thermal losses. Amount of air is some ten times the required stoichiometric volume. The too much fresh air-cools the flame and the combustion is then not optimal at all. The emission of Carbon monoxide (CO), Particle matter and other harmful gases is high in open fire stoves.

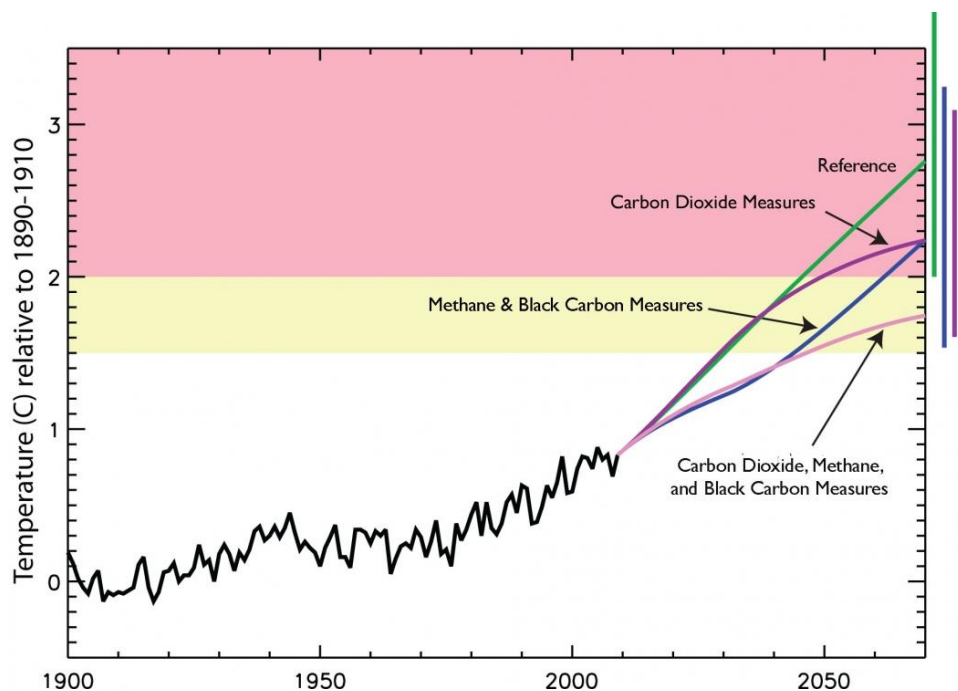


Figure 2-5. Comparison of temperature and CO emission from cook stoves between 1900 and 2050.

Open fire with chimney

Open fire with chimney is good in one way that it removes the generated smoke, particles and gases out of the kitchen in a proper way through the integrate chimney. This helps the kitchen environment to be clean from smoke. However, open fire with integrated chimney would draw too much fresh air towards the fire due to the draft effect. This cools the fire plume reducing the performance of the stove.

Dakota Fire Pit

Dakota Fire Pit, shown in fig. 2-6, is use to burn safely underground. To make this fire pit, two vertical shafts or holes are dug, and then are connected by a tunnel underneath. Then, the fire wood is placed in one tunnel or hole and ignited. The cook put is placed on the other vertical shaft and it creates a draft in the tunnel to keep the wood burn effectively. In this stove, the air flow by the draft is just enough (near stoichiometric). The fire will not be too cold and generation of soot is also low as the result of the enough air. However the earth might take too much heat and might reduce the efficiency.



Figure 2-6 Dakota Fire Pit confined fire in underground with J tube combustion chamber (source: survivopedia.com)

Non-Catalytic combustion Stoves

Non-Catalytic Stoves, shown in fig. 2-7, are designed with baffles and/or secondary combustion chambers, which route the burnable gases through the hottest part of the firebox and mix them with sufficient air to burn them more completely. They can attain up to four stages of combustion and completely burn the wood smoke before it escapes. Air flow is also here near stoichiometric flow.



Figure 2-7 Non-Catalytic Stoves with secondary air inlet and combustion chamber (source: popularmechanics.com)

Rocket wood Stove

Rocket stove, shown in **Error! Reference source not found.**, is one of the improved stoves esign for improved thermal efficiency and less emission. It. Rocket stove has 'L'-shaped a

combustion chamber made with 4 cm thick clay liner that has internal diameter and size depends on the size of the pot to be used.

There are different types of household rocket stoves in Ethiopia; simply classified as households and institutional rocket stoves. Household rocket stoves are shown in figure 2-6. Their design is the same but different in the existence of skirt they have 'L'-shaped combustion chamber made with 4 cm thick clay liner that has internal diameter and size depends on the size of the pot to be used. The clay liner is then covered with galvanized sheet metal on the outside. The fuel and air inlet of tikikil rocket stove, shown in fig. 2-8 (a) is located at its bottom and it has 11 cm x 11 cm dimensions. The thermal efficiency of rocket stove is claimed to be around 30 % according to test reports. Tikikil stove production is guided by GIZ Energy Coordination Office and the production is mostly made by local manufactures.

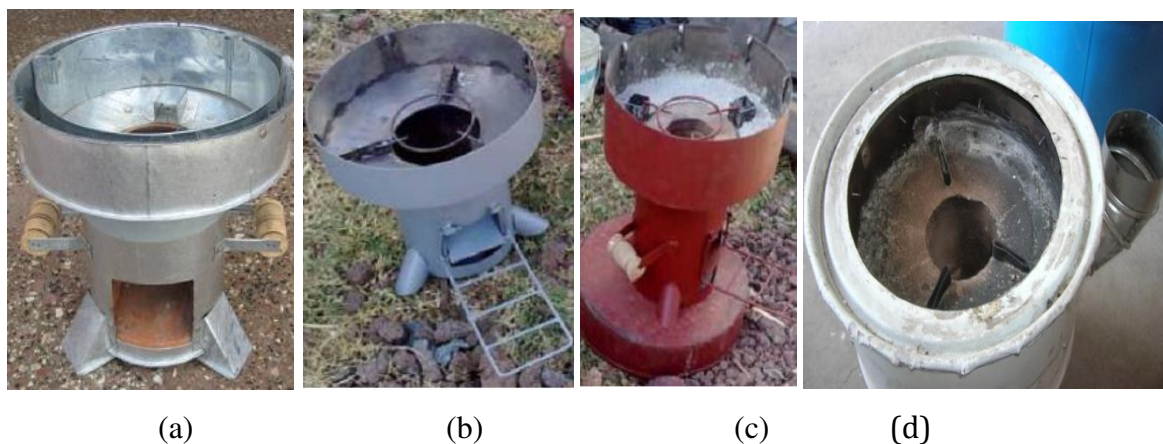


Figure 2-8 Rocket firewood cook stove or Tikil improved stove a) with double skirt , b) Deluxe single skirt 1,(c) Deluxe single skirt 2 and (d)rocker stove with short chimney (source: GIZ, 2010).

The dissemination of the stove is being done in Ethiopia mostly with governmental support and from international funding. Skirt in rocket stove design is considered to be important as it protects the heat loss from the pot and also to extract heat from the flue gas to skirt. The skirt is considered to be

Intuitional rocket stoves shown in fig. 2-9, are similar in design with household rocket stoves. Their difference is that they are built for large-scale intuitional cooking where large number of people are served such as in hospitals, universities, comps, etc. They are made from thick sheet metal (up to 2 mm) and they require experienced manufacturer. Its initial investment is high that

only few facilities could be able to install it in their kitchen. It has also a single pot seat that hinder it from being used in commercial scale.



Figure 2-9. Intuition rocket stoves with large size pots (without chimney left, with chimney right)on it (source: https://en.wikipedia.org/wiki/Rocket_stove)



Figure 2-10 Combustion chamber options for rocket stove (photo by MacCarty)

The possible combustion chamber for rocket stove is shown in fig. 2-10. Rectangular cross section assembled from several pieces, and a single piece of circular cross-section are the two possible options. The combustion chamber is mostly made from clay other similar materials might also be used in some circumstances.

Comparison of performance of rocket stove with other stoves

The comparison of the specific fuel consumption of various cooking stoves has been shown in figure 2-11, and it shows that improved biomass cooking stoves could cut the biomass consumption significantly in addition to the cooking time. When we compare the energy

required to boil one liter of water rocket stove consumes just 2470 KJ of energy compared to 6553 kJ consumed by open fire stove. When the time to boil to 2.5 liter of water rocket stove takes 10 liter whereas open fire takes around 30 minute.

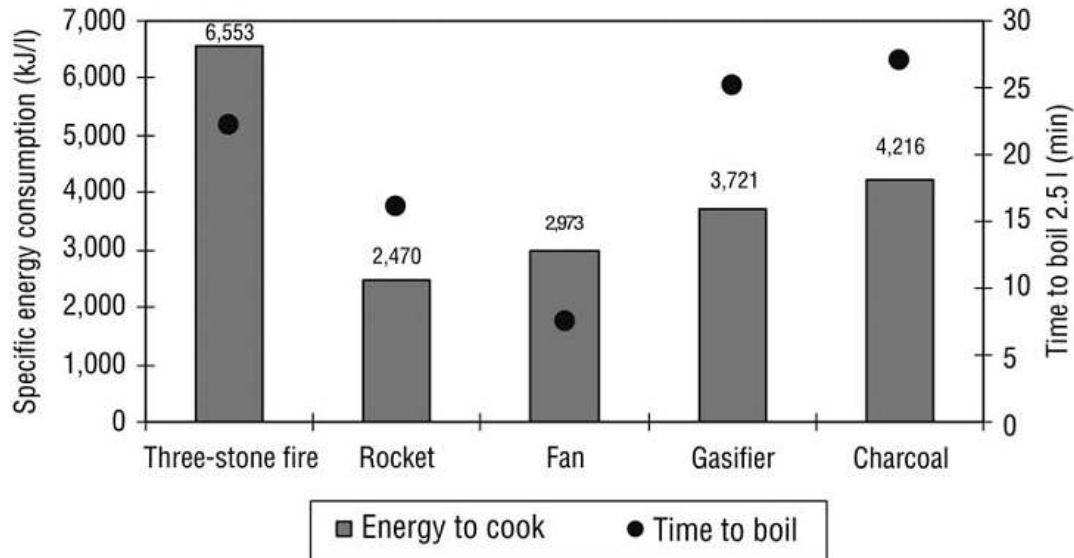


Figure 2-11 Specific energy consumption (energy consumed to boil 1 liter of water and then simmer for 30 min.) and time to boil 2.5 liter of water for the various stoves (source: MacCarty, et. al, 2008).

In addition to fuel saving to cook, and time rocket stove also cuts air pollution as of fig. 2-8; for example, the emissions are as low as 26% of the carbon monoxide (CO) and 60% of the particulate matter (PM) produced by the well-designed open fire stove (shown in fig.). Most importantly, it shortens cooking time by nearly 5 minutes for 5 liters of water, compared to the improved open fire stove. The overall global warming impact, or radiative forcing, including CO₂ considering the biomass was not harvested sustainably is shown above. It could be examined that rocket stove has the potential of reducing global warming compared to other stoves (MacCarty, 2008).

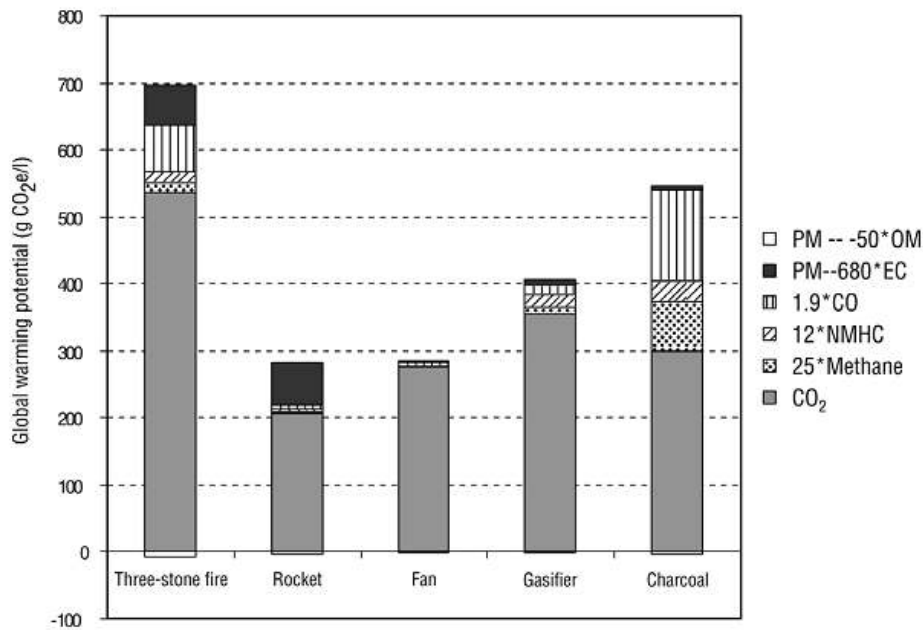


Figure 2-12 Total global warming impact, grams CO₂ equivalent on a 100-year time-frame, per liter of water boiled and simmered for 30 minutes (Source: MacCarty et al., 2008)

Effect of chimney in combustion and smoke exhaust in wood stoves

The combustion chamber of a stove is expected to be maintained with an optimum and good fresh air draft through the burning fuel. The effect of proper level of draft is to keep high temperatures in your stove providing high level of combustion. When too little air draft being pulled into the fire, it will result in incomplete combustion generating high smoke and creating excess charcoal. On the other hand, when too much fresh air draft occurs, it might create complete combustion but it will create too much air entrainment and cools the fire and the pot will not get enough heat from it (Bryden et al., 2005).

Therefore when the opening is optimum it will have advantage to the fire that it helps to reduce excess air and generate high temperature gas and improve heat transfer to the pot. This is the most important factor when considering to reduce fuel use in a cooking stove which at the end have environmental advantage. Improving the combustion efficiency with excess air might reduce pollution of the kitchen or even the environment, but it must be compromised with the other important factor save firewood.

The calculation of the exhaust chimney is made by the conservation of mass principle. That means smoke that leaves will be equal to the air flow plus the fuel flow rate. This design has been shown in Bryden, et al. (2005). They showed ways to increase convective heat transfer between that stove

and the hot gas: firstly, the flue gases scraping the combustion chamber should be as hot as possible to heat up the pot surface. Secondly, the hot gas flow velocity towards the pot should high as a faster flow the surface of the pot will enhance the heat transfer by convection. Their proposed result however did not consider about the expected resident time of the hot gas contact with the pot surface. Therefore, all those factors could be enhanced with optimization of chimney cross sectional area and height of chimney.

Figure 2-13 shows the comparison of fuel consumption of various stoves by each stove to boil and simmer a liter of water in a GIZ test. We can see that rocket stove with chimney consumes less fuel compared to other stoves. It consumes nearly half of what the traditional open fire stove would consume.

To comprehend the issue of draft in chimney, it is good to comprehend what draft is. Draft is the thing that we name the impact of how the fresh air streams up the fireplace or wood stove. Draft then is the mixture of volume flow rate, velocity, and static pressure weight of the exhaust gas. In addition, temperature of the gasses becomes an integral factor here also.

For matters of this dialog, fireplace draft is typically considered as the pace at which the fresh air go in to the combustion chamber and the flue gas gasses go up out of the chimney.

The reason smoke (or other vent gas) goes up the fireplace or the cook stove chimney is a result of the vacuum created in the stack. The vacuum created is in general with respect to the fresh air pressure in the house. This is similar to the phenomena that fresh air comes into a house.

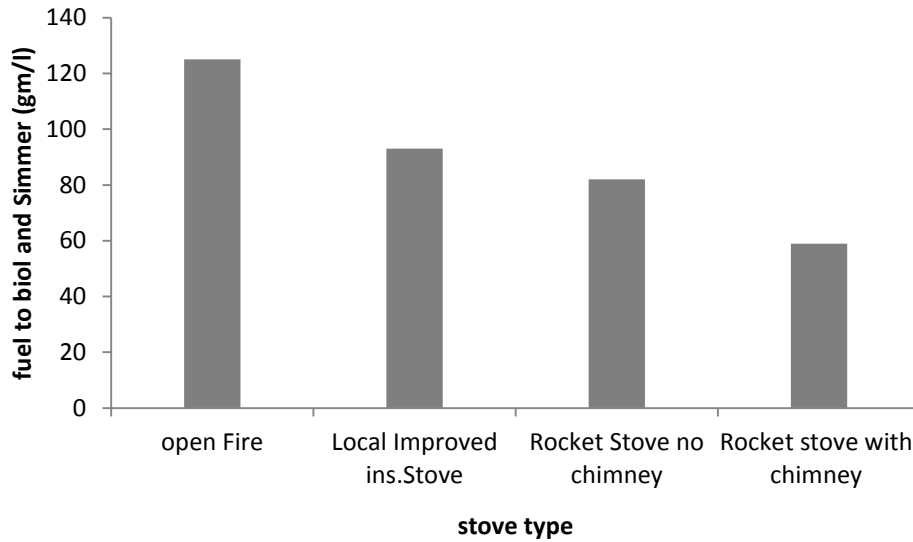


Figure 2-13. Comparison of open fire and different rocket stoves fuel consumption to boil and simmer one liter of water (source: GIZ, 2010)

Stove test for three type of stoves (fig. 2-13.) was conducted MacCarty et al. (2008). The test result compared to the three stone stove was displayed in fig. 2-14, and it shows that rocket stove with chimney could significantly reduce CO and PM emission.



Figure 2-14. Three “Improved” Rocket Stoves tested by MacCarty et al. (2008) at Aprovecho Research Center: (a) single pot b) two pot and c) rocket stove with chimney

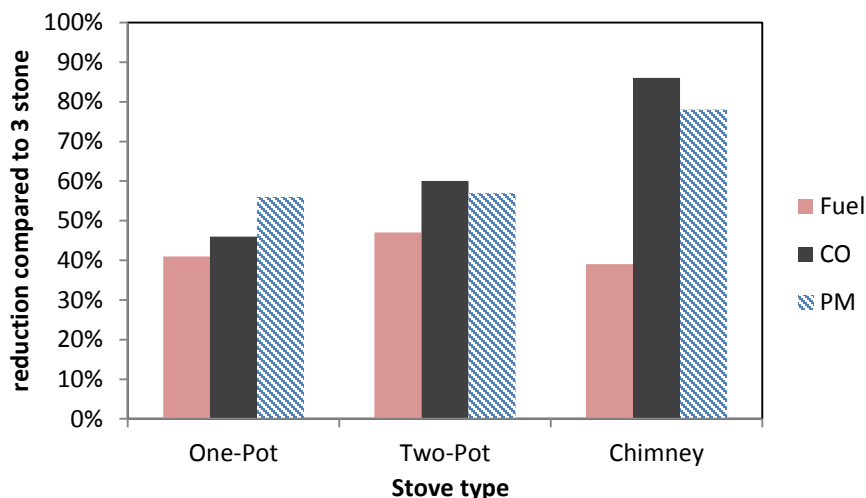


Figure 2-15. Three “Improved” Rocket Stoves tested by MacCarty et al. (2008) at Aprovecho Research Center: (a) single pot rocket stove b) two pot rocket stove c) rocket stove with chimney

Wood gasified stove

Wood gasifier stove is one type of improved biomass stove designed for complete combustion of fuel. In Gasifier stove the fuel is stacked and let to be maintained at an approximate 450-degree Fahrenheit temperature as it burns, thus creating massive amounts of smoky gases. This gas is then drawn away from the fire where it can be burned in a high oxygen environment in the gasifier's secondary combustion chamber as shown in fig. 2-16 (b).

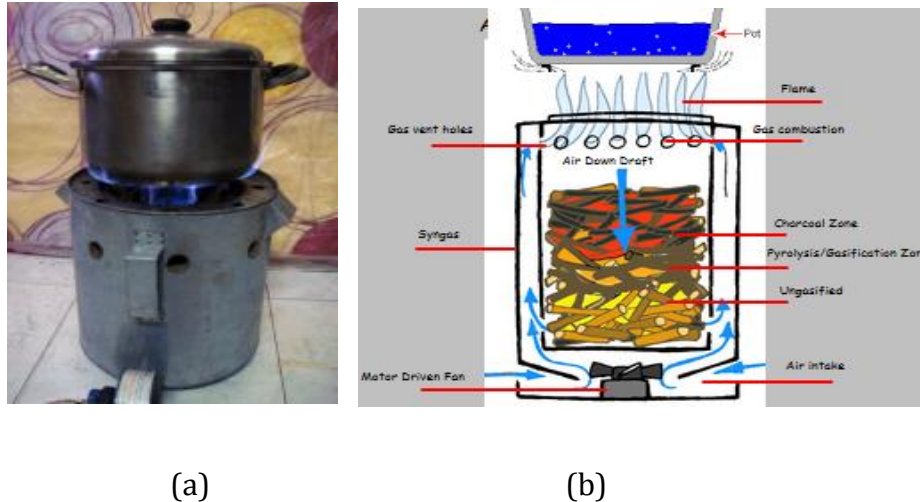


Figure 2-16 Wood Gasifier stoves (a) gasifier stove at operation b) gasifier stove detail sections
(source: envirotowns.com)

Mirt improved biomass cook stove

Currently there are various improved stoves being practiced by the public and the so called 'mirt' injera baking stove is one of them. Gonzye and fetenech also came as alternative baking stoves.

Mirt' injera baking firewood stove, shown in Figure 2-17. Mirt injera baking stove (a) the stove parts (b) the stove during baking injera has been developed by Ministry of Water and Energy to decades ago and has been used with in significant portion of the households.

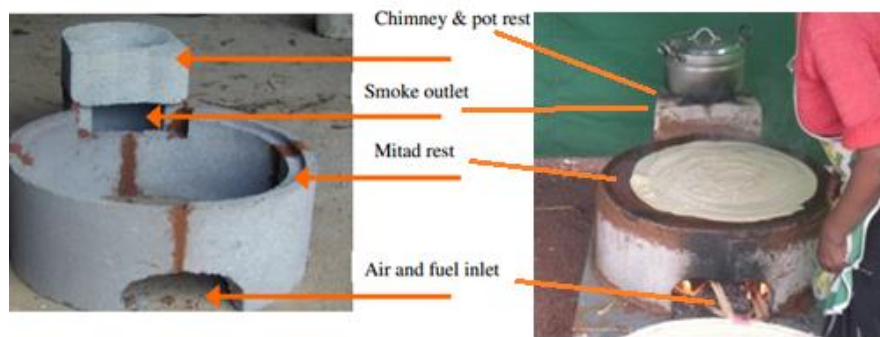


Figure 2-17. Mirt injera baking stove (a) the stove parts (b) the stove during baking injera

It is made of cement and sand mixture; it has some mold on which the stove is manufactured. Scoria also called red-ash is used as an element of the raw materials mixing with cement; river sand is also used as alternative to the scoria. The design of the mirt stove is it has a single lower opening for air and fuel (firewood) inlet. And it has a single smoke outlet. The smoke in most case is brought to another stove for cooking small items like water heating from the recovered heat which otherwise could be lost. In some cases the smoke is released through chimney to have smoke free kitchen.

The stove is basically cylindrically shaped enclosure whose axis is vertically oriented. With an opening around the bottom (fire-bed level) of its lateral face as air-fuel inlet and another of smaller size as exhaust outlet opposite but above the level of the fire-bed, this cylindrical enclosure is made when four mating pieces of walls match together to form a cylindrically shaped enclosure. The wall in some cases is reduced to three pieces. The cylindrical shaped enclosure is the place where combustion of fuel (firewood, agricultural residue, leaves, branches, cow dung, etc.) takes place. The confinement of the combustion chamber is the one that makes difference with the traditional three stone open stove. The top part of the walls of the enclosure has slotted rim that provide resting the clay plate called mitad upon which the injera is baked.

The combustion, air inlet and smoke exhaust mechanism in mirt stove is regulated by the proved openings as shown in fig. 2-18. At the back or the outlet of the smoke there are two other parts integrated ('O' and 'U'- chimney stack), one atop the other, attached with the cylindrical enclosure or the combustion chamber. Over these parts the heat recovery from the leaving smoke is done by using the exhaust smoke as a heat source for cooking in addition to the baking activity which is done on the main enclosure. In addition those chimneys are used to assist the stack effect. The U-shaped structure used to slow the speed of the smoke leaving and hence the combustion in combustion chamber is optimized. On the 'O' part of the stove the secondary cooking activity of wot cooking pot is placed. The 'O' chimney creates a draft for the stove by enabling sufficient level of pressure difference between itself and that of the combustion chamber. All those makes mirt stove a lot better than that of the traditional three stone open fire stove.

The stove testing and innovation is supported by non-governmental organizations like the German's NGO GIZ, Aprovecho, etc. Various stove test shows that mirt stove has 50% biomass saving potential than the traditional three stoves and its thermal efficiency is also rated as 18% (Energylopedia, 2014; Anteneh, 2011).

Depending up on the wall thickness the current types of mirt stoves are categorized in to classic mirt and slim mirt. The difference between the classic one and slim mirt is that slim mirt has thinner wall thickness of 4cm compared to 6cm of the classic's wall thickness. The advantage of the development of slim mirt is to overcome the challenge of the increasing cost of the raw materials as slim mirt uses lesser material due to its thinner walls (Anteneh, 2011). In turn the cost of the whole stove is reduced to have market benefit.

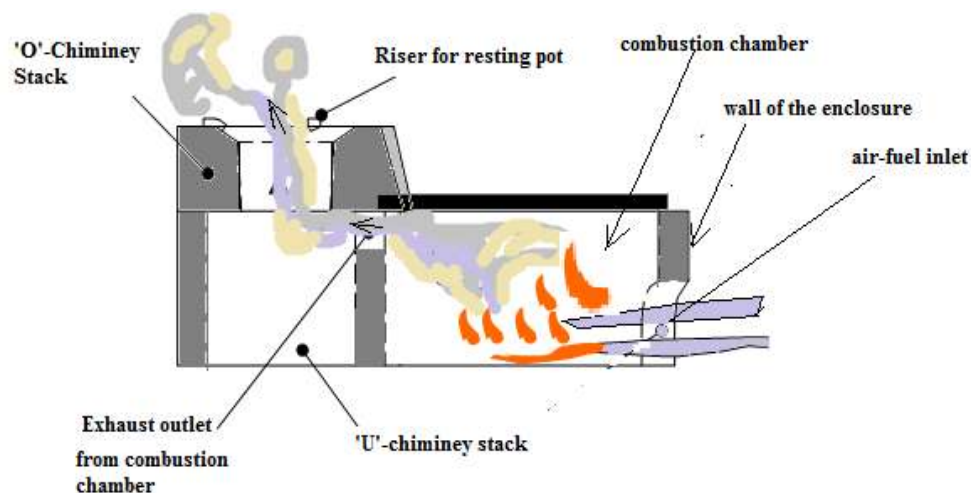


Figure 2-18 more detail from cross sectional view of Mirt Stove (modified based on Anteneh, 2011; energylopedia, 2014)

Gonzye and Fetenech (Upesi) improved biomass cook stove

(a)

Figure 2-19 (a) & b , are also another improved stove design to be used for baking injera or preparing wot, boiling water, coffee, tea, etc. Unlike Mirt stove, which is made from cement, they are made of burned clay, which is locally available material. Similar to Mirt, Gonzye is made of three to four partial circular walls but Gonzye does have pot or plate-rests. On the other hand, the Upesi stove consists of a single enclosed, cylindrical clay and a fuel inlet cut out of it. The stove test made by the Ministry of Water and Energy shows it has promising

energy saving capacity (energypedia, 2014). Several literatures show that Gonzye stove has thermal efficiency of about 16 to 20 percent (Taddese, 2013) and Upsi stove is also expected to be in the same range.



Figure 2-19 (a) Gonzye stove with injera baking plate on it and b) Upsi stove with a cooking pot on it (Source: energypedia, 2014).

Charcoal cook stove

Like firewood, charcoal is mostly used in households for cooking wot, tea and coffee, therefore improved charcoal stoves are also available in the Market. The advantage of using improved charcoal stoves is reducing the overall consumption (increasing the efficiency) and enhancing the quality of the indoor environment.

Figure 2-20 (a) shows the open charcoal stove (locally built). In this type of stove lots of heat is lost away from the pot by radiation and convection this leads to poor efficiency and high emission. The traditional Metal stove is better but it has also high heat loss due to the poor insulation. 'Lakech' clay stove, shown in fig 2-20 (c) is one of the improved charcoal stoves currently experienced by number of households in Ethiopia. The efficient charcoal stove was developed by adopting and modifying the Kenyan one. Lakech stove has relatively higher efficiency of its types. It is convenient to move and operate. It has door at the bottom and perforated charcoal rest, which facilitate airflow and ash removal. The same family stoves are available varying with construction type and size.



(a) (b) (c)
 Figure 2-20 charcoal stoves in Ethiopia: (a) open charcoal stove (b) traditional metal charcoal stove
 c) Lakech stove or Jiko

There are also improvement of charcoal stove towards rocket stove where the fuel inlet is under the chimney the same as rocket wood stove, to enhance the air draft through stack effect.

The design of EcoZoom's Jet jiko, as shown in fig. 2-21, began in 2012 in a cookstove laboratory in the USA run the Aprovecho Research Center. The performance of the stove is shown in table



Figure 2-21 The EcoZoom Jet charcoal-only cookstove (source:<http://ecozoomstove.com>)

Table 2-2 Preliminary Fuel Use and Emissions Test Results (source: *aprovecho.org*)

parameter	unit	Benchmark three stone Wood Stove	Charcoal rocket stove (without skirt)	Charcoal Jiko Stove	Wood Rocket stove (with Skirt)
Time to Boil	min	-	22	29	22
Fuel to Cook	g	850	604	675	733
Energy to Cook ^a	kJ	15,000	19,130	21,396	12,579
CO to Cook	g	20	12	74	15
PM to Cook	mg	1500	81	85	1289

^a Charcoal energy to cook = kg to cook *LHV

Table 2-2 shows the comparison of performance of charcoal rocket stove with other stoves. We can see that charcoal rocket stove has advantage on fuel saving, less time to cook and less emission of CO and PM compared to conventional charcoal stove.

Homemade Electrical baking and cooking stoves

Electric injera baking stove, shown in fig. 2-22, is another substitute for traditional three stone fire stove as it solves the problem associated with the rising price of firewood and the physical discomfort comes from the smoke of the three stone firewood stove. The state owned national electric power supplier, Ethiopian Electrical Power Corporation (energypedia, 2014), develops electric injera baking stove. The stove has imbedded electrical resisters in the plate or also called ‘mitad’. The plate used is the same as the one used in other fire stoves the old difference is the heat from fire wood is substituted by heat from electrical resister. It works with about 3.5 kW of electrical power consumption and it can have thermal efficiency of as high as 70 percent (energypedia, 2014). Currently it locally manufactured by most simple workshops and its market is rising as the public in urban area is rapidly shifting from firewood stoves to electric stove.



(a)



(b)

Figure 2-22 Locally manufactured Electric mitad or stove for injera baking (a) and electric cooking stove (b) in Ethiopia (source: energypedia.info)

Factory made and Sophisticated electric cooking stoves

In Ethiopia, few households with higher income tend to use sophisticated electric and gas cooking stoves and ovens. Those factory made electric or gas stoves are usually sold in electric retail shops. Those stoves could potentially save significant amount of energy; however, their higher cost is the drawback.

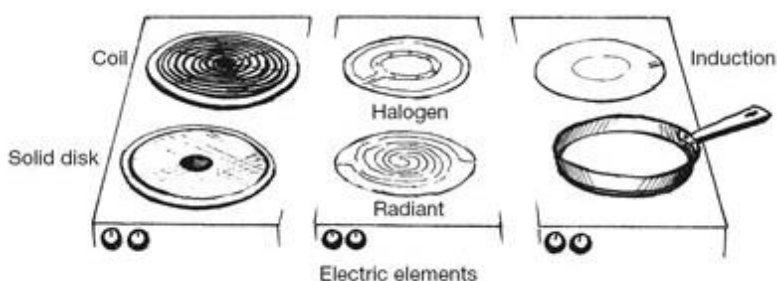
The stove shown in figure The best thing of sophisticated stoves is their available size of pot rest or plates. Electric stove hub, for example, can have up to four plates of various sizes. The user can therefore choose according to the size of the pan or pot. Figure 2-23(c) shows the types of sophisticated stoves types available.



(a)



(b)



(c)

Figure 2-23 Electric stove types used for cooking in Ethiopia: (a) coiled stoves and (c) electric hot plate or electric hub. c) More sophisticated induction stove tops (source: <http://smarterhouse.org/appliances-energy/cooking>)

Efficiency of Cooking Stoves

The main parameter that affects the amount of energy consumed in households is the stove efficiency. The stove efficiencies have been studied experimentally and table 2-2 shows the summary of list of efficiency of some stoves. The table shows the nominal³ efficiency, whereas the global⁴ efficiency is different (Alois, Martino, & Jannuzzi, 2005).

³ Only considers the energy conversion once the fuel reaches the consumer. It is defined as the percentage of the energy converted into heat. In this paper when “efficiency” is nominal efficiency

⁴ It also considers the energy loss in production, transport/transmission and final consumption of a the fuel

Table 2-3 Thermal Efficiency of Cooking Stove

Type of stove	Thermal efficiency (%)
Lakech Improved charcoal Stove ¹	20-25
Clay or mad charcoal stove ¹	29
Metal charcoal stove ¹	18
Electric cooking stove ²	50
Kerosene cooking stove	35-40
LPG stove	60
Traditional three stones open stoves	5-10
Mirt improved baking mitad stoves	18-25
Gonzye improved baking mitad stoves	16-20
Traditional enclosed mud stove (mitad)	10-13
Electric mitad stoves	70-80

¹ the loss of energy during charcoal production is not considered

²the loss of electricity during production and transmission is neglected

Source: Azemeraw (2013) and Amhara National regional State Energy agency

If global efficiency is considered for example for charcoal, the charcoal production efficiency is mostly less than 30%, and electricity generated from diesel generators could have global efficiency of around 40%, for coal it is around 30%, for hydro it could be as high as 85%. For electricity in addition to this, depending up on the grid there are transmission and distribution losses of about 15% to 20% (Alois, Martino, & Jannuzzi, 2005).

According to the design guideline proposed by size of the smoke chimney area is proposed to be equal to the

2.5.2 Lighting

Lighting is one sector that consumes significant portion of the residential energy in Ethiopia. The preference of energy for lighting is highly dependent on the level of electrification (Van Ruijven, et al., 2011). In rural and some urban areas where there is no access to electricity, candles, kerosene lamps, firewood and solar energy are the available options. In places where there is electricity, electric lamps are used. In Bahir Dar city, the lighting is based on electricity, and the available options for lighting are incandescent lamps, fluorescent tubes, and compact fluorescent lamps. The dominant share is taken by incandescent lamps because of their low initial cost compared to compact fluorescent lamps.

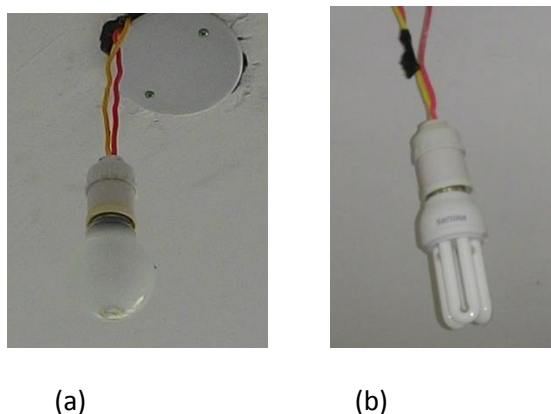


Figure 2-24 Common light bulbs in Bahir Dar (a) incandescent lamps (b) compact fluorescent lamp

2.5.3 Water heating

In developed countries, the majority of residential energy is consumed for space and water heating. However, in Ethiopia, due to the weather condition and income limitation space heating is not a common activity in Ethiopian households though charcoal stove is used in some cold areas in summer. In similar manner, space cooling is not a common activity in Ethiopian households. Water heating, however, is slightly more common in Ethiopian households compared to space heating; high income and modern households mostly use it though. The water heating is made usually by storage type electric boilers. Instant electric water heaters are becoming more common due to their energy saving and ease of use.

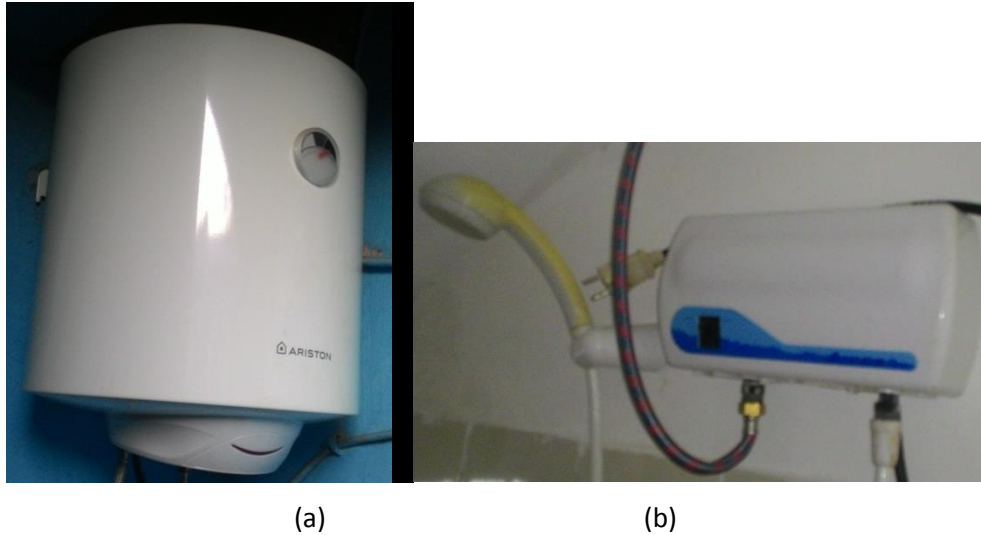


Figure 2-25. Electric water heating options in Bahir Dar (a) conventional storage type boilers (b) compact instant water heater

2.5.4 Appliances

Appliances like TV and its accessories, refrigerators, washing machines, etc. also consume significant portion of the residential energy in Bahir Dar. The ownership of appliances is highly dependent on various factors in addition to availability of electricity: income, educational level, and influence of neighborhood and more importantly on their settlement type (whether they are living in their own house or renting some one's house, apartment or single family house, etc.). The primary preference of households in Bahir Dar is to have TV and its complements. Secondly, households tend to won refrigerators as their income grows. Appliances like coffee grinding and juicer are becoming more common in Ethiopian households. Especially in apartments, where there is space and noise level limitation, households are forced to use more appliances like washing machines. Appliances like Vacuum clear, tumble dryer are not common in Ethiopian households, but they are penetrating to the market and will be common appliances very soon. In the future as the income level and the demand for comfort grows, it likely that households tend to own space cooling appliances like window or table fan, air conditioners, and other space cooling devices.

Chapter 3. Methodology

The study comprises primarily conducting a survey aiming to collect residential energy consumption patterns of 2014 other indicators like housing unit type, education, family size including preferences needed in policy setting and program implementation. The data is collected by interview type questionnaire within sample household. After obtain the data pre-analysis is conducted by MS excel by using is a set of equations designed to get results of average household energy consumption and per capita energy consumption. The end-use are categorized as cooking, water heating, refrigeration, lighting, TV, and other appliances. The total model is then performed By LEAP software performed based upon the disaggregate sample household's value. The modeling also includes forecasting the future energy consumption based up of various scenarios.

3.1 Description of the Study Area

The Energy modeling of residential sector is performed in Bahir Dar city, Ethiopia. Bahir Dar is the capital of Amhara National Region Stage (ANRS) situated at the north western part of Ethiopia and at southern part of Lake Tana, the source of the Blue Nile. Bahir Dar is also the capital of Bahir Dar specialized zone. The zone also includes rural villages, and small towns like Meshenti, Tis Abay, Zege and Zenzelima.



Figure 3-1 Map of Bahir Dar City (Source: wikipedia.org and Google map)

Bahir Dar is located at 11.7 °N and 36.8 °E and at altitude of 1800 m. Its population projection based up on the 2007 national census conducted by the Central Statistical Agency of Ethiopia (CSA), shows about 155,428 people. Bahir Dar is one of the fastest growing cities in Ethiopia; several sources estimate the annual population growth as 4.13%, and hence in 2014 the population of Bahir Dar city is estimated to be around 205,000. The city has various types of settlement. There are programs for distribution of lands to residents for construction of their own houses which results the rate of new house construction and expansion of the city to be fast. The older houses in Bahir Dar are mostly constructed with mud wall and corrugated steel roofs, although there are varicose house with brick wall which are mostly owned by high income people. There are also various illegal settlements so called 'moon houses'⁵, which are constructed without proper permission from the city administration. Those houses are unstructured and are constructed from simple materials like mud wall and steel roof, and mostly those residential houses are not connected to the public electricity grid, but they borrow electricity from the neighboring houses in some deal of monthly payment per light bulb.

Like any other cities in low-income countries, households in Bahir Dar have no access to natural gas supply network. The city has shortage in fuel supply and almost all of its fuel is based mostly on firewood and charcoal that come from neighboring villages. The firewood comes from nearby towns and islands located in Lake Tana⁶, and the wood is usually transported by car, boat and by using animals like donkey.

3.2 Data

The data that has been used in this study comes from two sources one from primary data collection and the other from secondary sources. The primary data is important at it is difficult to find sufficient data from other sources therefore the data is collected by using house to house interview type questionnaire and measuring their daily fuel consumption by portable weight scale. The secondary data is gathered from governmental and non-

⁵ Their name indicates that they are mostly constructed at night time without the knowledge and permission of the city administration

⁶ Lake Tana is the largest lake in Ethiopia and Bahir Dar is located at the coast of it. The lake has several islands, which have relatively better forest coverage.

governmental organizations' reports, from websites, from previous studies and reviewed articles.

This study employed both qualitative and quantitative research methods to cover the basic parameters in energy consumption. In order to obtain sufficient and representative data about the amount of energy consumption, types of fuel used, type of stove used and to know other factors that determine energy consumption pattern of the household.

3.2.1 Structured Household questionnaire

Questionnaire was prepared to collect the parameters about the current household energy consumption and expenditures. The questionnaire is interview type that had to be filled by experienced data collectors. Information included in the questionnaire are parameters like family size, monthly income, level of education, house type, fuel type, fuel amount, stove type, stove preference, fuel preference, electricity consumption. The entire questionnaire is designed to answer the questions like whether current household energy consumption looks like, what the main indicators that determine household energy fuel usage, and how long can the fuel or stove or device stay in order to shift towards future alternative technologies. The questionnaire is attached in appendix VI.

3.2.2 The methods sample size technique distributions

As the case study, carried out at Bahir Dar, all the 9 sub-cities (so called *kifleketema*) were selected. As the sub cities have different amount of households, different type of housing and settlement, the sample size varies proportionally with their household numbers. In addition the living standard educational level, income and type of building varies from sub cities to sub cities, the sampling have also taken these factors in to consideration. Considering these factors about 350 samples households in total were chosen randomly with in the sub-cities. Before the actual data collection training and pre-testing was collected. Later on the actual data collection is performed. The data collection was performed by three experienced statistician (data collectors). Weight gauges are used to measure as accurate as possible the amount of fuel consumption.

3.2.3 Market baseline survey

Market baseline study is also conducted to understand the available amount of fuel in the market and to know its price. The study was conducted on various markets and shops where the fuel supply is available. From the survey, the size and price of firewood is taken, the price of various available sizes of charcoal, availability of the fuel or supply consistency, etc. is taken.

3.2.4 Statistical Analysis

Preprocessing is conducted to change the row data to be able to use it in further modelling tool. The preprocessing is conducted by first filling the raw data in MS excel then the data is analyzed, converted to useful unit and means, standard deviations, confidence intervals and medians were calculated by stove type, fuel time, house type etc. Then the mean value was used for further analysis and forecasting.

3.3 Calculations

3.3.1 Calculation of average annual energy consumption of particular fuel per households

The energy consumption per household has two distinct parts: primary energy consumption and end use energy consumption. The calculation of the primary energy consumption per household was calculated by using the data collected usually in unit of kilogram, liter, etc. per month and then converting to energy using by multiplying this quantity with the LHV (MJ)/kg and converting it to annual energy intensity. On the other hand the useful annual average energy consumption per household by fuel is calculated by considering the conversion efficiency of the stove or the device on which it is being used this will lead to total energy consumption (SEI, 2011).

Calculation of household's primary annual energy consumption

In order to calculate the annual energy consumption of every sample household, first the amount of the annual fuel consumption of the households is obtained from the survey based up of the people's response; the amount is converted to mass except for electricity which is recorded time usage and wattage of the device. Afterwards the amount is

multiplied by the Lower heating value (LHV) of the fuel to obtain annual energy consumption.

$$E_{hs} = \frac{LHV_f * m_f}{3.6} \quad \dots(3.1)$$

Where

$E_{f,hs}$ household's primary annual energy consumption from a fuel (kWh/year)

LHV_f the Lower heating value (LHV) of the fuel (MJ/kg)

m_f amount of the annual fuel consumption by mass (kg/year)

The calculation of average household energy consumption of a fuel also called energy intensity is conducted by adding all the energy consumption of the sample households and dividing it with the number of the sample households (SEI, 2011; Taddese, 2013).

$$E_{f,hs,average} = \frac{\sum_1^n E_{hs_i}}{n} \quad (3.2)$$

Where

$E_{f,hs,average}$ average household energy consumption (kWh/year)

E_{f,hs_i} primary annual energy consumption of household i (kWh/year)

n number of sample households

The per capita energy consumption is calculated by dividing the average energy consumption of a fuel f calculate above in (2) with the average family number.

$$E_{f,per\ capita} = \frac{E_{f,hs,average}}{n_{fam}} \quad (3.3)$$

Where

$E_{f,per\ capita}$ average per capita energy consumption of a fuel f

n_{fam} average number of family per household

To calculate the city's each fuel's contribution for the total primary energy consumption of residential sector is done by multiplying the average household energy consumption of each fuel, as calculated above in equation (2), with the city's total number of households N .

$$E_{total,f} = (E_{f,hs,average})(N) \quad (3.4)$$

Where

$E_{total,f}$ total energy consumption of the city from a fuel type f (kWh/year)

N total number of the households in the city

f stands for a fuel type like charcoal, firewood, etc.

In order to calculate the city's total energy consumption or demand (E_{total}) the total energy consumption of each fuel (like charcoal, firewood, electricity, etc.) as calculate above in (4) is added as shown in (5).

$$E_{total} = \sum_1^n E_{total,f} = E_{charcoal,total} + E_{firewood,total} + \dots \quad (3.5)$$

Where E_{total} the total energy consumption

Calculation of average annual useful energy consumption

The above calculated parameters namely the household average, per capita and total energy consumption are calculated based up on the primary energy consumption. That means without considering the efficiency of the device up on which the fuel is used. Thus the calculation of the useful energy is made by considering the efficiency of the device like stove on which the fuel is used. Therefore the results obtained from equation (2) up to (5) for primary energy consumption of particular fuel is multiplied by the conversion efficiency namely thermal efficiency of the corresponding stove η_{st} .

For example, the household useful average annual energy is calculated as (SEI, 2011; Taddese, 2013):

$$E_{f,hs,average,useful} = \frac{\eta_{st} * E_{f,hs,average}}{100} \quad (3.6)$$

Where

η_{st} device or stove efficiency(%)

$E_{f,hs,average,enduse}$ useful household energy consumption of a fuel (kWh/year)

For per capita energy consumption the useful value calculation is done in similar manner.

$$E_{f,per\ capita,useful} = \frac{\eta_{st} * E_{f,per\ capita}}{100} \quad (3.7)$$

And the useful total energy consumption will be calculated as

$$E_{f,total,enduse} = \frac{\eta_{st} * E_{f,total}}{100} \quad (3.8)$$

Where

$E_{f,per\ capita,useful}$ useful per capita energy consumption from a fuel type f

$E_{f,total,useful}$ useful total energy consumption of a fuel f

Calculation of the share of households using various fuels

The percentage or share of the households using particular type of fuel was done by taking the number of households using various fuels and dividing to the total number of households. Assume X_i is the number of households using particular fuel f_i , and N is the total number of households, the share of the total number of households could be calculated as (SEI, 2011; Taddese, 2013)

$$Sh_f = \left(\frac{X_i}{N}\right)(100) \quad (3.9)$$

Where N is the total number of sample households, Sh_f the share of fuel type f (%), X_i is the number of households using particular fuel type f

3.3.2 Energy consumption for appliances and lighting

Energy consumption by appliances is calculated based on the approximate wattage of the appliances and the hours of daily usage that will be collected based on the response during the interview. The following equation shows the method to calculate the annual energy consumption of the fuel (electricity) per households.

The primary energy demand (UED) for lighting and for other appliances can be calculated as shown in equation (1).

$$PED_y = \frac{\sum_1^n (P \times t_d) \times d_y}{n} \quad (3.10)$$

Where: PED_y = the average yearly primarily energy intensity for the device (kWh/y) ; P = the wattage of the device (kW), t_d = average usage time per day (h), d_y =days used per year , and n = the number of households using the device.

The useful energy demand of the device will calculated as shown in equation (10)

$$UED_y = PED_y \times \eta \quad (3.11)$$

Where: UED_y =average useful energy intensity per year ; PED_y =primary energy demand per year; η =efficiency.

Table 3-1 shows the wattage for some appliances used in households. It could be seen that some appliances have high wattage while others have moderate or low.

Table 3-1 Wattage of various appliances

Appliance	Wattage (W)
Coffee maker	900-1200
Juice maker	300
Clothes washer	350-500
Clothes dryer	1800-5000
Water heater (family size)	4500-5500
Water pump (deep well)	250-1100
Personal Computer, in use/sleep	120/30
Laptop	50
Refrigerator average size	1000
TV 14"/ 19"/21"	90/100/120
Vacuum cleaner	1000-1440
11-W Compact Fluorescent Lamp (CFL)	11
15-W Compact Fluorescent Lamp (CFL)	15
60-W Incandescent Lamp	60
100-W Incandescent Lamp	100

Source: <http://www.absak.com/library/power-consumption-table>

3.3.3 Calculation primary fuel price

Like many other commodities the price of fuel is a major issue in the public and it seems to affect the fuel choice significantly. In order to know the price of the fuels several observations were conducted and the following is the average price of the fuel. The fuel supply is mostly from the farmers who usually carry fuels with in the neighborhood using their donkey. Sometimes the fuel mostly charcoal is bought from market.

The fuel price is calculated using the unit price of the fuel in the market and the lower heating value of the fuel. First, the market price is observed to know the average price of the fuel and the corresponding quantity. This price and quantity of different fuels can be found in Appendix [III](#).

The fuel price in terms of the energy unit can be categorized in to the primary fuel price and useful fuel price. The primary price is the price of the fuel without considering the efficiency of the stove upon which it is used. Whereas the useful price of the fuel considers the efficiency of the stove to account the actual amount of the fuel used. Such approach have been practices on other studies (Alois, Martino, & Jannuzzi, 2005; Taddese, 2013)

Calculation of primary fuel price

The primary fuel price is computed as

$$pr_f = \frac{(F)}{\left[(LHV) \left(\frac{1kWh}{3.6MJ} \right) \right]} \quad (3.12)$$

Where pr_f unit price of the fuel (ETB/kWh)

$F_{pr,m}$ market price of the fuel per kilogram (ETB/kg)

LHV Lower heating value of the fuel (MJ/kg)

$\frac{1kWh}{3.6 MJ}$ is used to account conversion of MJ in to kWh

Calculation of useful fuel price

The price of fuel based on its end use value is calculated in the same as the primary one, the difference is the end use it the consideration of the conversion efficiency of the stove up on which the fuel is used. Therefore, the efficiency of the stoves that are being practiced in the sample households is found from several sources. The equation that is used to calculate the end use price is (Taddese, 2013; SEI, 2011; Alois, Martino, & Jannuzzi, 2005):

$$pr_{f,useful} = \frac{(F_{pr,m})}{\left[(LHV) \left(\frac{1kWh}{3.6MJ} \right) \times \eta_{st} \right]} \quad (3.13)$$

Where $pr_{f,useful}$ the useful price of the fuel (ETB/kWh)
 $F_{pr,m}$ is the market price per kilogram of the fuel(ETB/kg)
 LHV Lower heating value of the fuel (MJ/kg)
 $\frac{1kWh}{3.6MJ}$ is used to account conversion of MJ in to kWh
 η_{st} efficiency of the stove upon which the fuel is used

3.4 Energy Modeling with LEAP

The model that was used to calculate the primary energy of the household, end use energy of the household and the per capita primary and end use energy consumption in the urban and rural area, energy consumption of, the share of the households using various fuels is done with systematic approach on LEAP software.

LEAP (Long-range Energy Alternatives Planning) was developed by Stockholm Environment Institute (SEI) is an integrated energy modelling tool that can be utilized to model energy demand, production, and resource analysis in all sectors of energy (Connolly, 2010). Although users from developed countries or those who are employed to companies in connection to developed countries are expected to buy the license, users from developing countries can get the license for free. Leap is one of the widely used energy modelling too in the world (Connolly, 2010). LEAP is usually suitable and can mostly be used to analyze the energy demand and supply of a system in particular nation or local area.

Leap initially requires information to be provided in structured manner as current account and then the analysis will be performed using an annual time-step. The time limit can extend to forecast for several years between 20 and 50.

The vital model in LEAP is an end use driven scenario analysis (Amare, 2007) (Taddese, 2013). Firstly, the so called reference scenario is created considering the current energy situation and the previous trend. This will be projected over time steps towards expected future changes based on the likely plans and growth trajectories. It is “business-as-usual”

scenario with implementation of anticipated and likely to be carried out projects and policies. Then an alternative scenario that simulates new policies measures can be developed (SEI-Boston 2005, SEI, 2006).

3.4.1 Data structures and input to LEAP under current account

LEAP requires the data to be entered in organized hierarchy or branches. In this model the data input to the leap is done in organized way of branch trees for the residential sector (other sectors like transport, commercial, agriculture, etc. are not included). As shown in fig. 3-2 the branch trees under residential sector are categorized into: Cooking, Baking, Lighting, Water heating, Refrigerators, and other electronic appliances. In order to simplify the analysis cooking is re-classified in to baking (which included baking *injera* bread) and cooking (which includes cooking *wot*, tea, coffee or other similar food items. *Afterwards*, the classification will be based on the types of fuels used such as wood, electricity, sawdust, cow dung, etc. Then the type of stoves or the devices on which the fuel is used is considered to make the final branching. Over the final branch the so called activity level and final energy intensity is filled as shown in Figure 3-2. The data is entered in tables that are used to provide expressions showing the value at the current Accounts (so called base year) values (SEI, 2011). Similar branching is made for other categories as shown in table 3-1.

In leap Energy demand, or consumption, refers to the energy requirements at the point of end-use for the various consumption activities in particular society. Demand for energy can be categorized in to various final energy-using sectors; those are industry, transport, households, services and agriculture. TED uses a similar structure to organize its demand-side technology information. Household is divided into end use sectors like cooking, lighting, appliances, water heating, space conditioning (heating and cooling) etc. (SEI, 2011).

The energy consumption in leap is specified as activity level (e.g. the share of fuel used for particular branch) and annual energy intensity of a fuel (energy use per unit of activity, e.g. energy used per household or per person). And then the energy consumption per branch is calculated as the product of an activity level and the annual energy intensity (energy use per unit of activity) (SEI, 2011).

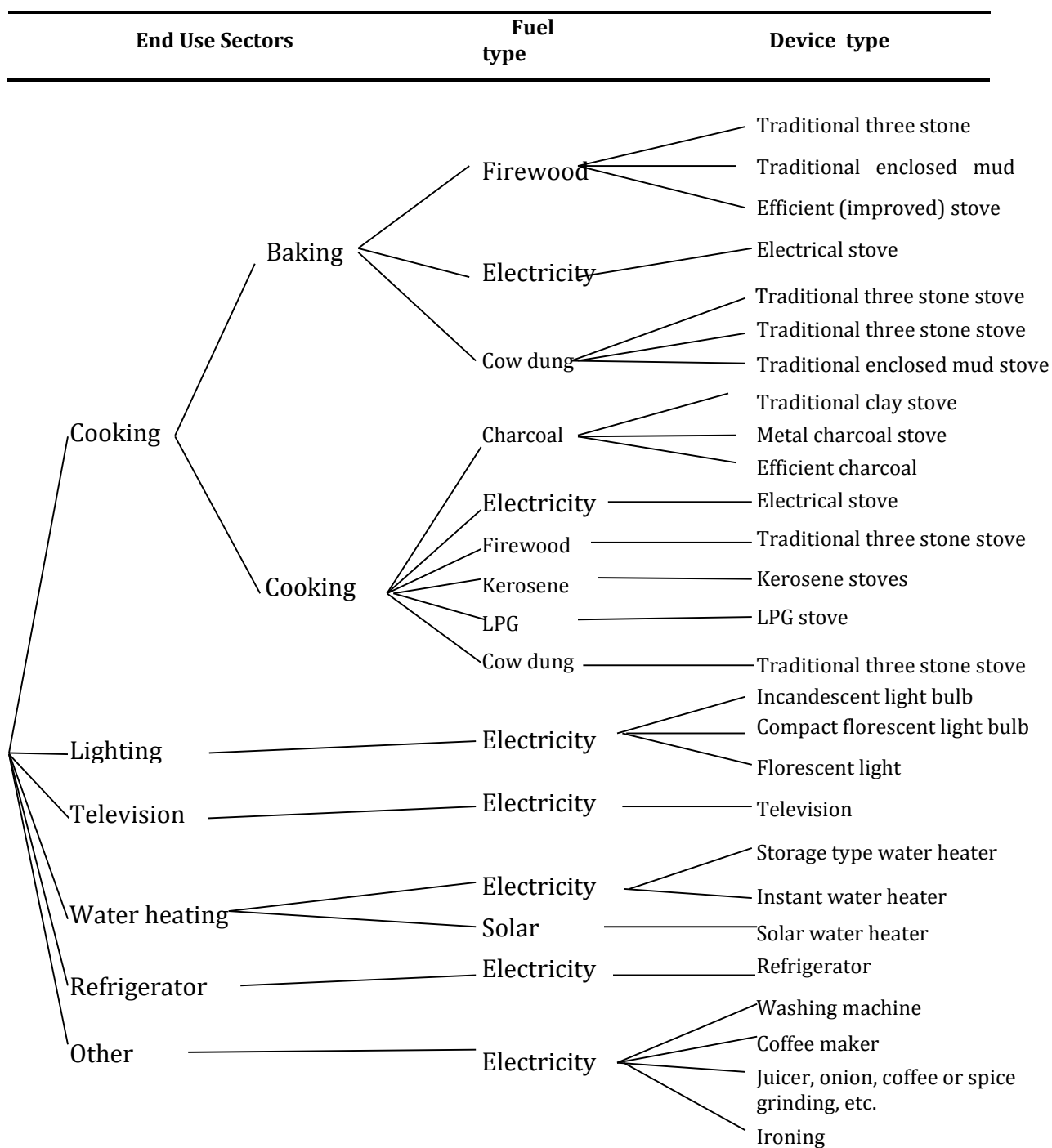


Figure 3-2 Data input tree structure to LEAP

The two parameters (activity level and energy intensity) are found from the pre-analysis of the raw data collected from the survey and performed on MS excel spread sheet. For example figure 3-3 show the entering of the parameters on the data entry table of the LEAP for wood fuel for baking. The energy intensity is entered as energy consumed per person.

Branch: Demand\Residential\Cooking\baking\Wood\Traditional Three stone Stove

Branch: All Branches Variable: Activity Level Scenario: BAU: Business as Usual (BAU)

Activity Level Final Energy Intensity Unscaled fuelshare All Variables

Activity Level: A measure of the social or economic activity for which energy is consumed. [Default="0"]

Branch	2014 Value	Expression	Scale	Units	Per
baking	100.00	100	Percent	Saturation	of People
Wood	63.20	Interp(2030,33.5)	Percent	Share	of People
▶ Traditional Three s	44.00	44	Percent	Share	of People
Traditional Enclose	16.00	16	Percent	Share	of People
Improved Efficient	40.00	40	Percent	Share	of People

(a)

Branch: Demand\Residential\Cooking\baking\Wood\Traditional Three stone Stove

Branch: All Branches Variable: Final Energy Intensity Scenario: BAU: Business as Usual (BAU)

Activity Level Final Energy Intensity Unscaled fuelshare All Variables

Final Energy Intensity: Annual final consumption of energy per unit of activity level. [Default="0"]

Branch	Fuel	2014 Value	Expression	Scale	Units	Per
▶ Traditional Three stone	Wood	757.52	757.52		Kilowatt-Hour	per Person
Traditional Enclosed m	Wood	645.83	645.83		Kilowatt-Hour	per Person
Improved Efficient stov	Wood	596.67	596.67		Kilowatt-Hour	per Person
Highly efficient stove	Wood	416.67	416.67		Kilowatt-Hour	per Person

(b)

Figure 3-3 data entry table in LEAP for Current Account: (a) activity level (b) final energy intensity.

3.5 Residential Energy Consumption Scenario Analysis

It is clear that one of the objectives of this project is to analyze the long-term energy consumption scenarios that might evolve through years. Using LEAP, the reference and the mitigation scenarios will be expressed.

3.5.1 Reference scenario

Reference scenario also called Business As Usual (BAU) scenario will be expressed considering things like expected changes like population growth, and growth of number of emerging new technologies, and growth people's interest for comfort, etc. The natural increase based on population growth and demographic changes such as household size and the changes in age groups is the first factor. Another factor is the rise of economic activity and development. It is known that the population of Ethiopia is growing at the rate of nearly 3% and due to urbanization the growth in urban areas is just less than 4%. The growth of the household number in Bahir Dar will be taken as 4% that will lead to growth in total energy consumption.

The increase of the income of the people will also affect the energy consumption both in terms of quantity and quality or the fuel type choice. It is common to observe newly built houses are both large in size and are designed to accommodate more appliances, as the result, it will lead to rise of the household's energy consumption.

In general the reference scenario does not consider any interruption of the policy makers or the change of the people's attitude towards reducing energy consumption as the result of policy change (Bundit & Pimporn, 2006; SEI, 2011).

In Bahir Dar, for the business-as-usual scenario, the share of each cooking fuel in the households continues along its historical trajectory. The share of electrical stoves will grow at the rate that depends on the number growth of household income. It also depends on the growth of the population. Usually in Bahir Dar newer buildings are becoming more electrified (with more electrical appliances and stoves) than the older buildings. The business as usual scenario is constructed based on the help of previous trend of the household energy consumption in Ethiopian urban households studied by various scholars (Azemeraw, 2013; Taddese, 2013; Alem, Beyene, Köhlin, & Mekonnen, 2013).

The data from 2005 is used to construct the BAU Scenario as the previous historical data and the trend in coupled with the current data collected through the survey is entered and the simulation in LEAP. The main parameter that affects the growth of the total consumption is the growth of the household population that is accounted in the model.

Table 3-2 Growth rate of the fuel consumption in Business as Usual Scenario

Activity	Fuel	Current share	Growth rate of the share (for Business as Usual)
Cooking 'Wot' , tea, coffee	Electricity	18.3 %	5.10%
	Charcoal	79.1 %	-2.00%
	Cow dung	0.3 %	-2.50%
	biogas	0	0
	LPG	0.3 %	-2.50%
	Kerosene	0.5 %	-9.60%
	Wood	1.5 %	0.00%
Baking 'injera'	Cow dung	1.8%	-1.20%
	saw dust	1.5%	0.10%
	Electricity	33.5%	4.10%
	Wood	63.2%	-3.90%
Lighting	Incandescent Light (60 and 40 W)	41.3	4.20%
	Compact Florescent Light (11-15 W)	58.1	-6.60%
	Florescent Light	0.6	-2.50%
Refrigerator	Medium size	43.2%	3.1%
TV and its accessories	Medium size	82.7 %	0.7%
Water Heater	Solar water Heater	0	0
	Instant water Heater	15	4.40%
	Storage boilers	85	-1.20%
Other appliances		5.3	6.9%

The other parameter is the growth of the people income which intern lead to more comfortable life this leads to fuel shift and also simulated in the model. Thus the residents will own more appliances having lager size and in fact larger power consumption. Table 3-3 shows the parameters that determines the future energy consumption in Bahir Dar city.

3.5.2 Forecasting of future energy consumption based up on reference scenario

Forecasting can be performed in analysis in various time series functions, for which historical data is required to be filled either in the form of historical data using the interpolation steps or approximate smooth function. The forecasting of the future values will be performed using various methodologies. The data to be supplied in those functions should be with time-series data, containing of pairs of year/value data and percentage growth rate parameter that is given after the last data year. For example the calculation of the value in any given year can be made by linear interpolation of a time-series of year/value pairs as follows.

The final optional parameter to the function is a growth rate that is applied after the last specified year. If no growth rate is specified zero growth is assumed (i.e. values are not extrapolated). Each intermediate year's value is calculated as follows:

$$Value_{iy} = Value_{fy} + [Value_{ey} - Value_{fy}] * \left[\frac{year_{iy} - year_{fy}}{year_{ey} - year_{fy}} \right] \quad (3.15)$$

Where:

iy = the intermediate period, the value of which is to be interpolated.

ey = the end period used as the basis for the interpolation.

fy = the first period used as the basis for the interpolation.

When growth rate per year is known then the value can be calculated as:

$$Value_{iy} = Value_{fy} + Value_{fy} \left(\frac{GR}{100} \right) \quad (3.16)$$

Where

GR is growth rate (%)

iy = the intermediate period, the value of which is to be interpolated.

fy = the first period used as the basis for the interpolation.

3.5.3 Mitigation scenario

Considering the various changes in policy from the stakeholders or the intervention measures the mitigation scenario will be constructed.

It is known that energy consumption depends on factors like income level, people's fuel preference due to social reasoning, resource availability, and environmental condition. Therefore the primary step would be to know the baseline trend (based on baseline) scenario, and mitigation trend (based on mitigation scenario) (SEI, 2011). This will be followed after find the existing energy demand for the households based on the current account.

The policy change that is considered in this study is has three categories. Firstly shifting towards more energy efficiency appliances to reduce the energy conversion losses, provide opportunities to change from one kind of fuel to another. Enhance the public awareness stimulate people's usage of renewable energy resources.

Improving energy consumption efficiency for cooking

It is known that the majority of the energy in households is consumed for cooking which is conducted on inefficient stoves. Introducing new efficient stoves would save ample amount of energy and reduce the CO₂ emission. There are several types of stoves that have been developed by several institutes.

Introducing fuel shift

Fuel shift is one solution to reduce the amount of energy consumption. It is known that the majority of the households in Ethiopia use sources of transitional biomass fuels like firewood for cooking and converting this to electricity not only improve the consumption efficiency (as electrical stoves are more efficient) it will also reduce the GHW emission as electricity in Ethiopia is produced from hydropower.

Using efficient light bulbs

Lighting is also one of the activities that consumes significant amount of energy in Ethiopia. However it is obvious that the majority of the households use the less efficient incandescent light bulbs. Therefore the replacement of those light bulbs with more efficient compact florescent light bulbs is vital. This would save significant amount of energy of the households.

Introducing and expanding usage of solar power

Due to its strategic geographical location, Bahir Dar has huge potential of solar radiation. This potential would be used either by PV cell to convert it to electricity or for Solar Water Heating. Solar Water Heating would effectively cover the household's hot water demand including for shower. Therefore the new policy should focus on it.

Energy Efficient Appliances

Although energy consumption due to appliances in Bahir Dar is small compared to other at this time, its value is expected to be projecting fast. Thus the share appliances in the total household energy consumption will be significant. The government, therefore, is expected to act to introduce policy to have appliances with low energy consumption or more efficient types.

Chapter 4. Results

4.1 Characteristics of Sample Households

The total number of households covered by the survey was 350 of the Bahir Dar city households, and due to the sampling technique used it could be stated that the data obtained can represent the energy consumption situation in Bahir Dar.

Table 4-1 some common parameters of the households involved in the survey in Bahir Dar

Stratum	Sub-stratum	Households	Percentage (%)
Education level	No education	70	20.29
	Primary school	40	11.59
	Secondary school	108	31.30
	Diploma/vocational	46	13.33
	Degree and above	81	23.48
Family size	Maximum	12	
	Minimum	1	
	average	4.8	
	Variance	3.41	
Type of the house	Steel roof /mud wall low quality	63	18.37
	Steel roof /mud wall medium quality	158	46.06
	Modern villa	70	20.41
	Apartments	17	4.96
	Single bed room	35	10.20

4.2 Characteristics of Household Energy Consumption

The fuel types that are commonly used for household cooking in the case study city include firewood, charcoal, cow dung, crop residue, BLT, electricity and Kerosene. The energy consumed in households is mostly for cooking, lighting, refrigeration, TV, water heating and for running other appliances. Figure 4-1 shows the total energy consumed in Bahir Dar city in 2014 is about 320 GWh. The major portion of this which means more than 87% or

more than 300 GWh is consumed annually for cooking purpose; and the cooking is mostly done by the traditional fuel sources such as charcoal and firewood. More than 186 GWh of the cooking energy comes from charcoal and 118 GWh from fire wood the remaining small amount of energy for cooking comes from electricity and very small amount from Cow dung. The remaining end-use activities use electricity: to list just less than 23 GWh of electricity is consumed for TV and about 24 GWh consumed for refrigerator and about 13 GWh is consumed for lighting. Energy consumed for water heating is just less than 3.5 GWh and other appliances consume about 0.37 GWh of electricity per year.

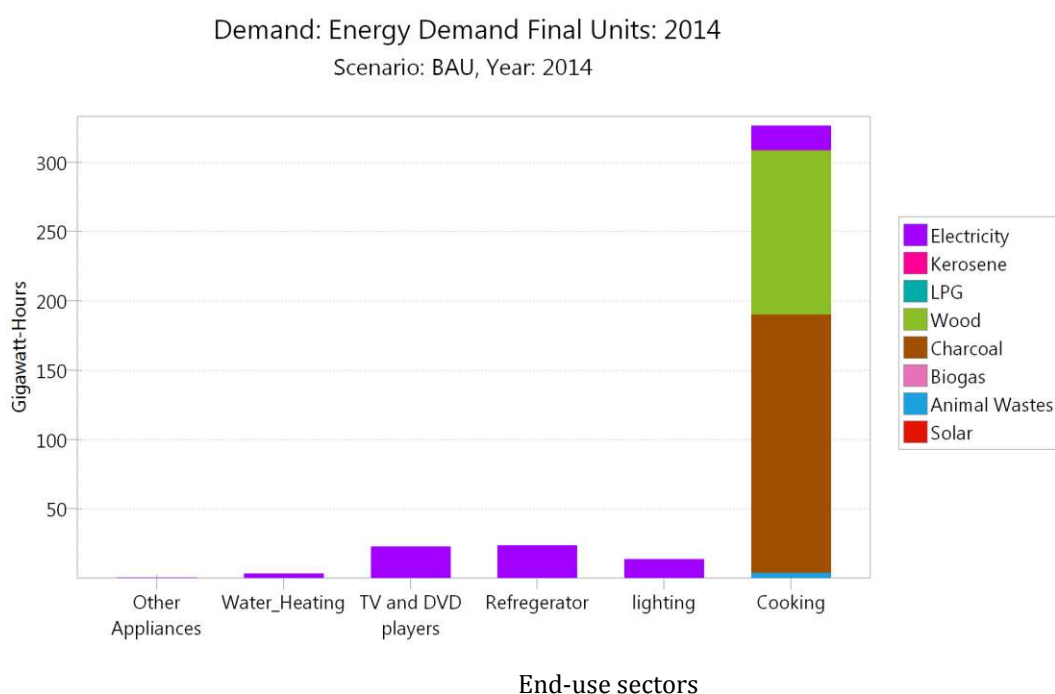


Figure 4-1 Residential energy consumption for various end use activities in Bahir Dar Ethiopia, in 2014

The comparison of energy consumption in Bahir Dar based up on various fuels is presented in table 4-2. It shows the total energy consumption of the fuels, the share, energy intensity or the energy consumption per household and the energy consumption per capita. It could be seen that charcoal is the dominant energy source with energy intensity or per household of nearly 3388.9 kWh/yr, and it accounts for nearly 48% of the energy demand of Bahir Dar households in average. Fire wood is also consumed in significant portion with about 2146.5

kWh/yr and it covers about 30 % of the households energy demand in average. Electricity is about 1475.6 kWh/yr and is 20.8 % the total energy consumption of the households. The energy intensity from other fuels like kerosene, LPG and animal wastes is marginal.

Table 4-2 Energy consumption from various fuels in Bahir Dar city, in 2014

Fuels	Total for Bahir Dar		energy intensity (per households) (kWh/yr)	per capita (kWh/yr)
	Quantity (× 10 ⁶ kWh/yr)	Share of total (%)		
Charcoal	186.39	47.7 %	3388.9	706.02
Wood	118.05	30.2 %	2146.5	447.2
Electricity	81.16	20.8 %	1475.6	307.42
Kerosene	0.66	0.2 %	12	2.5
LPG	0.3	0.08 %	5.47	1.14
Animal Wastes	3.86	1 %	70.18	14.62
Biogas	-	-	-	-
Solar	-	-	-	-
Total	390.42	100%	7098.72	1478.86

Source: author's own survey analysis

4.2.1 Energy consumption for baking

When the share of households that use the energy sources or fuels is compared, it is dominated by firewood. It could be examined in Figure 4-2 that more than 63% of households use firewood for baking injera, while more than 33.5% use electricity for coking. The percentage of households that use saw dust or cow dung is less than 2% each.

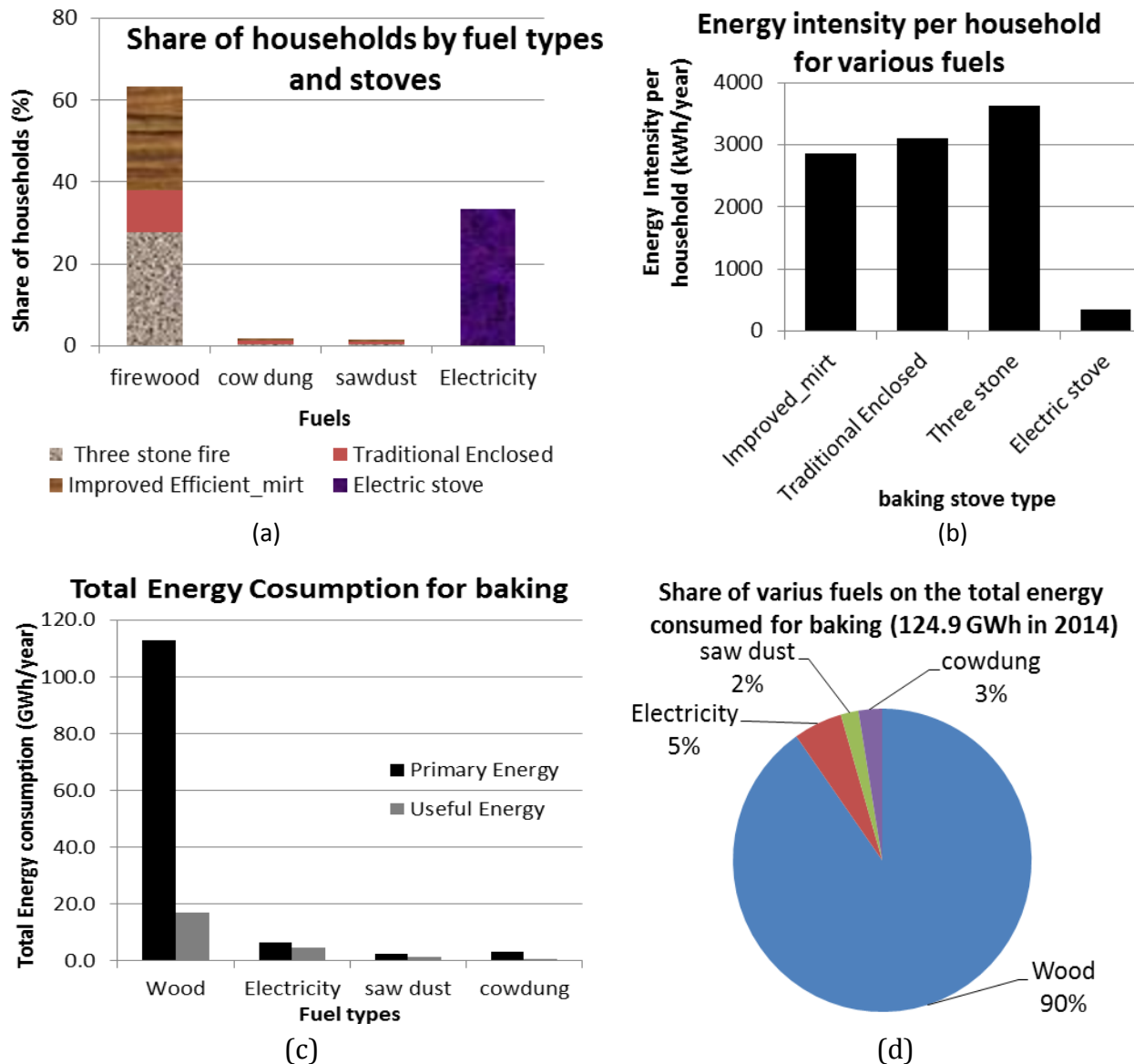


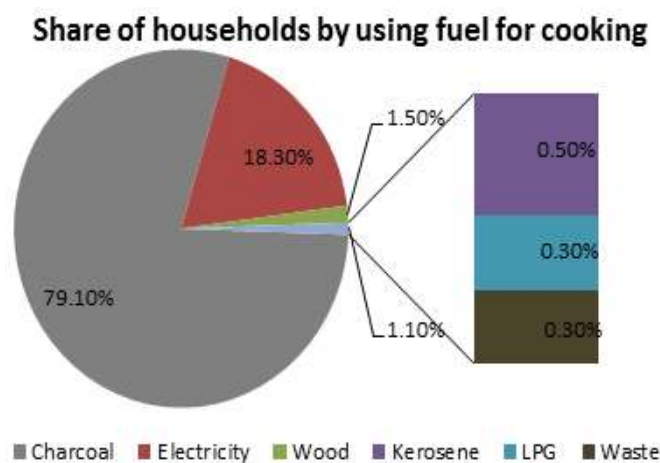
Figure 4-2 Households primary Energy Consumption for Baking (a) share of households by fuel usage with corresponding stoves, and (b) energy intensity per household for baking with various stoves c) comparison of primary and useful energy (d) Share of fuel in the total energy consumption (Fuel mix) in Bahir Dar in 2014.

4.2.2 Energy consumption for cooking

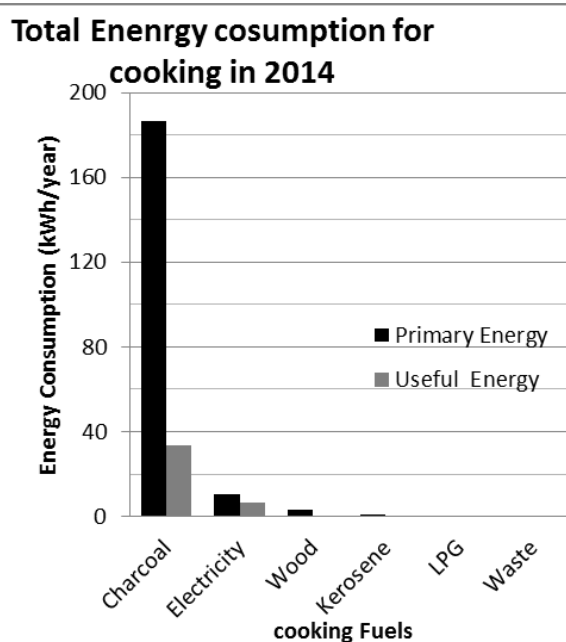
Cooking wot is another activity just like baking that also consumes considerable amount of energy. Unlike baking, the popular fuel that are used by the households to prepare wot, tea and coffee are charcoal, electricity and kerosene. As shown in Figure 4-3 (a) more than 79% of the households use charcoal for preparing wot, tea and coffee. More than 18 % of

the households use electricity for cooking. About 2.5 % of the households use firewood, waste kerosene and LPG for cooking purpose.

Figure 4-3 (b) shows the primary and useful energy



(a)



(b)

Figure 4-3 (a) Share of Households in Energy Consumption for Cooking and boiling (b) the quantity of energy consumed for cooking from the major fuels

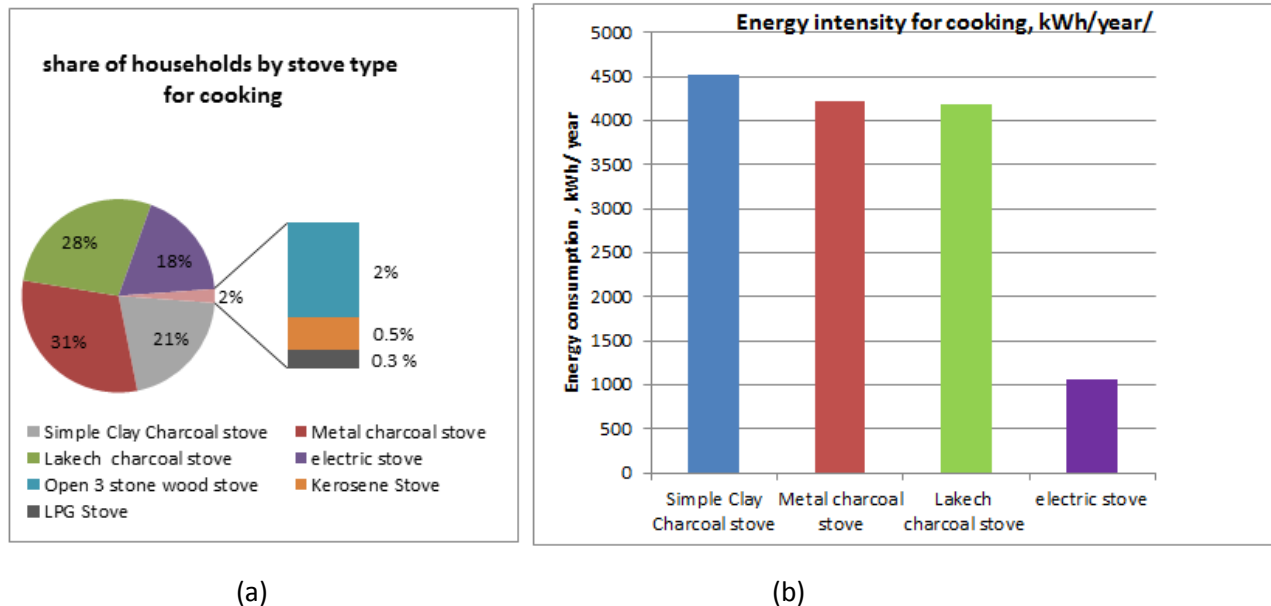


Figure 4-4 (a) Share of households by stove types, (b) Annual energy intensity of households by stove types being used for the fuels

Figure 4-4 shows the share of households by stove types used for cooking and boiling and the annual energy intensity of the households by stove type. We can see that most households (31 %) use metal charcoal stoves and 28 % of the households use improved charcoal stove called Lakech stove. Around 21% of the households use simple clay charcoal stoves and 18 % electric stoves for cooking. We can see that the household energy intensity

4.2.3 Energy consumption for Lighting

Lighting is one of the energy demanding activities in households. Its energy source is electricity for grid connected households. The survey shows that three types of electric lamps have been used in households namely: incandescent light bulb, compact florescent lamps (energy saving light bulbs or lamps) and florescent light bulb. The data presented in figure 4-5 (a) shows that the majority of the households use either incandescent light bulbs or compact florescent light bulb, about 41 % of the households use the former one and 58 % use the later. The share of households that use florescent light is less than 1%. The total energy consumed for lighting in Bahir Dar is shown in figure 4-5 (b) and it shows that of the total of about 13.6 GWh/year, more than 10 GWh/year is consumed by households with incandescent light bulbs and about 3.5 GWh/year in households with compact florescent light bulbs the remaining for florescent light bulb.

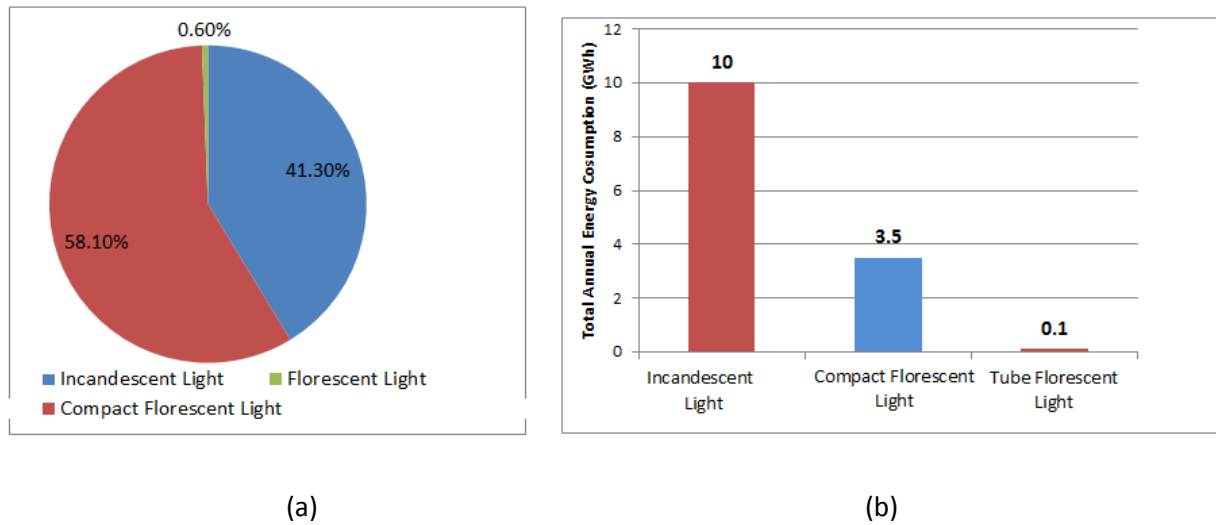


Figure 4-5 (a) share of households using various lamps for lighting, (b) annual total energy consumption in Bahir Dar for lighting using different light bulb

There is huge potential of energy saving by just using the compact fluorescent Lamps (CFL) instead of the conventional incandescent lamp which consumes four times more energy. The main reason why the residents prefer the incandescent lamp is due to its availability at low cost. CFL lamps are slightly expensive but they last longer. There is overall cost benefit in using CFL lamps

4.2.4 Energy consumption for water Heater

Although energy consumption for water heating is low compared to other activities like cooking as shown in above figure 4-1 it is important analyzing it as its share is expected to growing faster. The result displayed in figure 4-6 shows that about 85 % of the households use conventional storage type electrical boiler while 15 % use instant water heater for shower purpose. Currently in Bahir Dar the share of solar water heaters for water heating is zero.

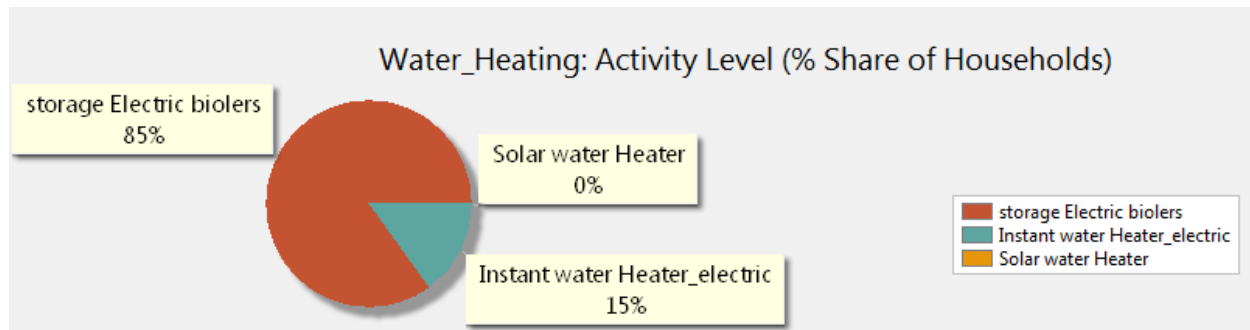


Figure 4-6 share of households with various types of water heating devices

Shifting from conversional storage type boilers to instant water heaters would save significant amount of energy.

4.2.5 Energy consumption for appliances

Appliances are one of the energy consumers in the households, although the amount is not as much as other activities like cooking. The data provided on *table 4-3* shows the share of households using various appliances and the average amount of energy consumed per household per appliance per year. The table shows that more than 82% of the households own TV and its accessories that consume about 500 kWh/year and more than 43% of the households own refrigerator and about 1000 kWh/year per household. Other appliances like Washing machines, coffee grinding or making machines, baking oven, juice making machine and ironing machine are rarely owned in households, about 0.24% , 1.9 % , 0.5 % , 1 % , 1.7 % of the households use each devises respectively. Their energy consumption is based on electricity and it is between 90 and 160 kWh/Year/ per households.

Table 4-3 Share households using various appliances and their Energy Intensity

Appliances	TV	Refrigerator	Washing machines	Coffee Grinding Machine	Baking Oven	Juicing Machine	Ironing
Share of households using %	82.7	43.2	0.24	1.88	0.5	1.0	1.68
Annual Energy Intensity of the appliances (kWh/Y)	500	505	125	90	105	150	160

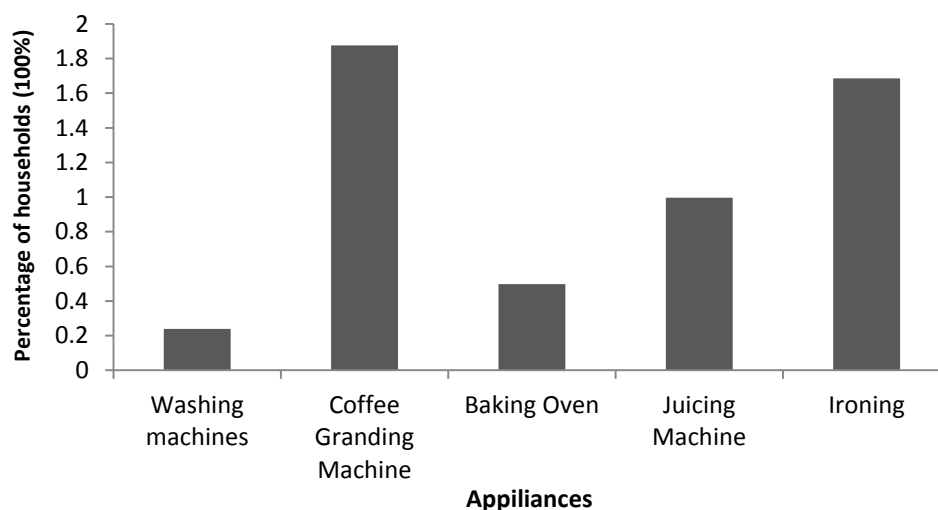


Figure 4-7 Household percentage share of auxiliary households appliances

Figure 4-7 shows the total household percentage utilization of the auxiliary appliances is only 5.8%. This of various appliances used in the city. About 2% of the households use coffee grinding and ironing while just 1% of the households use juicing machine. Washing machine and baking oven is the other appliance that is used in households but their share is very being low less than 0.5 %.

Although the share of households owning such appliances is low at the moment, due to the moment created by economic growth and need for making life easier, there is the share is

expected to grow dramatically. As a result effective policy measures have to be designed to introduce energy efficient appliances and also to create awareness about efficient appliances and their benefit.

4.3 Effect of House Type for Energy Consumption Pattern

Figure 4-8 and 4-9 shows the comparison various types of buildings considering the share of fuel type used for baking and cooking activity respectively. Figure 4-8 shows that modern settlements namely modern villa and apartment use more electricity for baking than any other types of houses, 70% of the former and 88% of the later households use electricity for baking. The share of households of moderate quality houses type that use of electricity is about 30% and more than two third of the households in this type of buildings using firewood. Traditional fuels are more common in low income houses and single room accommodations. Nearly 95 % of the single room and 87% of the low quality house residents use firewood for baking activity the consumption of electricity for baking in those households is very low.

Figure 4-9 shows the comparison of various types of buildings by the share of fuels used for cooking. Similar to baking case, for cooking case, the modern residents rely more on electricity. More than 75 % of the residents in apartment buildings use electricity and more than 54 % of the residents in modern villa use electricity. However the amount of charcoal consumption is still significant, in modern villa, more than 43 % of the residents use charcoal for cooking. Charcoal is more used in low income and single room houses. In almost 80% of the average quality house and 90% of the low quality houses charcoal is the fuel used for cooking and in single family houses, who do not have much option for fuel mix, all of the sample households found using charcoal. Firewood is not usually used for cooking purpose as it is suitable for baking only.

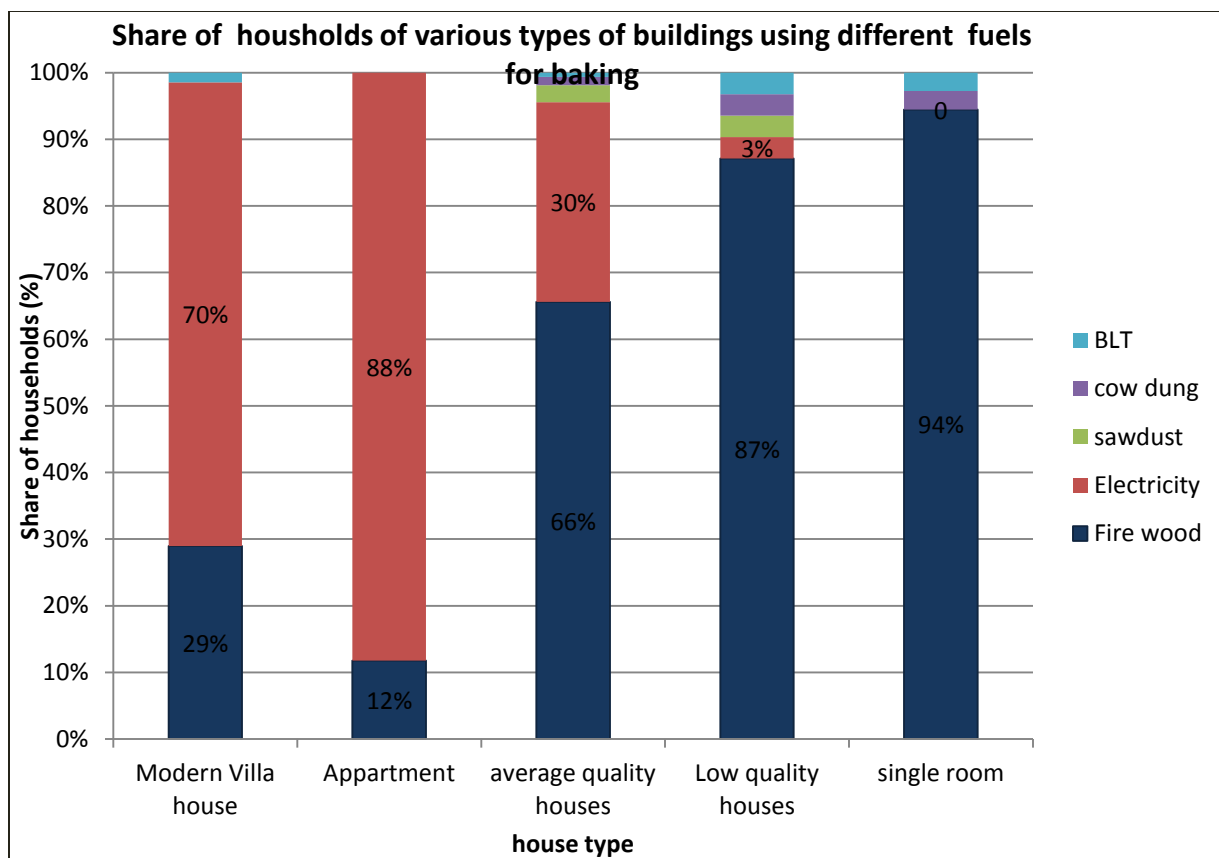


Figure 4-8 Comparison of types of houses with the share of fuels used for baking

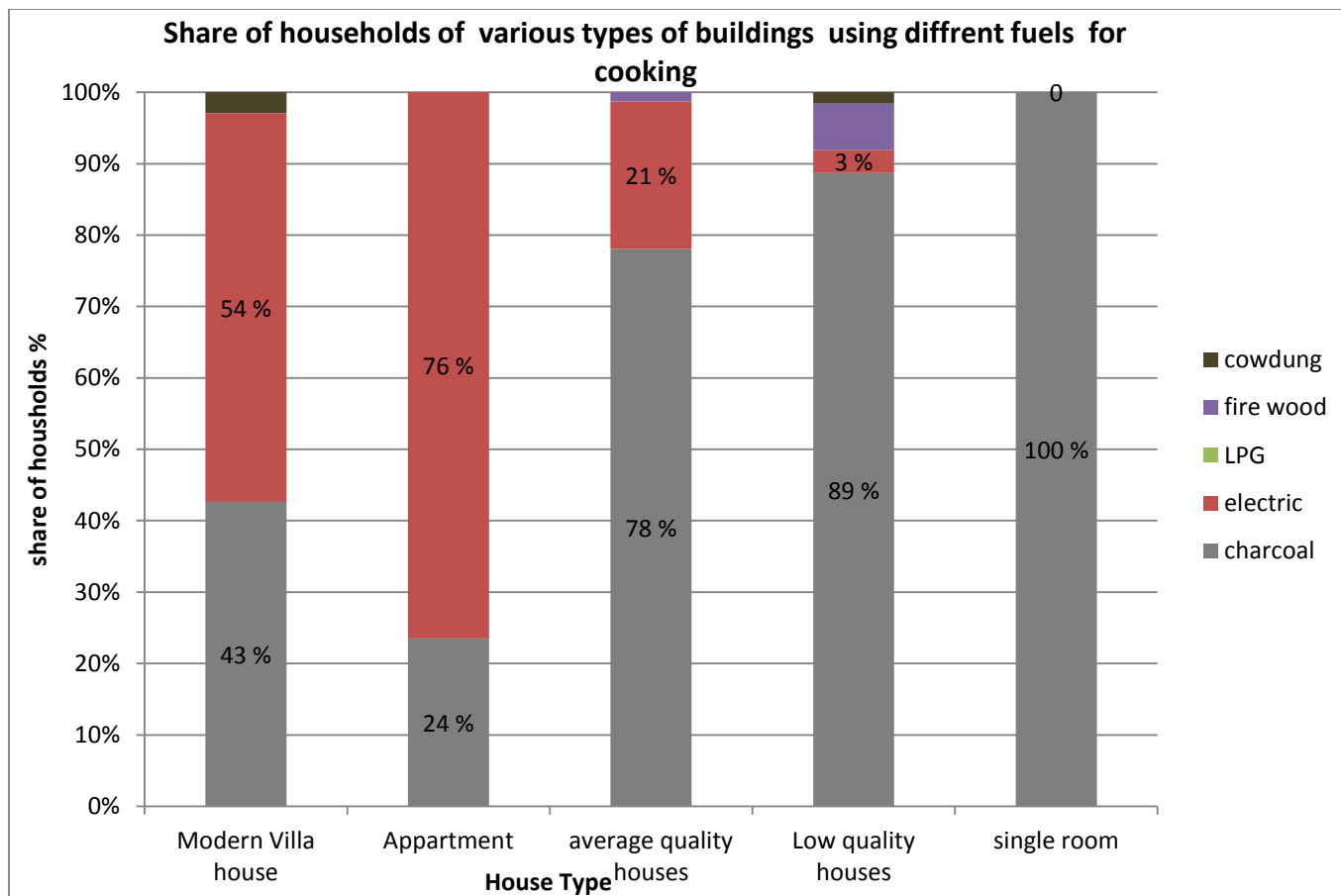


Figure 4-9 Comparison of types of houses with the share of fuels used for cooking

Figure 4-10 shows the variation of stove usage in households of various house types. We can see that residents in modern villa houses and apartment buildings tend to use more modern type of stoves in terms of efficiency. In contrast, in low income type houses and single family houses the usage of traditional type stove is apparent. Figure 4-10 also shows the variation of stove used for cooking in various types of buildings due to usage of electricity the ownership of electrical stove is apparent in modern houses. Single room residents owning more metal charcoal stoves and traditional three stone stoves, they own less modern stoves.

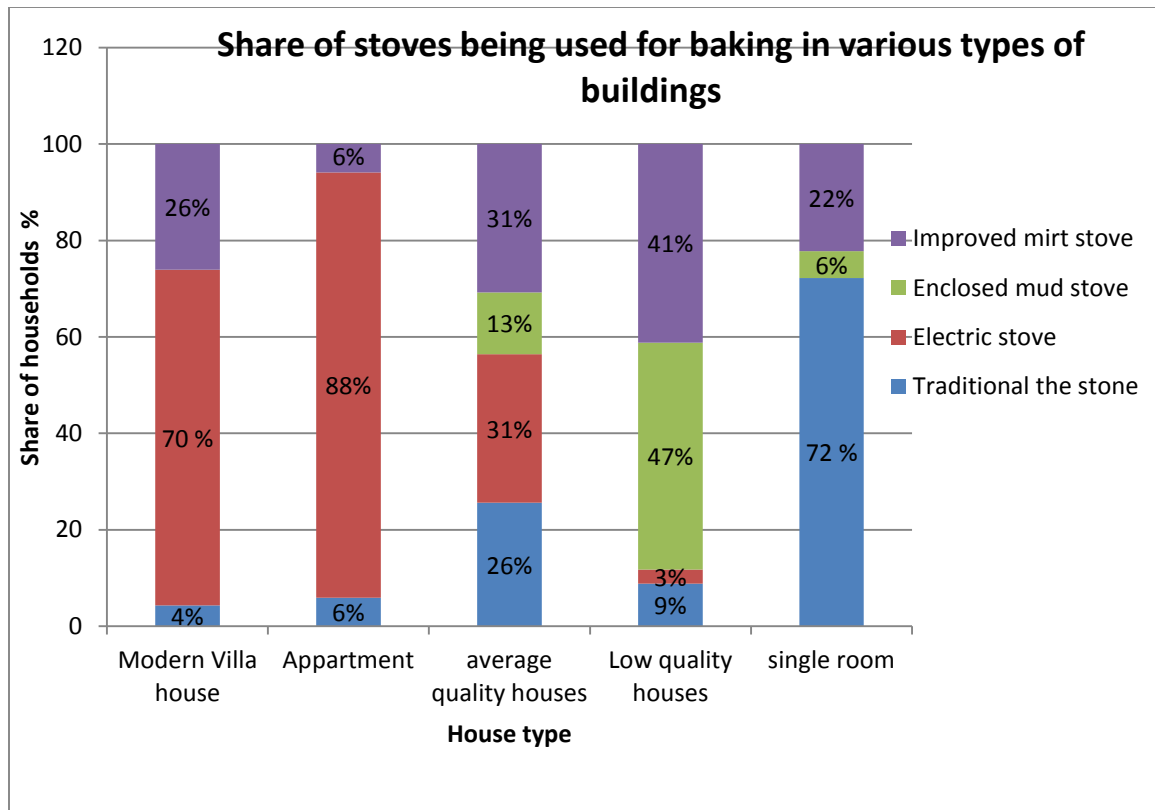


Figure 4-10 Comparison of types of houses with the share of stove type fused for baking

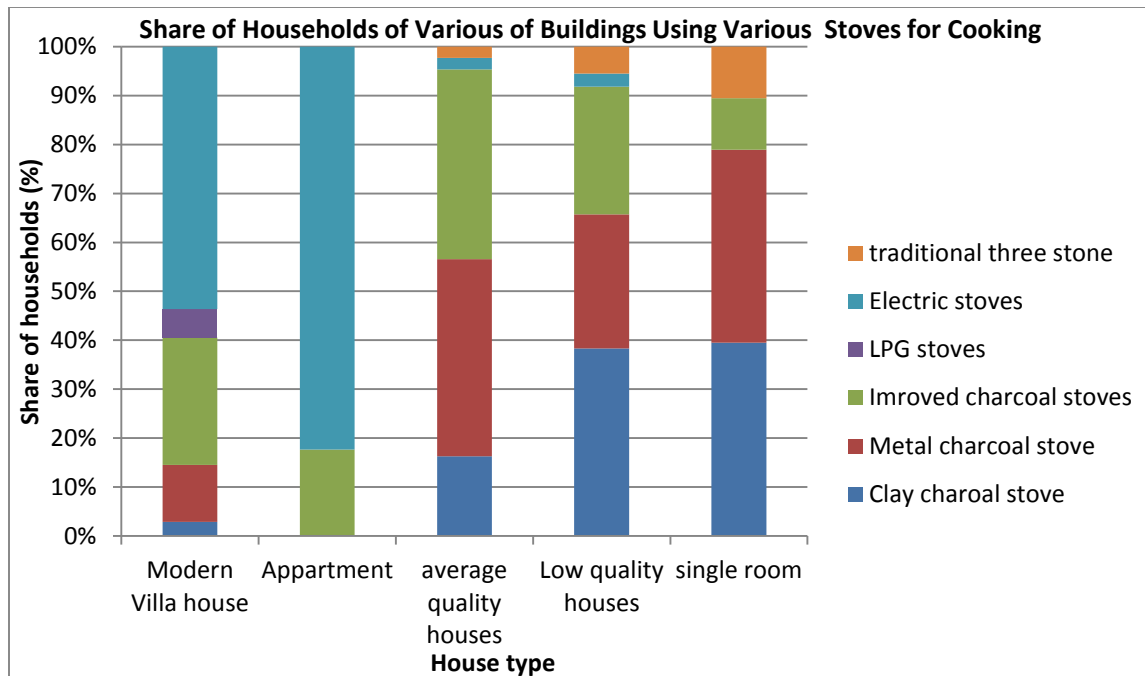
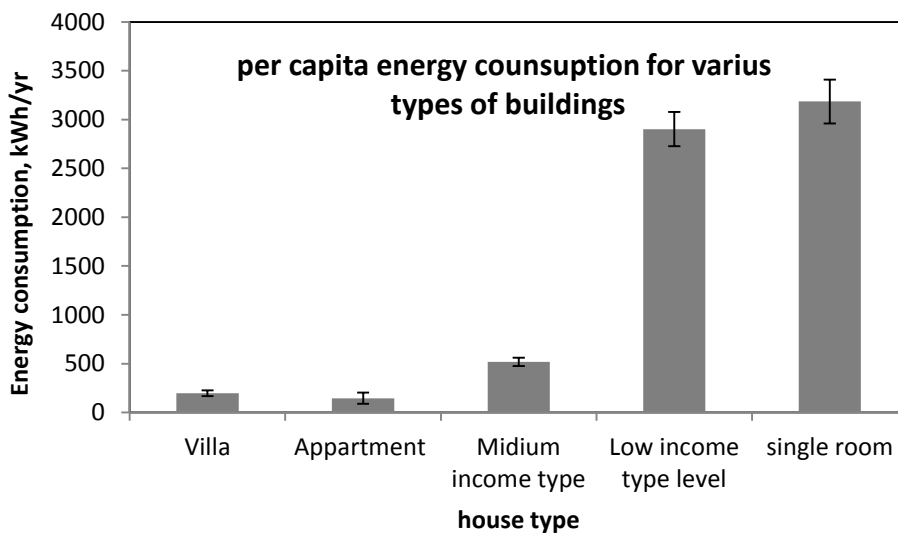


Figure 4-11 Comparison of types of houses with the share of stove type used for cooking



4.4 Economic Analysis

4.4.1 Prices of different fuels in Bahir Dar

Figure 4-12 shows the price of various fuels in Bahir Dar, Ethiopia in 2014. The graph shows that LPG, with price of about 3.5 ETB/kWh is the most expensive fuel source. The other fossil fuel source, kerosene, is also expensive with price of almost 1.7 ETB/kWh. The

three mostly used fuels, electricity, fire wood and charcoal are in the same price rage. Electricity has an average price of 0.42 ETB/kWh which makes it one of the cheapest fuels among them, whereas charcoal and firewood have an optimum price nearly 0.55 ETB/kWh each. Cow dung and sawdust are slightly cheaper.

Fuel price in Bahir Dar, birr/kWh

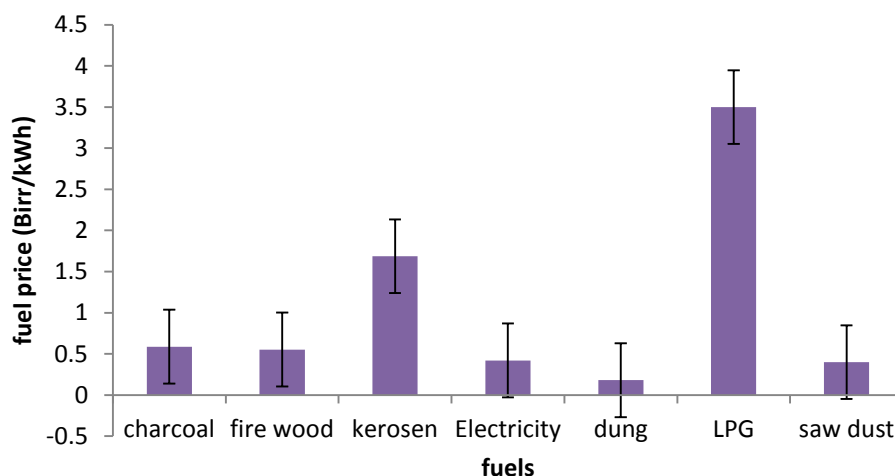


Figure 4-12 various fuel price in Bahir Dar⁷

4.4.2 Comparison of annual household energy expenditure

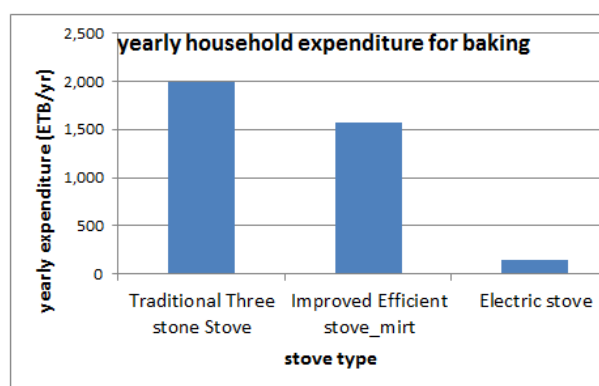
Table 4-4 shows comparison of household expenditure for various fuels. The table contains average, household's consumption for various fuels and the corresponding unit price of the fuel, both will give average annual expenditure of various fuels. We can see that due to large amount of consumption charcoal accounts for the highest expenditure with value of about 1991.53 ETB/year/household. The second fuel with large expenditure is firewood with average house hold expenditure of about 1185.85 ETB/year/household. The average house hold expenditure for electricity is about 619.77 ETB/year. The expenditure for other fuels is negligible as they are used in few households. The total average household annual expenditure for energy in Bahir Dar is about 3849.13 ETB/year.

Therefore, we can see that the usage of electricity is advantageous over fire wood or charcoal. However still there are very few households those use electricity for cooking.

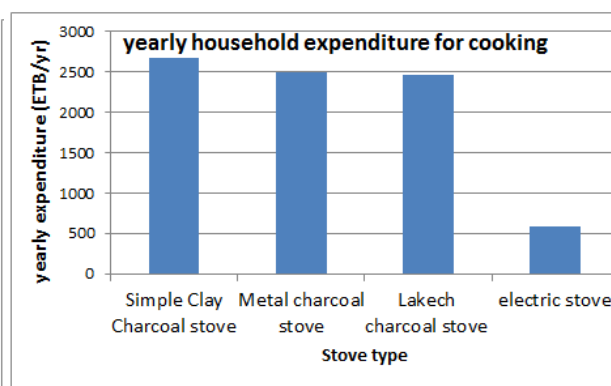
⁷ Exchange rate 1 USD = 20.3 ETB

Table 4-4 Comparison of annual household energy expenditure

fuel type	Fuel Price (ETB/ kWh)	Primary average	
		consumption of households (KWh/year)	Annual expenditure (ETB/year/household)
Charcoal	0.59	3388.91	1991.53
Wood	0.55	2146.36	1185.85
Electricity ⁸	0.42	1475.64	619.77
kerosene	1.69	12.00	20.25
LPG	3.50	5.45	19.09
cow dung	0.18	70.18	12.63
Total		7098.55	3849.13



(a)



(b)

Figure 4-13 comparison of average yearly expenditure of households for various fuels based on various types of stoves, (a) for baking (b) for cooking. Source: own interpretation of the survey result

⁸ Electricity price is estimated by taking average tariff for correspond average consumption

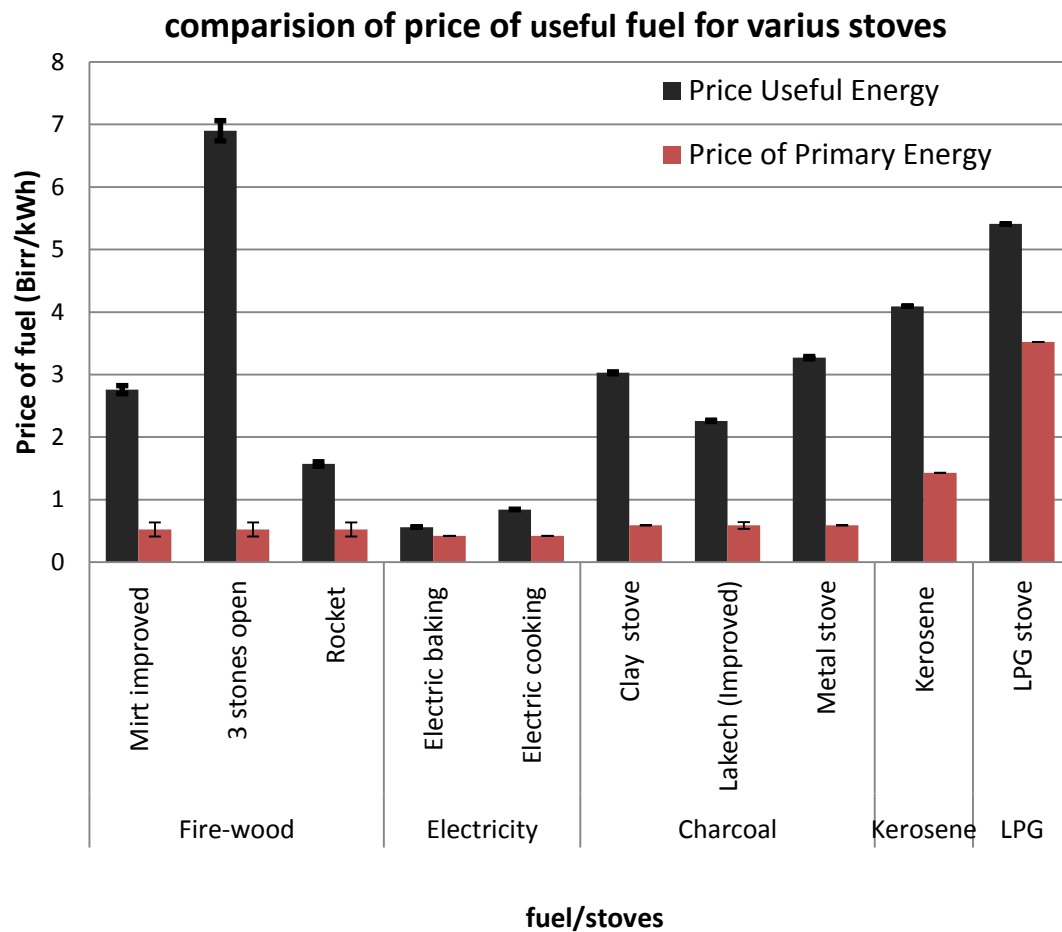
Traditional stove which is usually perceived among households as economical tends to be the most expensive due to its low efficiency. In contrast electric stove, as the result of its high efficiency and ease operation, is found out to be far cheaper.

4.4.3 Price of Useful Energy

Table 4-5 shows the comparison of the useful energy and primary energy for various fuels considering the types of stoves used in households. The useful energy price depends on the efficiency of the stoves upon which the fuel is used, therefore, fuels used on efficient stoves have better or lower useful price than those used on lower efficiency stoves. For example, the useful energy price of firewood used on improved mirt stove has a price of about 2.76 ETB per useful KWh, whereas firewood used in traditional open three stone stoves has a price of nearly 7 ETB/ kWh. Whereas electricity, which is used on stoves with better efficiency, has a price of useful energy about 1 ETB/kWh, this makes it the cheapest fuel in terms of useful fuel price. Similarly, charcoal used on low efficient Metal stove have useful fuel price of about 3.3 ETB/kWh and when used on Lakech improved charcoal stove its useful price becomes as low as 2.27 ETB/kWh. All in all the stove efficiency has high impact on the price of useful energy of the fuel.

Table 4-5 Price of various fuels available in the market of Bahir Dar, Ethiopia

Price of Primary energy				Stove efficiency (%)	Type of stove	Price of Useful energy (ETB/kwh)
Type of fuel	ETB/kWh					
	FUEL Price (ETB/kg)	FUEL Price (ETB/kWh)	Varian- ce			
Fire- wood	1.84	0.552	0.013	20	Mirt improved stove	2.76
				8	Open 3 stones	6.9
				35	Rocket stove	1.57
Electricity	0.42	0.420	0	75	Electric baking Stove	0.56
				50	Electric cooking stove	0.84
Charcoal	4.57	0.588	0.003	20	Clay stove	3.03
				26	Lakech (Improved)	2.26
				18	Metal stove	3.27
Kerosene	15	1.43	0	35	Kerosene stove	4.09
LPG	45	3.52	0	65	LPG gas stove	5.41
Cow dung	0.5	0.18	0.01	8	3 stones fire	2.25
				20	Mirt improved	0.9



4-14 Comparison of primary and useful price of various fuels available in the market of Bahir Dar, Ethiopia

4.5 Assessment of Energy Saving Potential in Households Based on the Long-Range Energy Consumption Scenarios

The modelling of energy consumption is done based on the two scenarios the Business as usual scenario also called Reference Scenario and Mitigation Scenarios. Business as Usual scenario (BAU) or Reference Scenario a scenario that shows the consumption pattern where no or little policy intervention is made to change in the consumer behaviors. The results shown in the following sections are obtained for those scenarios. For example we can see the result displayed in fig. 4-14 on which the difference in energy consumption between the businesses as usual and mitigation scenarios in 2030 is apparent.

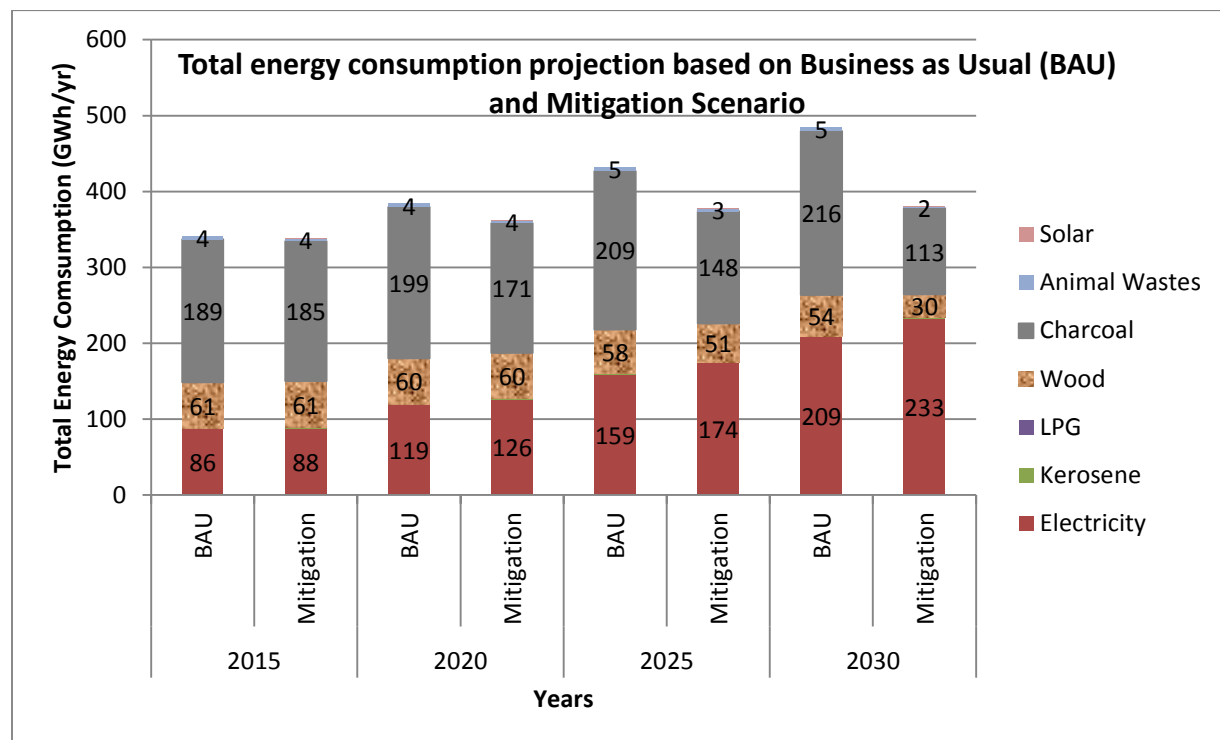


Figure 4-15 Comparison of BAU (business as Usual Scenario) and Mitigation scenario for energy consumption projection between 2014 and 2030 in Bahir Dar city

The figure shows that for Bahir Dar city in 2030 the energy consumption with Business as Usual Scenario (BAU) will be around 500 GWh/year. However, it could be seen that with

effective implementation of appropriate policy measures, the energy consumption in mitigation scenario shows the total energy could be kept as lower as 380 GWh/year as value that saves 100 GWh/year as compared by BAU scenario.

Figure 4-15 shows the comparison of growth rate of energy consumption for Business as Usual and Mitigation Scenarios between 2014 and 2030. We can see that the growth rate of energy consumption in Business as Usual Scenario will maintain constant rate of 2.5 % each year. For mitigation scenario the consumption growth rate will decline will start to be negative by 2030.

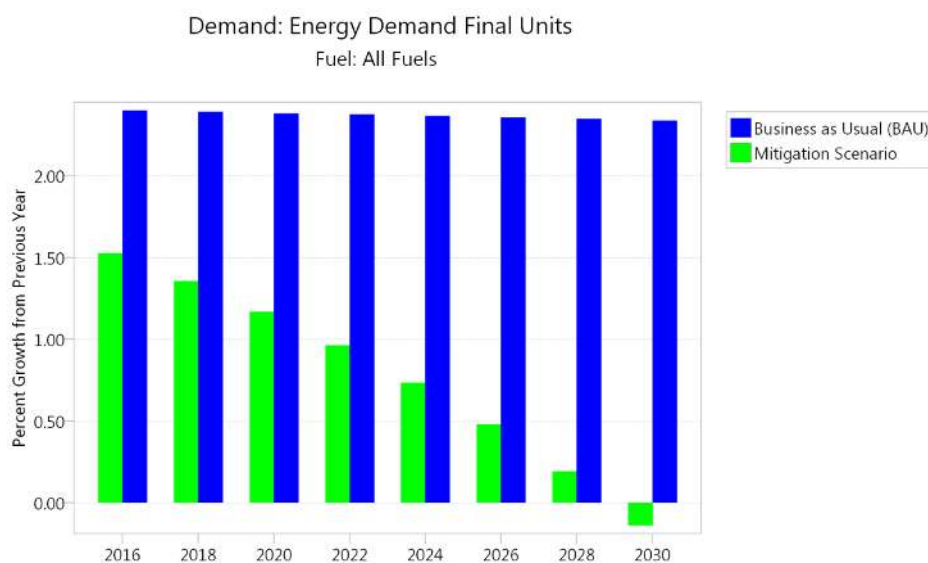


Figure 4-16 comparison of growth of energy consumption between 2014 and 2030 for Business as Usual (BAU) Scenario and Mitigation Scenario

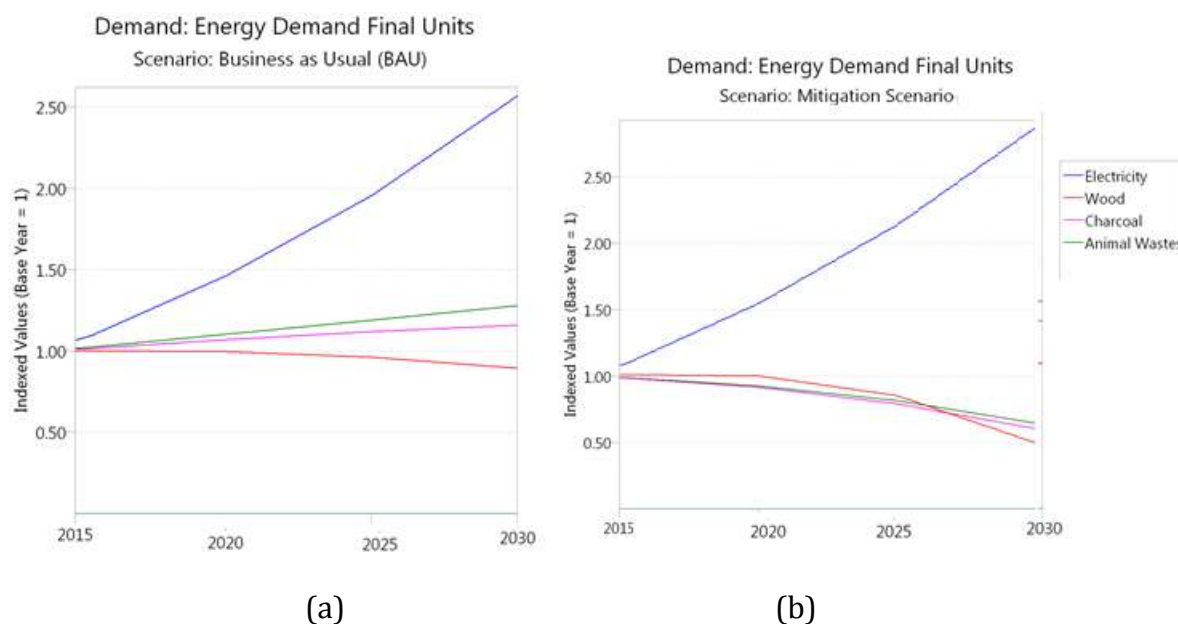


Figure 4-17 comparison of projection of fuel consumption between 2015 and 2030 for Business as Usual (BAU) Scenario (a) and Mitigation Scenario (b)

Figure 4-16 shows the comparison of the projection of the fuel consumption between 2014 and 2030 based up on Business as Usual (BAU) Scenario figure 4-16 (a) and Mitigation Scenario figure 4-16 (b). Both in the business as usual and mitigation scenario the electricity consumption will increase nearly threefold. The fuel wood consumption will slightly be reduced for business as usual scenario while it is reduced by half for mitigation scenario. Similarly, the consumption of the other traditional biomass fuel charcoal will slightly increase for business as usual scenario while it is reduced by more than 40 % for mitigation scenario. The reduction of the consumption of traditional fuels, charcoal or firewood, is mainly due to fuel shift to electricity.

Moreover, the consumption of the fossil fuels such as kerosene and LPG is expected to decline to be very small amount by the end of the forecast period 2030 both for Business as Usual (BAU) Scenario and mitigation scenario. In contrast new fuel types like solar energy will immerge in to the household energy mix with Mitigation Scenario.

4.5.1 Scenarios for cooking

Figure 4-17 and 4-18 shows that energy growth in Bahir Dar city for cooking between 2014 and 2030 based on Business as Usual (baseline) and Mitigation Scenarios respectively. The result displayed in figure 4-17 indicates that if things continue in the current way, the energy consumption for cooking will grow from 270 GWh/yr of the 2014 value to more than 340 GWh/yr in 2030, and charcoal will still be the major energy source during this period. However, due to economic growth, there will be fuel shift from charcoal to electricity, and then the consumption of electricity will grow significantly. However, there will be fuel shift from firewood to electricity therefore the wood consumption will shrink slightly from 60 GWh/yr to less than 55 GWh/yr between 2014 and 2030. Due to fuel shift the consumption of electricity for cooking will increase from 17 to 58 GWh/yr between 2014 and 2030 for Business as Usual Scenario (BAU).

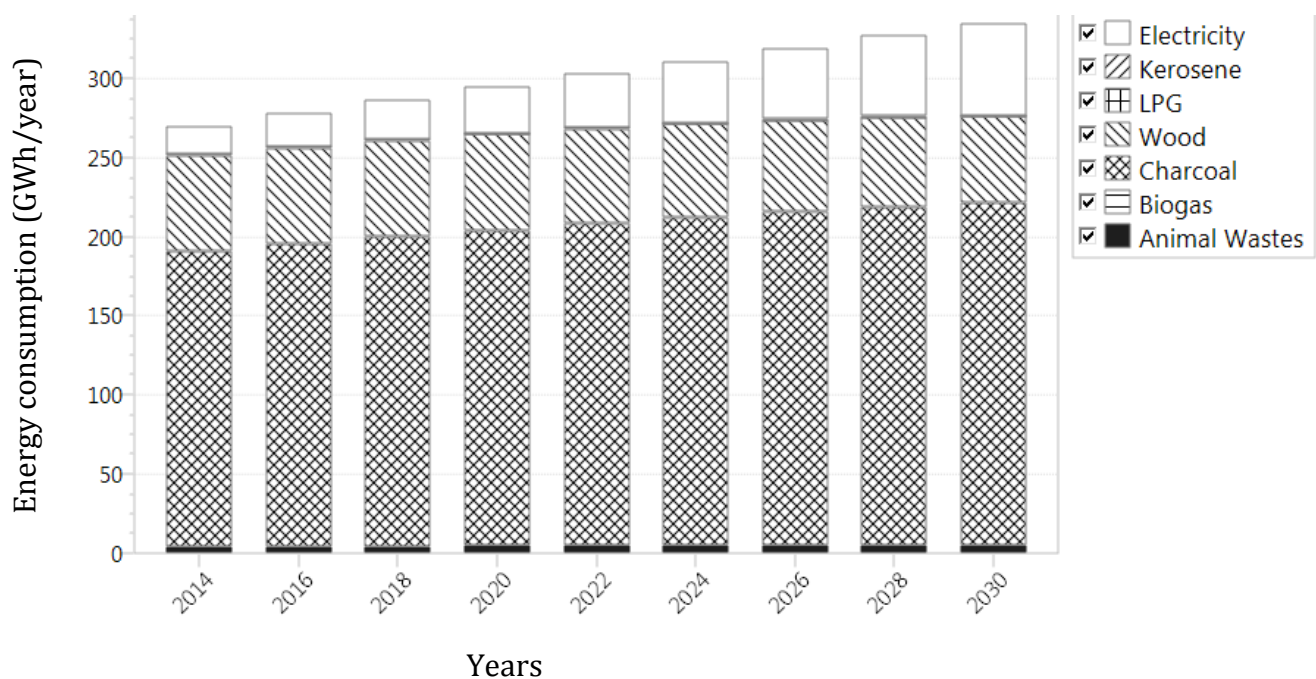


Figure 4-18 Forecasting of household energy consumption for cooking sector with Business as Usual Scenario

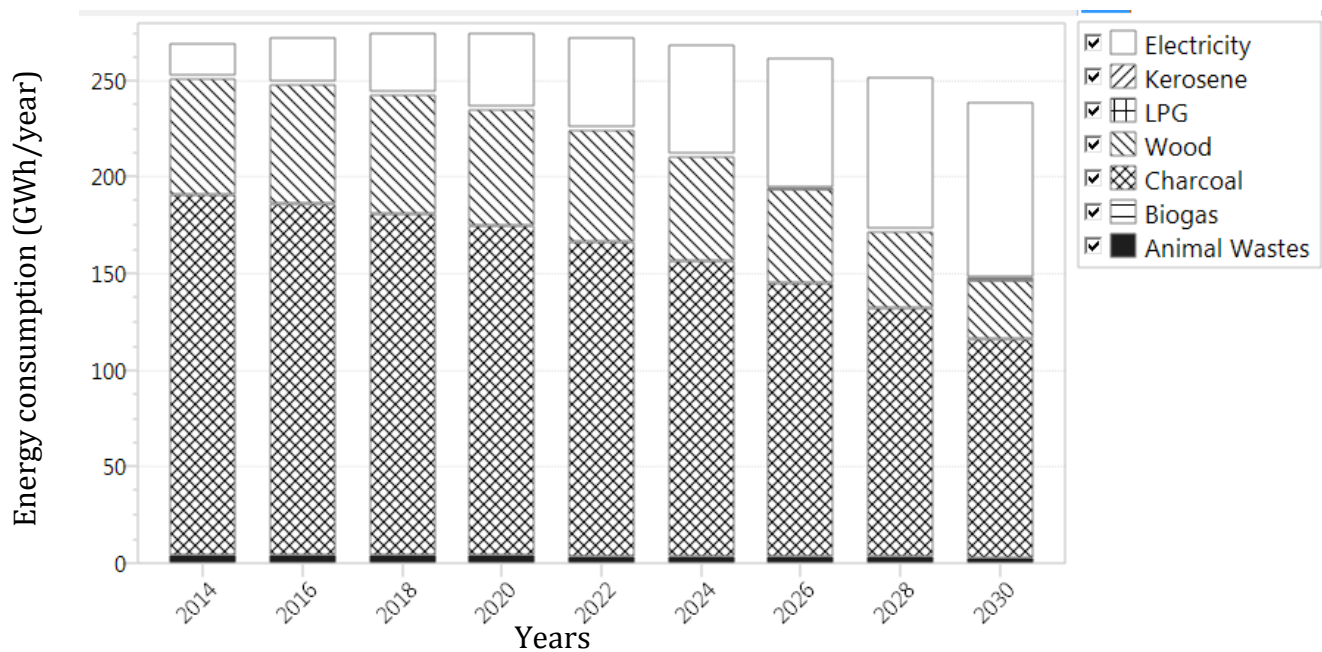


Figure 4-19 Forecasting of household energy consumption for cooking sector with Mitigation Scenario

Figure 4-18 shows the forecasting of residential energy consumption for cooking in Bahir Dar between 2014 and 2030 based up on mitigation scenario. This could show that with few policy adoptions, the energy consumption for cooking could drop significantly from 270 GWh of the 2014 value to just 240 GWh in 2030. The energy consumption of firewood and charcoal drop as the result of efficient technology adoption and fuel shift to electricity. In 2030 more than one third of the energy for cooking would come from electricity which is used on more efficient stoves than the traditional firewood that is mostly used on randomly made inefficient stoves. The graph shows that on effective implementation of energy efficiency measures along with fuel shift to electricity, energy consumption for cooking using charcoal could drops from almost 190 GWh in 2014 to nearly 110 GWh in 2030. Similarly, there is significant amount of reduction of firewood usage from 60 GWh to just above 30 GWh/yr.

4.5.2 Scenarios for Water heating

Water heating for shower is also another sector that consumes significant amount of energy in Bahir Dar, as the income of the people increases its consumption is expected to increase. Figure 4-19 shows the comparison based on Business as Usual (BAU) scenario and Mitigation Scenario for energy consumption for water heating between 2014 and 2030. With just using more instant water heater, which are more efficient than storage type boilers, and introducing solar water heaters, the energy consumption for water heating could be reduced by at least 3 GWh/year from its value of 24 GWh/year that would happen with BAU in 2030. As water heating for shower is perceived as luxurious activity it would be difficult to expect significant portion of the households to own solar water heaters unless sort of policy intervention like provision of incentive is made.

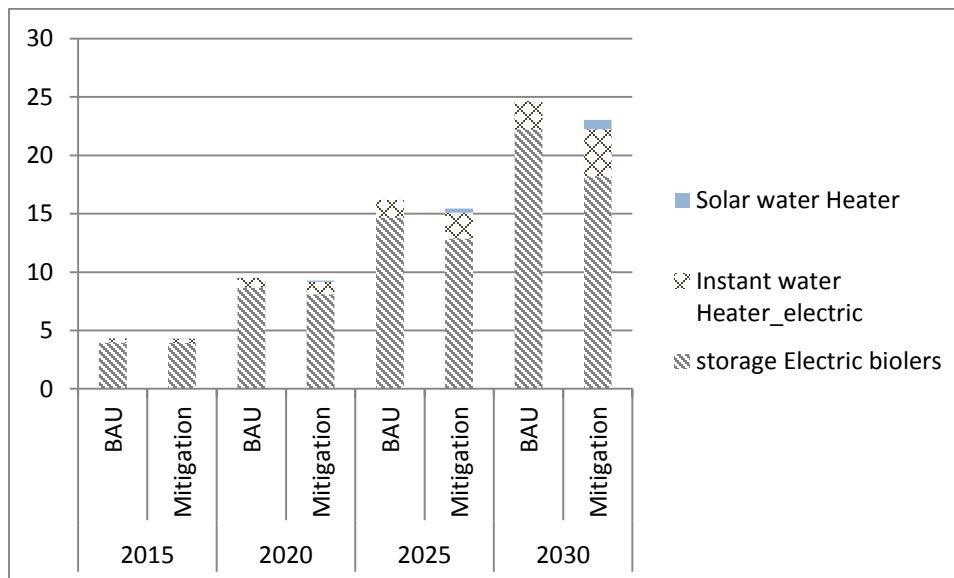


Figure 4-20 comparison of BAU and mitigation scenario for water Heating

4.5.3 Scenarios for Lighting

Lighting could also be an opportunity for energy saving in households. Figure 4-20 shows the comparison of business as usual and mitigation scenario for lighting. Business as Usual Scenario the demand will increase from 14 GWh of its 2014 value to more than 22 GWh in 2030. However, when we consider the mitigation scenario, higher penetration of efficient

light bulbs would save significant amount of energy and would limit the 2030 demand to nearly similar to the 2014 demand, having high growth rate of urban households.

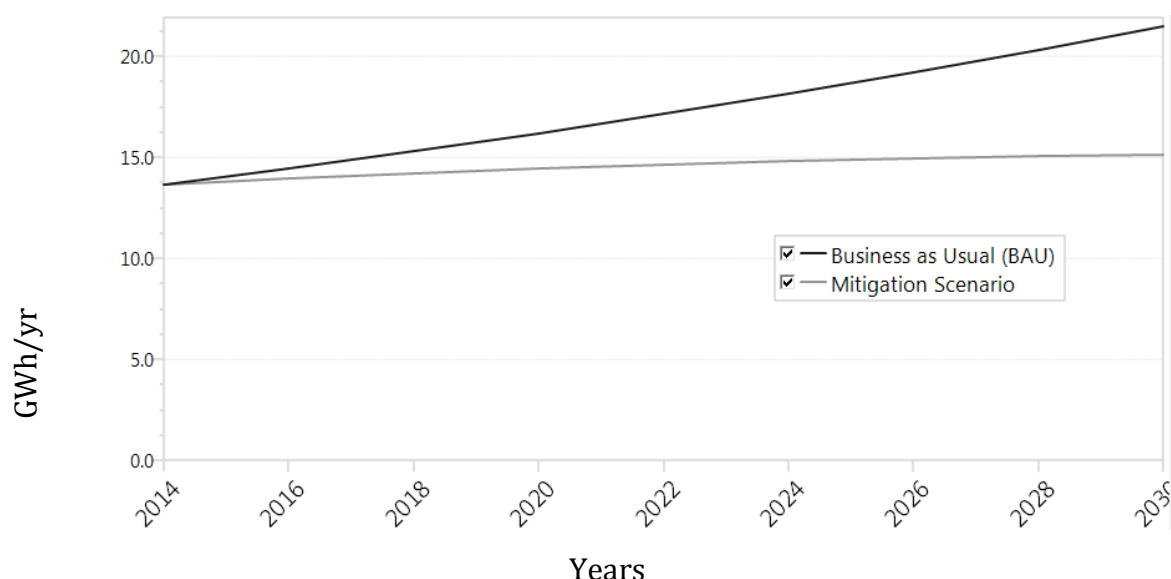


Figure 4-21 comparison of BAU and Mitigation Scenario for lighting

4.5.4 Scenarios for appliances

Figure 4-21 shows the projection energy required for refrigeration with business as usual and mitigation scenarios. The business as usual scenario holds the energy intensity of the refrigerators to be kept constant at their 2014 value until 2030. Therefore, the total energy demand for refrigerators will grow due to the increase in the ownership of refrigerators. Although new owners tend to by larger size of refrigerators this will be compensated with the fact that the efficiency of newer refrigerators is generally getting better than that of existing refrigerators (Swisher, 1994). Thus, with BAU scenario, the energy consumption for refrigerator grows from 12 GWh in 2014 to nearly 30 GWh in 2030.

The mitigation scenario is based up on the ideas that new model refrigerators will be forced to be more efficient by regulation. The efficiency of the appliance will be improved slowly over time to make refrigerators in 2030 to consume 20 % less energy

compared to the existing ones of the same size . Therefore, the mitigation scenario will lead to save around 6 GWh of energy in 2030 compared to the BAU scenario.

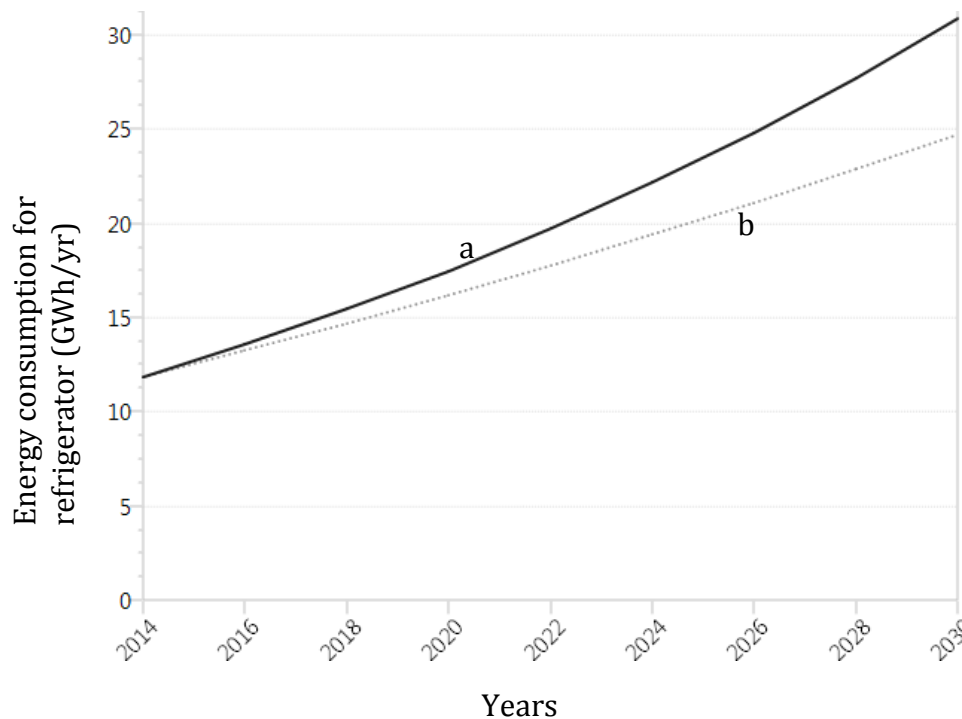
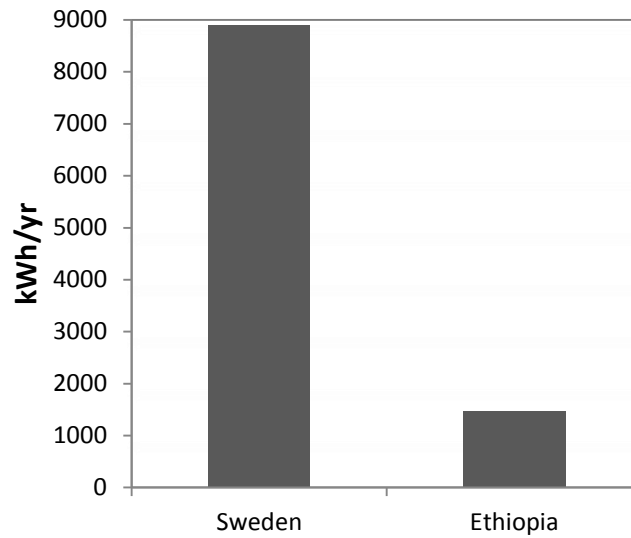


Figure 4-22 Comparison of (a) Business as Usual Scenario (BAU) and (b) Mitigation Scenario for appliances (refrigerator)

4.6 Comparison of the Result with Sweden

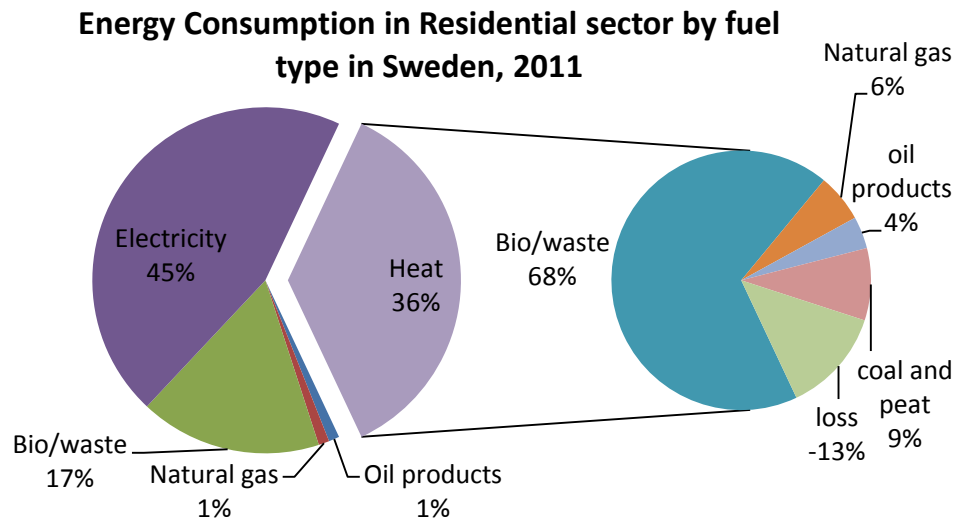
A typical Swedish single-family home utilizes around 900 kWh of energy annually from various sources compared to 7098 kWh of Bahir Dar Ethiopian. When we consider the energy consumption in Swedish household sectors around 60 percent is utilized for space heating and around 20% is utilized for hot water. The remaining 20% is utilized for lighting; appliances other household electronics including cooking. Cooking consumes just 19 percent is used to heat up water; the remaining 56 percent is utilized for space heating.



4-23 Comparison of per capita residential energy consumption in Sweden and Ethiopia. For Sweden the 2011 data is considered, and for Ethiopia the data from Bahir Dar city is taken in 2014.

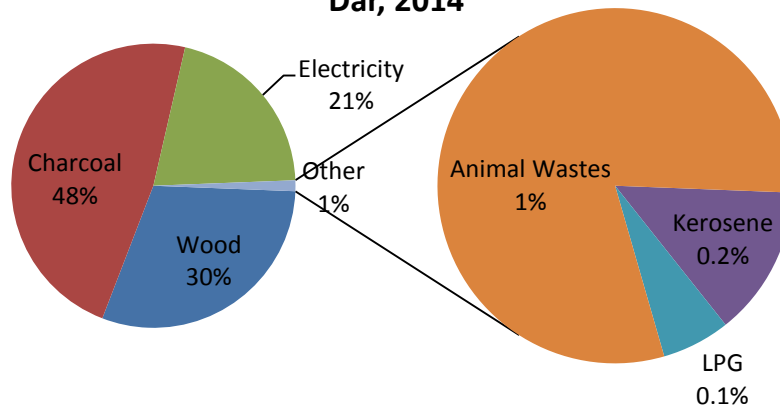
Figure 4-23 shows that more than 45% of the residential energy in Sweden comes from electricity and around 17% comes from bio/waste, and 36 % of the residential energy comes as heat. District heating in Sweden uses bio/waste in significant portion (over 68%). The contribution of conventional energy sources like natural gas, coal, and oil is relatively small in district heating only about 20% of the energy for heating comes from such energy recourses.

In Bahir Dar, Ethiopia as shown in figure 4-23 (b), about 80% of the energy comes from traditional biomass (48% charcoal, 30% firewood and 1% animal waste).



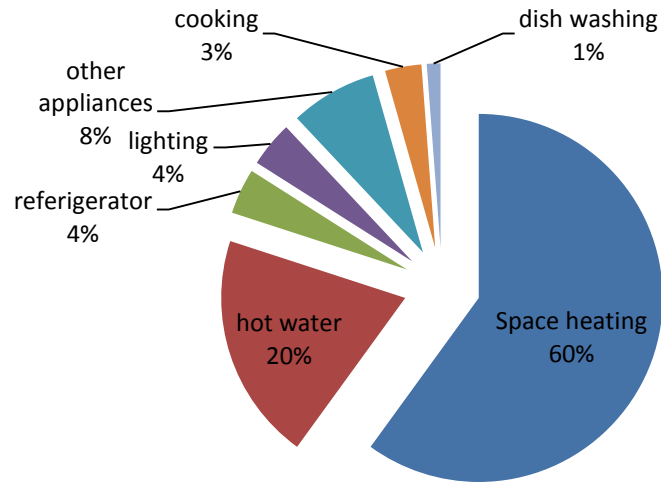
(a)

Fuel -mix in residential Energy Consumption, Bahir Dar, 2014

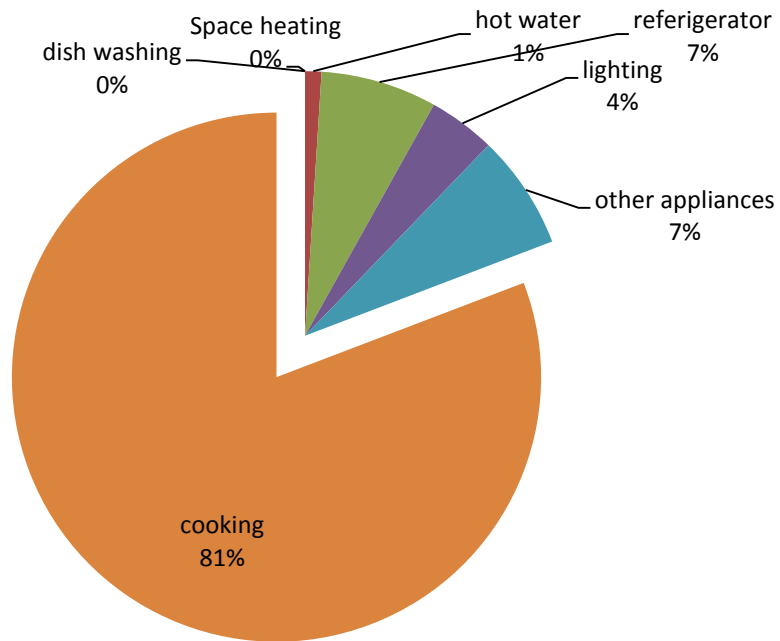


(b)

4-24 Energy Consumption in Residential sector by fuel type in (a) Sweden, 2011 and (b) Bahir Dar, Ethiopia, 2014 (source: for Sweden modified from Kander & Stern, 2014)



a) Seden



b) Ethiopia

Figure 4-25 comparison of energy consumption in Swedish and Ethiopian households. The residential energy consumption in Sweden shown in (a) is dominated by heating sector, while in Ethiopia (b). it is dominated by cooking sector. Energy consumption in Swedish dwellings (source: www.eon.se),

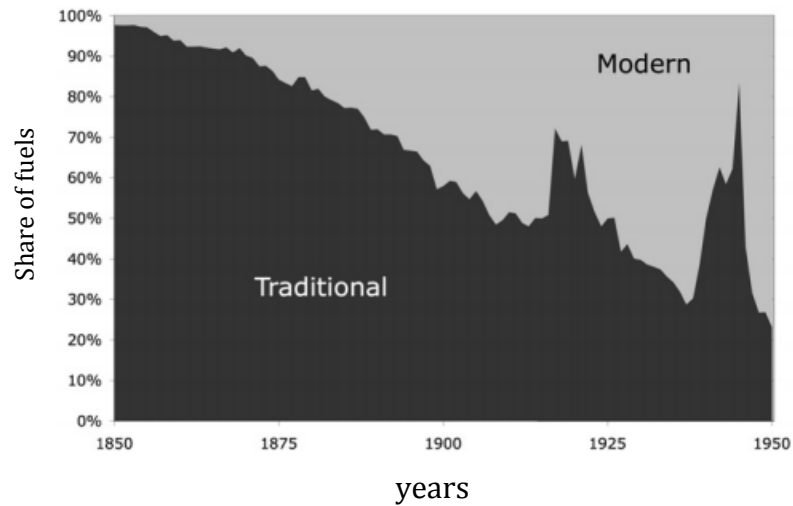


Figure 4-26 Trend of shares of traditional and modern energy carriers in Sweden (source: Kander & Stern, 2014)

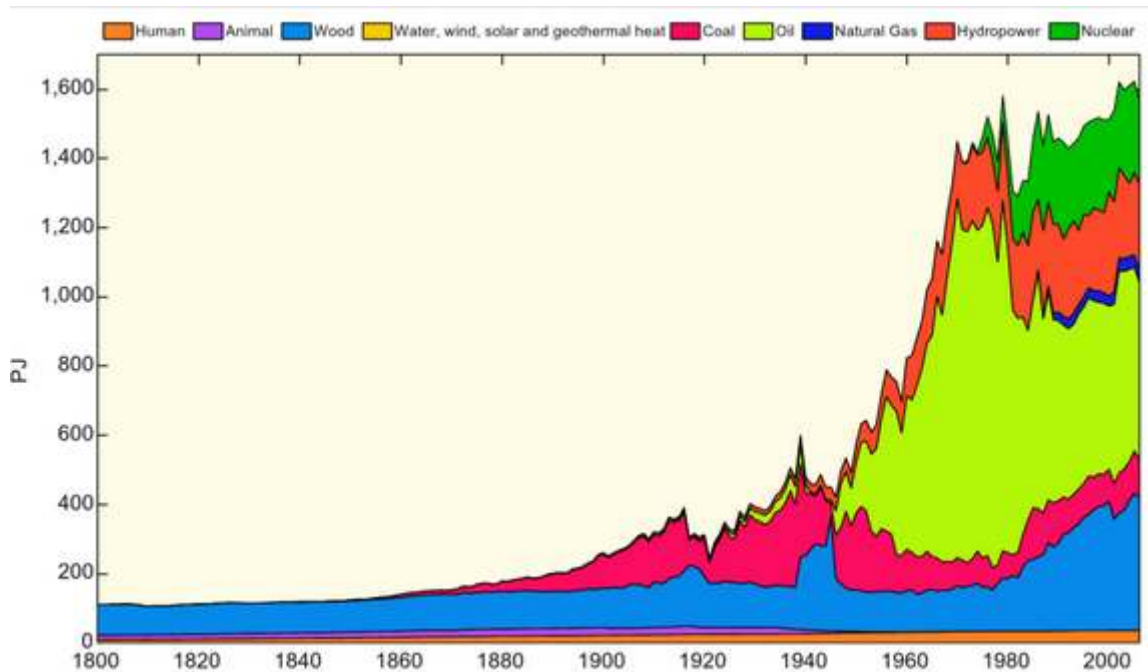


Figure 4-27 Trend of the quantity of energy used in Sweden by energy carriers between 1800 and 200 (source: Cleveland & Morris, 2012)

Figure 4-26 and figure 4-27 show the trend of energy sources in Sweden from mid-19th century until mid-20th century based on the study of Kander & Stern (2014). The graph shows that before mid-19th century, the energy sources are mainly traditional biomass like

firewood and animal waste; pretty much the same as what we have in Ethiopia currently (Figure 4-1 Residential energy consumption for various end use activities in Bahir Dar Ethiopia, in 2014). The shift to modern energy sources occur in significant rate and keep the share of traditional energy sources low in mid-20th century. Currently the share of energy sources is entirely from modern and the issue now is to enhance energy efficiency.

Sweden has goal to reduce the use of energy per heated unit area in dwellings by 20 percent in 2020 and by 50 percent in 2050. Sweden has a building regulation for limiting the power demand of electrically heated houses and for improving energy efficiency and (Perman, 2011). Owners of single-family houses have changed from using oil-fired boilers to using more district heating, heat pumps and bio-fuels up to solar collectors for heating. In addition, better insulation thickness of wall and ceiling are among the methods in reducing the energy consumption. Sweden has the largest number of heat pumps of the member countries of the E U (Perman, 2011).

The values presented at figure 4-28 shows the reduction in energy intensity for space heating and water heating per square meter of the dwelling in Sweden. This reduction is merely due to energy efficiency. This reduction is continuing and it will play major role on total energy consumption reduction.

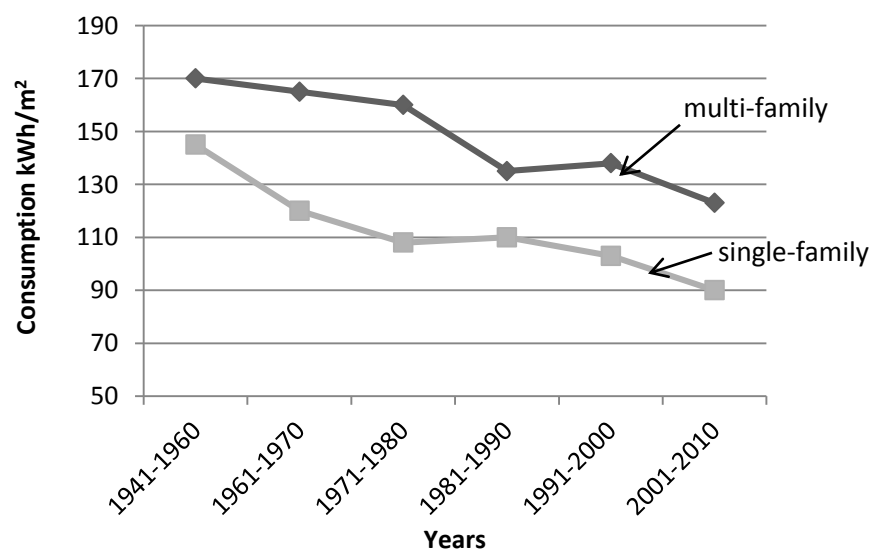


Figure 4-28 Trend of average energy consumption for space heating and water heating in Sweden kWh/m² (source: modified based on data taken from vmisenergy, 2012)

The space heating energy demand from multi-family dwellings was about 170 kWh/m² in 60s and reduced to nearly 120 kWh/m² in 2010, which is about 1 kWh/m² reduction per each year. Similarly for single family dwellings space heating energy demand 145 kWh/m² and reduced to nearly 90 kWh/m² in 2010, which is about 1.1 kWh/m² reduction per each year.

Ethiopia would learn from Sweden in various ways, as space heating in Sweden is the complement of cooking in Ethiopia. Such reduction would mean

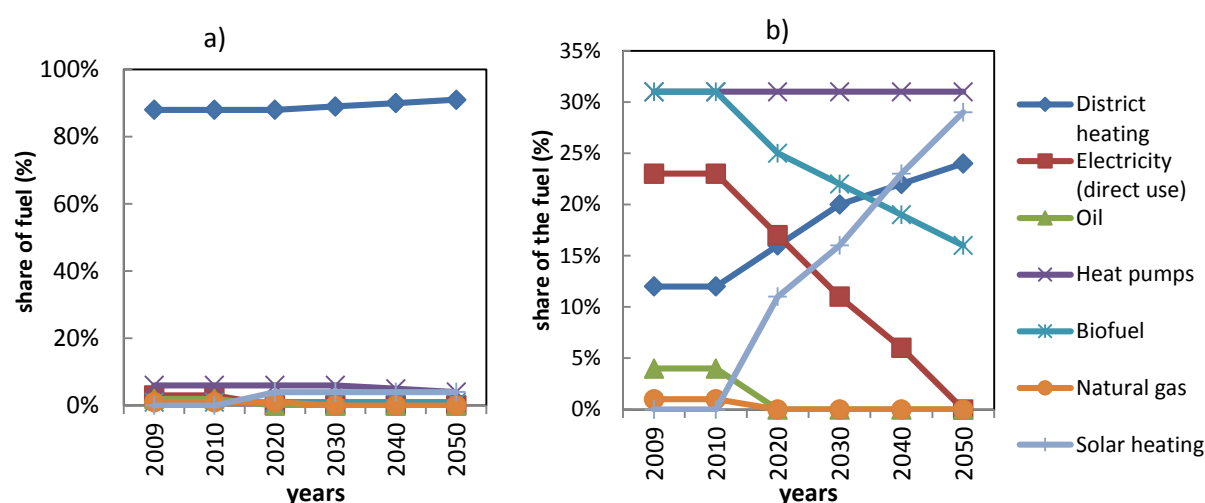


Figure 4-29 share of various energy source used for residential heating, (a) for apartment buildings b) for single family houses. District heating is the dominant heat source in apartment buildings while in single family houses heat pump dominates with solar heating replacing other energy sources.

Figure 4-29 shows the share of various energy sources for space and water heating purposes in Sweden between 2009 and 2050 based up on data from Swedish Energy Agency (2011). The figure 4-29 (a) shows that the share of district heating⁹ in the energy carrier grows as the result of the fuel shift from natural gas oil. Direct using electricity for heating over the period until 2050 the share of the solar energy (as a heating fuel) grows slightly to cover nearly 5 % of the heating energy demand by 2050.

⁹ District heating is hot water coming from industrial processes waste heat and are used for heating houses.

For single-family houses, (fig. 4-29 b) heat pumps and biofuels are mostly used as a direct energy source for heating in residential sector. The share of biofuel drops between 2010 and 2050 while the share of district heating in 2050 becomes twice as big as that of the 2010 value. The share of solar energy also grows significantly and to be leveled with that of heat pump in the energy mix.

Overall the household energy consumption in Bahir Dar, Ethiopia is highly dominated by traditional biomass resources and this trend will continue for some decades unless effective policy measures are implemented. The draw back in the current consumption is its poor conversion efficiency (10-20%). Fuel shift towards electricity would be an effective alternative, but this still will be limited with the power available on the grid. Improving the efficiency of the wood and charcoal stoves would save significant amount of biomass resources.

Chapter 5. Discussion

The objective of this study is to statistically and computationally investigate the energy consumption pattern in single-family residential buildings in Bahir Dar, Ethiopia. The study was conducted by taking Bahir Dar city as a case study, which could represent the majority of the urban areas in Ethiopia and perhaps some developing countries. On the study, exploring the current energy consumption concerning the amount and type of fuel and projecting the future energy consumption based upon various scenarios is conducted. The trajectory of the change of the energy consumption through time is highly important to understand the future energy profile both in terms of quantity and in terms of quality.

The result shows that the majority of residential energy consumption in Bahir Dar city is spent for cooking sector. Its fuel mix is dominated by traditional biomass; most households in Bahir Dar use firewood for baking, and charcoal for cooking activity. Electricity is also consumed for preparation of food in significant portion of the sample households. Households that use modern fuel like electricity tend to spend less energy per capita per household than the traditional biomass fuels. This is due to the fact that electric stoves have higher thermal efficiency compared to the biomass stoves (Here the electricity generation efficiency is not considered, even if we consider it, for hydropower plant the conversion efficiency is relatively higher unlike other conventional power generations.) The other reason that electric stove (also the same for gas stove)

In spite of the improved cook stove campaign most households are still using traditional three stone stove. Most attempts were on improving efficiencies-efficiency improvement is not the only improvement of cook stoves. The other very important factor in improving efficiency are the public perception in adoption rates, usability, cultural acceptance, performance, size and durability all affect this. Designing stoves with local end users helps to produce more culturally acceptable products that a one size fits all stove from a factory in China won't achieve

This study reveals that the share of kerosene and LPG as cooking fuels in households being currently low compared to other fuels. This indicates that several people seem to shift

from kerosene or LPG to charcoal and electricity as the result of the rise of fossil fuel price and inconsistent supply.

Similar studies conducted by several scholars have also shown the dominance of usage of traditional fuels in urban areas of Ethiopia. Study conducted by Taddese (2013) in Mekelle city, Ethiopia, using similar approach shows that the majority of households use traditional biomass intensively.

When we compare the urban household energy consumption of Ethiopia based on this study with other countries the difference is apparent. Study conducted by Mohamed (2010) on the household energy consumption of West Bank and Gaza Strip showed that around 65 % of the total energy is consumed in households. However his study showed that the dominant fuel used for cooking in Palestinian Territory being LPG the percentage of households using it estimated around 98.8%,. Around 10% of the households used wood as their cooking fuel, and the same share (10.0%) of the households used electricity for cooking, while 0.6% of the households used kerosene burner (Mohamed, 2010).

There is difference in energy consumption both in quantity and quality between different types of houses. However when we compare the modern and the low income house types, the majority of the modern high income house type (villa houses) found to be using more modern fuel basically electricity for their cooking activity, the low income households in contrast found to be using more of traditional biomass. The modern villas and apartment buildings which are mostly occupied by high income level occupants tend to consume more electricity than traditional fuels. Especially in apartment buildings, where using traditional fuels is not convenient (due to space limitation) the residents prefer to use electricity for cooking than firewood or charcoal. In contrast, in low income type buildings there is more consumption of traditional fuels although long-term shift from traditional fuels to modern fuels is also apparent.

Energy consumed for electronic appliances is currently low in Bahir Dar however the share is growing fast which indicated the demand for electricity will grow in the future. Thus, enhancing efficiency of appliances is essential for diminishing power utilization. Most developing countries, energy naming that advises buyers about the energy utilization of

appliances are standard practice and mandatory. Marking such as energy star will make people aware. This will reduce the power consumption of the appliances still giving the required output. In Ethiopia, considering the growing number of users of consumer electronics, such move would have high impact overall energy efficiency measures.

There should be target for increasing proportion of share for energy efficient appliances in the market to cope with the rising number appliance of users which in turn raises the electricity demand. With increasing energy efficient appliances, the overall consumption will be managed.

The result showed that lighting is consuming significant amount of energy mainly electricity in Bahir Dar. The study showed that significant portion of households use normal incandescent light bulb. However, other types if those households use Compact Fluorescent Light bulbs or Energy Saving Light bulbs, which have less power consumption compared to the incandescent light bulbs, huge amount of electricity would be saved. The reason for people to prefer to use incandescent light is its low capital cost compared to that of Compact Fluorescent Light bulbs or Energy Saving Light bulbs. However, at the end, due to its energy saving and longer life, compact fluorescent light bulbs is cheaper than incandescent light bulbs. The main challenge for compact fluorescent light bulbs to break the market is the availability of Chinese made poor quality bulbs that fail shortly. Apart from this, if effective quality control is made to keep only the best quality light bulbs in the market coupled with some incentive in tax reduction, huge amount of electrical energy would be saved.

Another energy saving end-use activity is water heating. The result shows that, in 2014 Bahir Dar, water heating for shower purpose is relatively low to less number of households owning water heater as the result of the financial limitation and the naturally gifted weather with optimum temperature. However, with the rising number of more modern houses, the share of using hot water for shower is growing significantly. The water heating is mostly done by electricity over conventional storage type electric boilers while the share of other types of boilers like instant water heaters is less. However instant water heaters are more efficient, so if the public tend to use those instead of the conventional storage type, there will be significant amount of energy saving. In addition, Bahir Dar city has high

solar energy potential and if solar water heaters are used there will be significant saving of electricity from its value of 25 GWh/year that is expected in 2030 in BAU scenario. The adoption of solar water heaters, however, will be hindered by their high initial cost and low public awareness. This would be overcome by some type of incentive (like loan or discount) is made by the government and by providing public campaign (Han, Mol, & Lu, 2010).

Making the comparison of price of various fuels is one of the objectives of the study. The price is classified into primary and useful fuel price per kilo watt-hour of energy (fig. 4-12 & fig. 4-14). In terms of the primary fuel price, LPG was found to be the most expensive fuel with almost double the price of kerosene which is the second expensive fuel. Other fuels like firewood, charcoal and electricity are cheaper, considering their primary fuel price. However, with useful fuel price, fire wood used on three stone stove is found to be the most expensive fuel on the other hand electric used for cooking but it was the cheapest in terms of useful fuel price. This is because traditional three stone stove has very poor efficiency which at the end cost a lot of money compared to using electric (fig 4-13 a). Fire wood on traditional three stone (open fire) stove is used in low income households (fig. 4-8 & 4-9). Nowadays the public understands that using electricity being cheaper than wood or other biomass resources and has high motivation to make fuel shift. The problem, however, is the inconsistent supply of electricity (both in terms of frequent power cut and insufficient power capacity) and the required high initial investment for purchasing and installing baking and cooking electric stoves. But as the income of the households grow and coupled with the rising biomass fuel cost and the expected reduction in the local electric stoves, the likelihood of fuel shift is higher.

Therefore, there are two options to reduce the price of fuels per kWh for wood energy. The first one is decreasing the cost of the fuel at the market –which is less likely to be possible. The second option is perhaps increasing the efficiency of an device and, then the energy useful energy delivered per unit of wood or the fuel used is increased. Thus, the focus should be improving the fuel efficiency. For example, if the average thermal efficiency of the stove is increased from 10% to 20%, this indirectly is reducing the fuel price.

Figure 4-15 to 4-20 show the comparison of business as usual and mitigation scenario in forecasting the future energy consumption for Bahir Dar city. The results indicated the

amount and fuel mix of energy consumption in Bahir Dar is changing in alarming rate. In the business as usual scenario significant growth is expected that will lead the energy consumption in Bahir Dar to be as high as 500 GWh/yr being dominated by traditional biomass fuel sources like firewood and charcoal. The mitigation scenario which shows the possibility of containing or reducing the energy consumption up to 2030 compared to the 2014 value. The mitigation scenario requires effective engagement of various stakeholders like the policy makers, the public and the energy sector including effective management of the fuel shift (essentially from traditional to modern fuels like electricity). This reduced the energy consumption for cooking sector a lot. This is because when firewood or charcoal is used most of the energy is wasted due to poor efficiency of the devices (stoves) but electricity is used on better devices (stoves). Therefore, the overall energy consumption for cooking drops due to fuel shift.

The study demonstrates a decent long-run energy move prospect in Bahir Dar, Ethiopia. As household income improves, households switch to cleaner fuels. This agrees with other studies such as Nnaji et al., (2012) and the energy ladder theory. However the fuel shift is only towards electricity, for example from charcoal to kerosene for cooking wet and from firewood to electricity for baking injera, as implied by the energy ladder hypothesis. The example of vitality move is that, over the long term, people will shift to electricity more this might be burden for the available grip power. Accordingly, if Ethiopia figures out how to exploit its naturally gifted hydroelectric power potential and supply power at reasonable cost to the individuals, fuel shift could be effective and then significant number of the energy related issues could be resolved.

As fuel shift may not be solely effective due to various reasons including availability of consistent electricity and cost of electricity, enhancing the cooking efficiency even the firewood stoves and charcoal stoves is vital option. The current move from traditional stoves to cleaner and energy efficient stoves is important. However hoping to shift just to electricity in the future when things are fulfilled, people still use traditional biomass on non-efficient stoves is expected to use it for longer period. Therefore, promoting efficient way of consuming biomass for cooking is important.

The result of household energy consumption in Bahir Dar, Ethiopia was compared with that of Sweden as a case study western country that effectively implemented energy efficiency measures in residential sectors. Here Bahir Dar could not represent the rural part of Ethiopia but still possible to compare with the national data available for Sweden. While the Swedish per capita residential energy consumption is dropping, in Ethiopia the value is increasing. In Ethiopia the continued usage of inefficient stoves and traditional biomass coupled with high rate of growth in ownership of household appliances like refrigerators, washing machines, large sized TVs , coffee grinders, juicers, etc. cases increasing trend for energy consumption. In Sweden, on the other hand, the dominate sector for residential energy consumption is space and water heating, so the energy efficiency measures target in reducing those aspects. There might be split chance in reducing energy consumption from appliances this would by providing policy restriction in newly diffused appliances, but the effusion of appliances might be fast as households already have sufficient appliances. In Ethiopia, however there is more room to reduce household energy consumption from newly introduced appliances by setting regulation to reduce their wattage. This will be huge benefit in terms of reducing long-term energy consumption.

Chapter 6. Conclusions

With the growing of population number and the improvement of living standards the share of residential buildings' energy consumption has increased in a great amount in urban buildings of developing nation, Bahir Dar city, Ethiopia. This fact has made energy consumption in urban residential sector to be one of the major parts of the total consumption. As the result, this sector has concerned and lead for several studies to be made in various countries. This study has aimed to model energy consumption in residential buildings of Bahir Dar city, Ethiopia. The survey shows that the per capita energy consumption in Bahir Dar city is about 1478.86 KWh/year giving the total energy to be around 330 GWh/ year in 2014. The majority of the total energy in household being consumed for cooking. The type of fuel used for this is traditional biomass, which affects the environment and the health of people.

The study analyzed the projection of energy consumption based up on various scenarios. The analysis result of the study with the a Business as Usual Scenario shows that as the result of the development of the economy, couples with the population with intense rate of urbanization, the demand for energy is increasing very fast while the energy supply is going short as the result of the unsustainable use. The expected long term energy solution through business as usual scenario is energy shift from traditional biomass towards electricity providing availability of electricity. In addition the rise in people income and the need for comfortable and better life will lead to possession of more household appliances for the residents. This will lead to more demand to electricity and the supply may not be sufficient. The 2030 residential energy consumption according to the BAU scenario would be nearly 540 GWh/year, which is nearly 40% more compared to the 2014 value of 390 GWh/year.

The mitigation scenario included several possible scenarios including energy efficiency measures, fuel shift to electricity, and control of appliances energy consumption, and using more solar Energy technologies with Solar Water Heaters and Solar PV. The result shows the possible energy reduction in the future, which is useful in several aspects. It could be seen that if the policy improvements are implemented by 2030, the current energy consumption of 390 GWh/year of the 2014 value could be lowered by 20 % in 2030 to be

as lower as 300 GWh/year. This figure is nearly 45 % reduction of the 540 GWh/year consumption values that could be done in 2030 based on business as usual scenario.

Therefore, that study shows that with effective implementation of the energy efficiency methods would save more than 240 kWh/year in 2030 which is nearly 45 million kg of firewood equivalent per year. The policy makers therefore should strive to implement the policy measures that could effectively reduce the energy consumption in the residential buildings.

The limitation of the study is uncertainty and lack of similar consistent previous historical data the few studies conducted previously could not reveal the end use energy consumption with in various end use categories of the energy consumption sector in the households. The estimation of electrical consumption of the appliances and the light bulbs is based up on the response of the households, as the result the model may not accurately predict the energy consumption. Lack of clearly stated governmental policy over residential energy consumption is also the challenge for mitigation scenario and the model couldn't include sufficient amount of the government plans which could affect the result.

Similar studies in the future may stress more on obtaining previous data, if possible the study could be conducted over years to use own primary data, or else obtaining as many years' data as possible to forecast the future energy consumption accurately. More scenarios such like emerging new types of homes, the rise in number of unstructured home (illegal settlement), the rise in price of electricity would also be included in the scenario model. Rural areas, which represent the major portion of the country, could also be included in future studies. In addition, It could be important to develop a data base of energy efficiency indicators for Ethiopia or even in Africa, comparable to the European Union's program called 'ODDYSEE' that shares information, statistics and achievements of several EU countries.

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I also want to thank the data collectors who showed accurate professionalism in obtaining the data and their applied their effort to approach the households. My gratitude also goes to the 350 households who opened their door and be involved in the study.

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Appendices

Appendix I: Various end use thermal efficiency

Table A-1 various end use thermal efficiency¹⁰

Type of stove and mitad	Thermal efficiency (%)
Lakech Improved Stove	20-25
Clay or mad charcoal stove	29
Metal charcoal stove	18
Traditional enclosed wood stove	10-13
Electric stove	50
Kerosene stove	35-40
LPG stove	60
Open fire three stones	5-10
Mirt mitad	18-25
Gonzye mitad	16-20
Traditional enclosed mitad	10-13
Electric mitad	70-80

Source: Ethiopia alternative development technology and Amhara energy agency office

¹⁰ This is efficiency of energy conversion just on the stove. Any loss of energy during production of the fuel is not considered: e.g. electricity from hydropower up to 95 %, wind turbine up to 35 %, nuclear coal oil, gas turbine power plants up to 35 to 45 %, geothermal or solar up to 18%. Charcoal has also very low production efficiency about 12 % on traditional kiln.

Appendix II Energy Contents of various fuels

Table A-2 Calorific values (Energy Contents) of Domestic Fuels

Fuel Type	Lower heating Value (LHV)
Wood	13.9 MJ/kg
Cow Dung	10 MJ/kg
Agricultural Residue	13.5 MJ/kg
Charcoal	28 MJ/kg
Kerosene	44 MJ/kg
LPG	46 MJ/kg

Source: UNHCR

Appendix III Cost of various fuels in market

Table A-3 Price of various fuels available in the market of Bahir Dar, Ethiopia

Type of fuel	Price of Primary energy ETB/kWh			Stove efficiency (%)	Type of stove	Price of Useful energy (ETB/k wh)
	FUEL Price (ETB/ kg)	FUEL Price (ETB/ kWh)	Variance			
Fire- wood	1.84	0.552	0.013	20	Mirt efficient baking	2.76
				13	Traditional Enclosed mud stove	4.25
				8	Open 3 stones	6.9
				35	Rocket (Tikikil) cooking stove	1.57
Electrici ty	0.42	0.420	0	75	Electric injera baking Stove	0.56
				50	Electric cooking stove	0.84
Charcoal	4.57	0.588	0.003	29	Clay stove	2.03
				26	Lakech	2.26
				18	Metal stove	3.27
				24	Merchay	2.45
Kerosen e	15	1.43	0	35	Kerosene stove	4.09

Type of fuel	Price of Primary energy ETB/kWh			Stove efficiency (%)	Type of stove	Price of Useful energy (ETB/k wh)
	FUEL Price (ETB/ kg)	FUEL Price (ETB/ kWh)	Variance			
LPG	45	3.52	0	65	LPG gas stove	5.41
				8	Open 3stones	2.25
Cow dung	0.5	0.18	0.010	13	Enclosed wood Stove	1.38
				13	Enclosed mitad	1.38
				20	Mirt stove	0.9
Saw dust	0.40	0.01		8	Open 3stones	5
				13	Enclosed mitad	3.08
				20	Mirt stove	2

Appendix IV Trends of energy consumption

Table A-4 Trend of share of fuels in urban areas of Ethiopia (source: Azemeraw, 2013)

Cooking Fuel	1986	1998	200	2004	2011
Wood	61.7	62.9	57.9	66.7	65.67
Charcoal	4.3	5	8.3	7.8	18.19
Electricity	2.7	3.8	2.2	2.4	6.41
kerosene	18.9	17.2	21.5	14.1	5.11
LPG	1	2.5	1.4	2.7	1.09
Biogas	0	0	0	0.0	0.00
Cow dung, saw dust, crop residue, etc.	11.4	8.5	8.7	6.2	3.53

Appendix V LEAP model and results

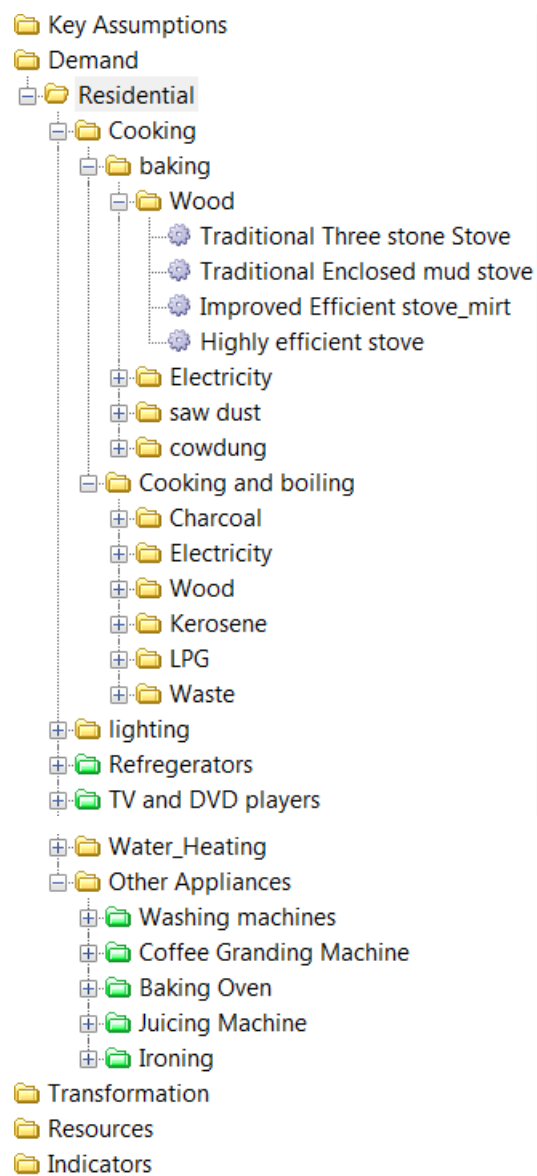


Figure A-1 Data input tree structure to LEAP

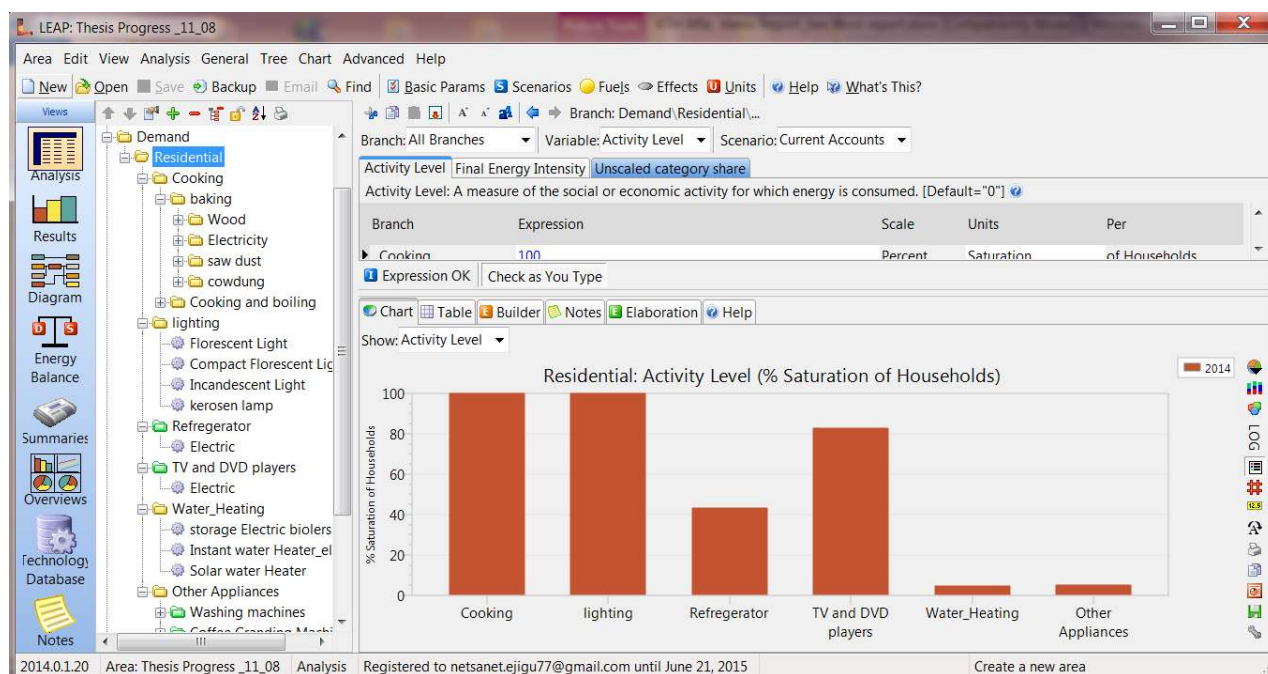


Figure A-2 the leap model

Table A-5 Cost analysis of light bulbs

	LED	CFL	Incandescent
Light bulb projected lifespan, hours	50,000 hours	10,000 hours	1,200 hours
Watts per bulb (equiv. 60 watts), watt	10	14	60
Cost per bulb	\$35.95	\$2.50	\$1.25
KWh of electricity used over 50,000 hours	500	700	3000
Cost of electricity (@ 0.025per KWh)	\$25	\$17.50	\$75
Bulbs needed for 50k hours of use	1	5	42
Equivalent 50, 000 hours bulb expense	\$35.95	\$12.50	\$52.50
Total cost for 50k hours	\$60.95	\$30.00	\$127.50
Total cost for 50k hours per household	\$304.75	\$150.00	\$127.50

Source: Author's modification based upon data from <http://eartheasy.com/>

Appendix VI Questionnaire in local language (Amharic)

ቀን-----

መለያቁጥር-----

አጠቃላይ ጥያቄ ስለቤት ሀይል ፍጆታ (አጠቃቀም)

1. የቤተሰብ ብዛት (አሁን በቤት የሚኖሩ የቤተሰብ ብዛት ወላጆችንን ጨምሮ)-----?

2. የቤት ዓይነት

ሀ/ ቪላ ቤት ለ/ኮንዶሚኒየም/አፓርትመንት/ቤት ሐ/ መካከለኛ አንጨት ቤት

መ/ ዝቅተኛ አንጨት ቤት ሠ/ ጎጆ ቤት ረ/ አንድ ክፍል ቤት/ዶርም/

3. ጠቅላላ የቤተሰብ ወርሀዊ ቢስንትነው-----? (በአጠቃላይ ወርሀዊ ገቢ፣ ክፍያዎች፣ ቤት ክራይ፣ ከንግድ፣ ወዘተ)

4. ቤተሰብ አስተዳዳሪ የትምህርት ደረጃ፣ የሚሰጥ የባል-----
(ምሳሌ፡- 7ኛ ክፍል፣ ዲፕሎማ፣ ድግሪ ወዘተ)

5.1. የሐይል አጠቃቀም ጥያቄ ለእንጀራ

I - እንጀራ ለመጋገር ምን ዓይነት ምድጃ ይጠቀማሉ? ሀ/ ባለ ሶስት ጉልቻ ምጣድ ለ/ ኤሌክትሪክ ምጣድ ሐ/ በጭቃ የሚሰራ ባህላዊ ዝግ ምጣድ መ/ ምርጥ ምጣድ (ማገዶ ቆጣቢ ምጣድ) ሠ/ ሌላ ዓይነት ካለ-----

II- እንጀራ ለመጋገር ምን ዓይነት ማገዶ ይጠቀማሉ?-----

ሀ/ እንጨት ለ/ ኤሌትሪክ ሐ/ ሰጋቱራ መ/ ኩብት ሠ/ የእንጨት ቅርንጫፍ ረ/ የእርሻ ትርፍ ራፊ ቅተላቅተል ሰ/ሌላ-----

III- እንጀራ ሲጋግሩ በየስንት ቀን ይጋግራሉ?-----

(ምሳሌ በ4 ቀን አንዴ፣ በ5 ቀን አንዴ ወዘተ)

IV - በአንድ ጊዜ ምን ያህል ማገዶ ይጠቀማሉ?-----

(ምሳሌ 1ኪ.ግ በአንዴ፣ አንድ እስር እንጨት በአንዴ ወዘተ)

V - በአጠቃላይ ማገዶ ይገዛሉ፣ የገዙትን ማገዶ ለምን ያክል ማገዶ ይጠቀማሉ?

(ምሳሌ አንድ እስር እንጨት ለ 2 ሳምንት፣ ለወር፣ አንድ ኬሻ ኩብት ለወር፣ አንድ ጋሪ እንጨት ለ2 ወር ወዘተ)

VI - ኤሌክትሪክ ምጣድ ለመጋገሪያ የሚጠቀሙክ ሆነ እና ለስንት ሰዓት ይጋግራሉ?

5.2. ለወጥ (ሻይ፣ ቡና ወዘተ)

I - ለወጥ ምን ዓይነት ምድጃ ይጠቀማሉ? ሀ/ የሸክላ ክሳራ ምድጃ ለ/

የብረት ፌርኔሎ ክሳራ ምድጃ ሐ/ ላቀች ክሳራ ምድጃ መ/ ኬርስንጋዝ ምድጃ ሠ/

ፈ/ ኤሌክትሪክምድጃ ሰ/ ባህላዊ 3 ጉልቻ

ሽ/ ሴላ _____

|| - ለወጥወይምቡናምንጻይነትማገደጠቀማሉ? ሀ/ ከሰል ለ/ ኤሌትሪክ ሐ/ ሰጋቱራ መ/ ኩብት ሠ/የእንጨትቅርንጫርእናየእርሻተረፈምርት ረ/ ኬሮሲን (ነጭ)
ጋዝሰ/ ጊዮንጋዝሽ/ሌላ _____

III - በቀንምንያክልጊዜወጥይሰራሉ_____?እናቡና፣ ሻይያፈላሉ?

IV - በአንድጊዜምን ያክል ማን ይጠቀማሉ? _____

V - በአጠቃላይ ማግደ ይዝሉ; የገዙትን ማግደ ለምን ያክል ማገደ ይጠቀማሉ?

(ምሳሌ 100 ኪ.ግ በወር፣ 1 ማዳበሪያ ክሰል በሳምንት፣ ½ ኪሻ (50 ኪ.ግ)፣ በወር ¼ ኪሻ (25 ኪ.ግ) በወር፣ 25 ሊትር ሥሃብ በወር ወዘተ)

VI በቀንበየትኛውሰዓትምግብይዘጋጃል? _____ምሳሌ /12-1:00፣3:00-4:00
ወዘተ/

VII- ኤሌክትሪክም ድጃለወጥማብስያየሚጠቀሙ አንድ ጊዜ ለስንት ሰዓት ይጠቀማሉ? _____

6. ስለምድጃስመራረጥ

1. ከላይ የሚጠቀሙትን የምድጃ አይነት ለምን ይመርጡ ታል?

ሀ) ምድጃውስለሚመች ለ) ርኅሽስለሆነ ሐ) ማገደስለሚቆጥብ መ)

ጭስስላሌላው.

ሠ) የለመድነው-ስለሆነ ረ)ሌላአማራጭስለሌለ ሰ)ሌላ

II. ከላይ የሚጠቀሙትን ማገደዳይነት ለምን ይመርጡታል?

ሀ. ርኅሽሰለሆነ ለ. በቀላሉገብያላይስለሚገኝ ሐ. ስለሚበረክት

መ. የለመድነው ሠ. ሌላአማራጭስለሌለ ረ. ሌላካለ-----

III. ከላይ የሚጠቀሙበትን የምድጃ ወይም የማገዶ አይነት ወደሌላ የመቀየር ግለጥሞች አለዎት

✓ አዎ ሀ. ወደዘመናዊ ማገዶ ቆጣቢ ምድጃ ለ. ወደ ኤሌክትሪክ ምድጃ

ሐ. ወደባህላዊዝግምድጃ ሙ. ሌላካለ-----

✓ አንቀይርም ምክኒያት ሀ/የምንጠቀመውተመችቶናል ለ/ሌላአማራጭአናውቅም

ሐ/ሌላለመጠቀም አቅማችን አይፈቅድም

ሠ/ሴላክማራጨአናገኝም

7. ስለዓየርንብረትመዛባት (Climate change)

በቤት-ወስጥ በምንጠቀመው የሀይልነዳጅ ምርጫ ዓይነት በተያያዘ ችግር ለምሳሌ ስለ ዓየር ንብረት የዓለም መቀነት መጨመር፤ ስለ ነዳጅ ዋጋ መጨመር፤ ስለ ጸንመጨፍ ጨፍ፤

ስለጤናችን ርዕዮተ-ሀገን ዝቅርታ ምን ድንኳን ነው?

ሀ/ በጣም

ለ/ መካከለኛ

ሐ/ ምንም

መ/ ሌላመልስካለ

8. ስለኤሌትሪክአጠቃቀም

I - በቤት-ውስጥምንዓይነት-መብራት-ትጠቀማላችሁ? _____

ሀ/ ፍሎረስንት

ለ/

የሀይልቆጣቢ (ኢነርጂይዘር) አምፖል

ሐ/ኖርማልአምፖልባለ 60ሻማ ወይም 40ሻማ

መ/ ሌላ _____

II - በቀንለምንያክልሰዓት-ያክል-መብራት-ይጠቀማሉ? _____ (ለምሳሌ ለ4 ሰዓት፣ ለ5ሰዓት፣ ከምሽቱ 1:00 ሰዓት-እስከ 5:00 ሰዓት-ወዘተ)

III - በቤት-ውስጥስንት-አምፖል-ትጠቀማላችሁ? _____

9. የተለያዩኤሌትሮኒክስ

I - በቤት-ውስጥቴሌቪዥንትጠቀማላችሁወይ? _____

ሀ/ አዎ

ለ/ አይደለም

ቴሌቪዥንየሚጠቀሙ-ከሆነምንዓይነት-ይጠቀማሉ? _____ (14 ኢንች፣ 21 ኢንች፣ 24 ኢንችወዘተ)

I የዲቪዲዩክርትጠቀማላችሁወይ? _____

ሀ/ አዎ

ለ/ አይደለም

II - ቴፕሪከርደርወይምዲቪዲዩሜወቻወይምአምፕሊፋየርትጠቀማላችሁወይ? _____

ሀ/ አዎ

ለ/

አይደለም

III - ካውያይጠቀማሉወይ? _____

ሀ/ አዎ

ለ/ አይደለም

መልሱአዎከሆነምንዓይነት-ካውያይጠቀማሉ _____

ሀ/ ኤሌትሪክ

ለ/ የክሰል

IV- የውሃማሞቂያ(ቦይለር)ይጠቀማሉ? _____

ሀ/ አዎ

ለ/ አይደለም

አዎከሆነምንአይነተቦይለርይጠቀማሉ? ሀ/ኖርማል

ለ) ቲኒሽ (ማጠራ-ከምያየለላት፣

Atmor)

V - ሌላበቤት-ውስጥምንዓይነት-የኤትሪክመሳሪያይጠቀማሉ? ሀ) ፍሪጅ፣ ለ) የሥጋመፍጫ ሐ)

የውሃማፍያ መ) የቡናመፍጫ፣ መጭመቂያ፣ ሽንኩርት-መፍጫሠ) ልብስማጠቢያረ) ዳቦመጋገርያለ/

ኮምፒውተርወይላትቶፕ ረ) ሌላ-----

10. ማገዶአሰባሰብ

I. የሚጠቀሙትንማገዶከየት-ይሰበስባሉ? ሀ) ግዥ ለ) ከግልይዞታ ለ) ከመንግስት-ወይምከህዝብደን መ) ሌላ----

II. ማገዶውንማንይሰበስባል? ሀ) አባወራ(አባት) ለ) እማወራ(እናት) ሐ) ሴት-ልጅ መ) ወንድ-ልጅ ሠ)

ሌላ-----

ማገዶውንለመሰብሰብስንት-ሰዓት-ይፈጃል? _____